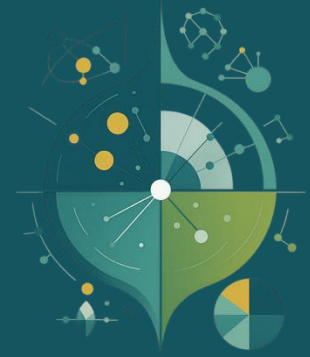




Analysing community data

Manuel Steinbauer

Science School of Quantitative Ecology 2025

bit.ly/SSoQE



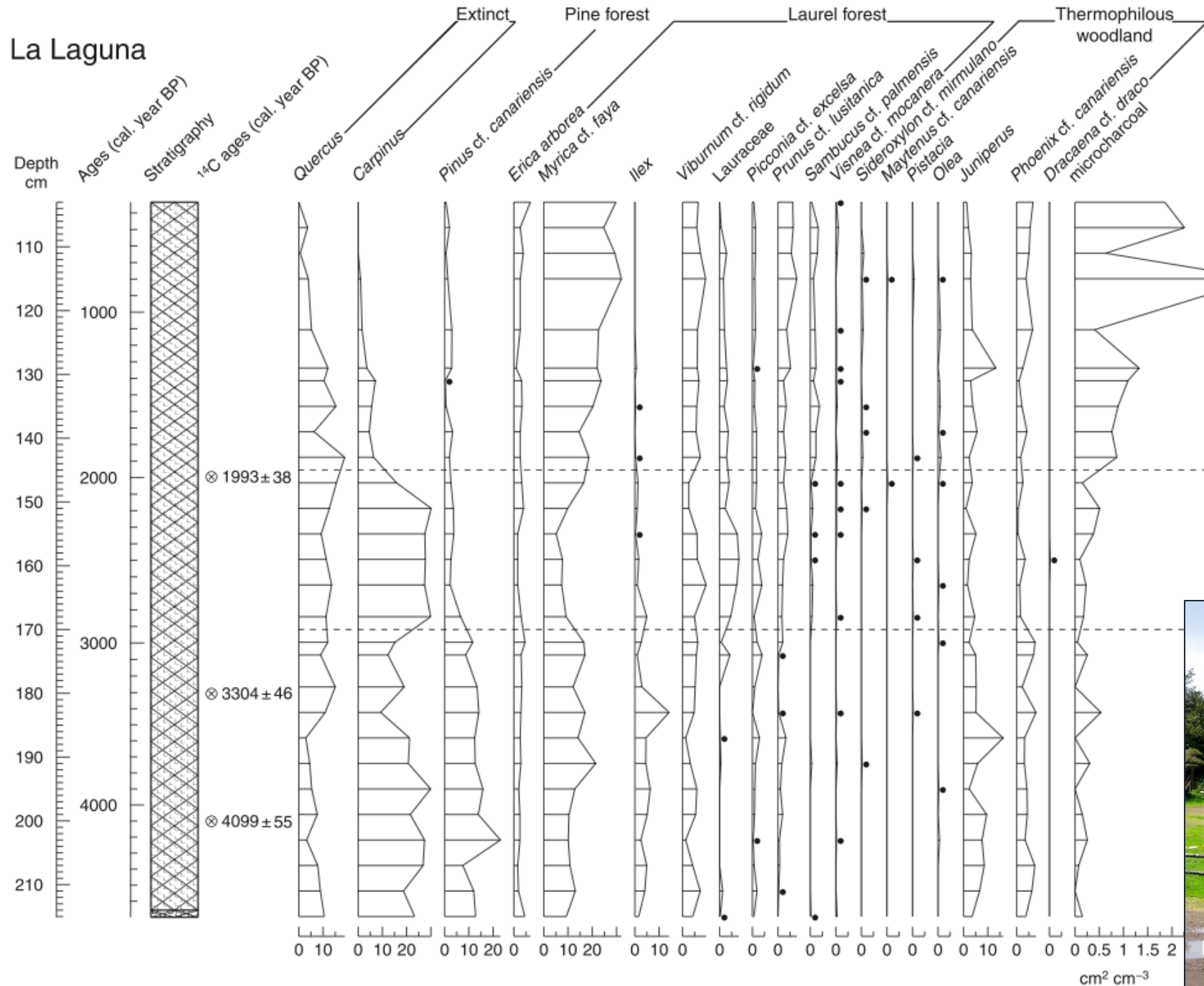
Aims

1. Understand the structure of **species community data**
2. Be able to interpret **Ordination and Cluster** analyses
3. Be able to **build a simple Ordination** in R.
4. Know **resources** for future analyses of community data

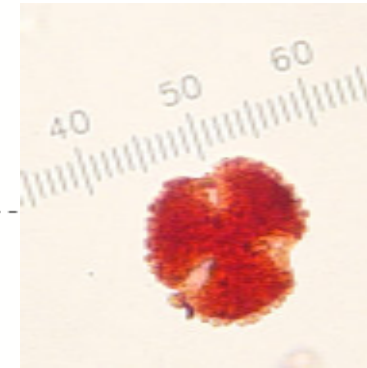
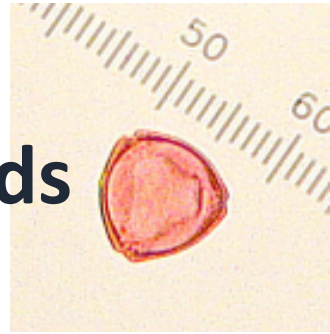


Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de

La Laguna

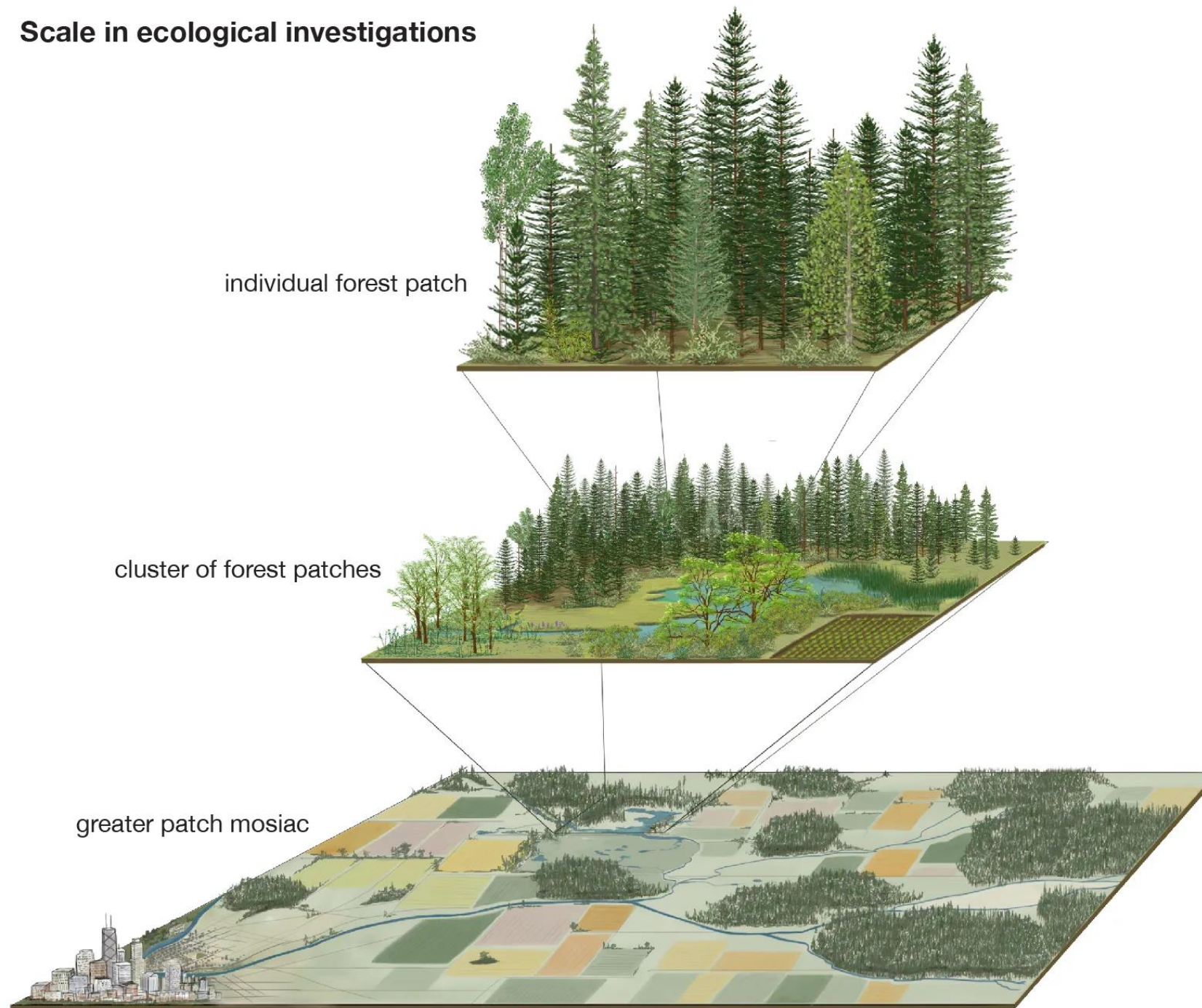


Tenerife, Canary Islands

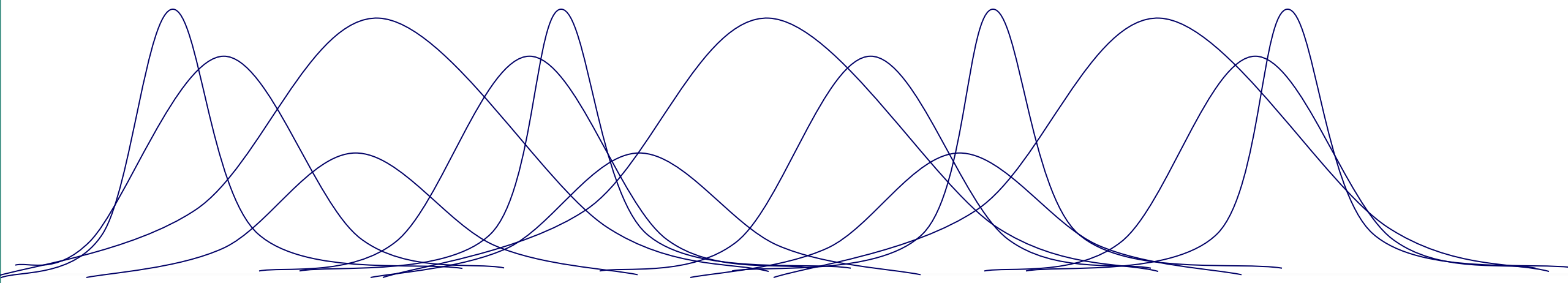
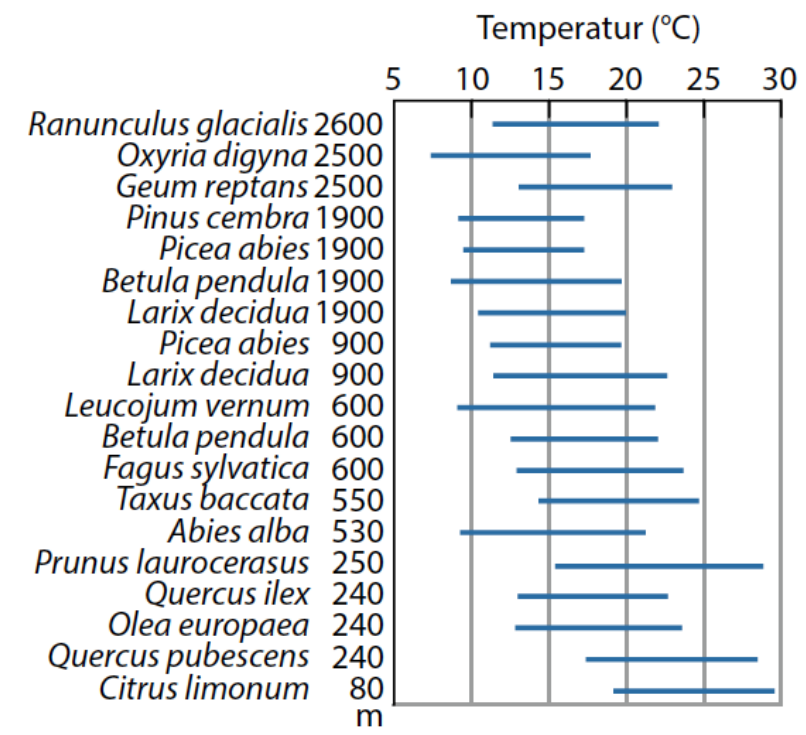




Scale in ecological investigations



Environmental niche

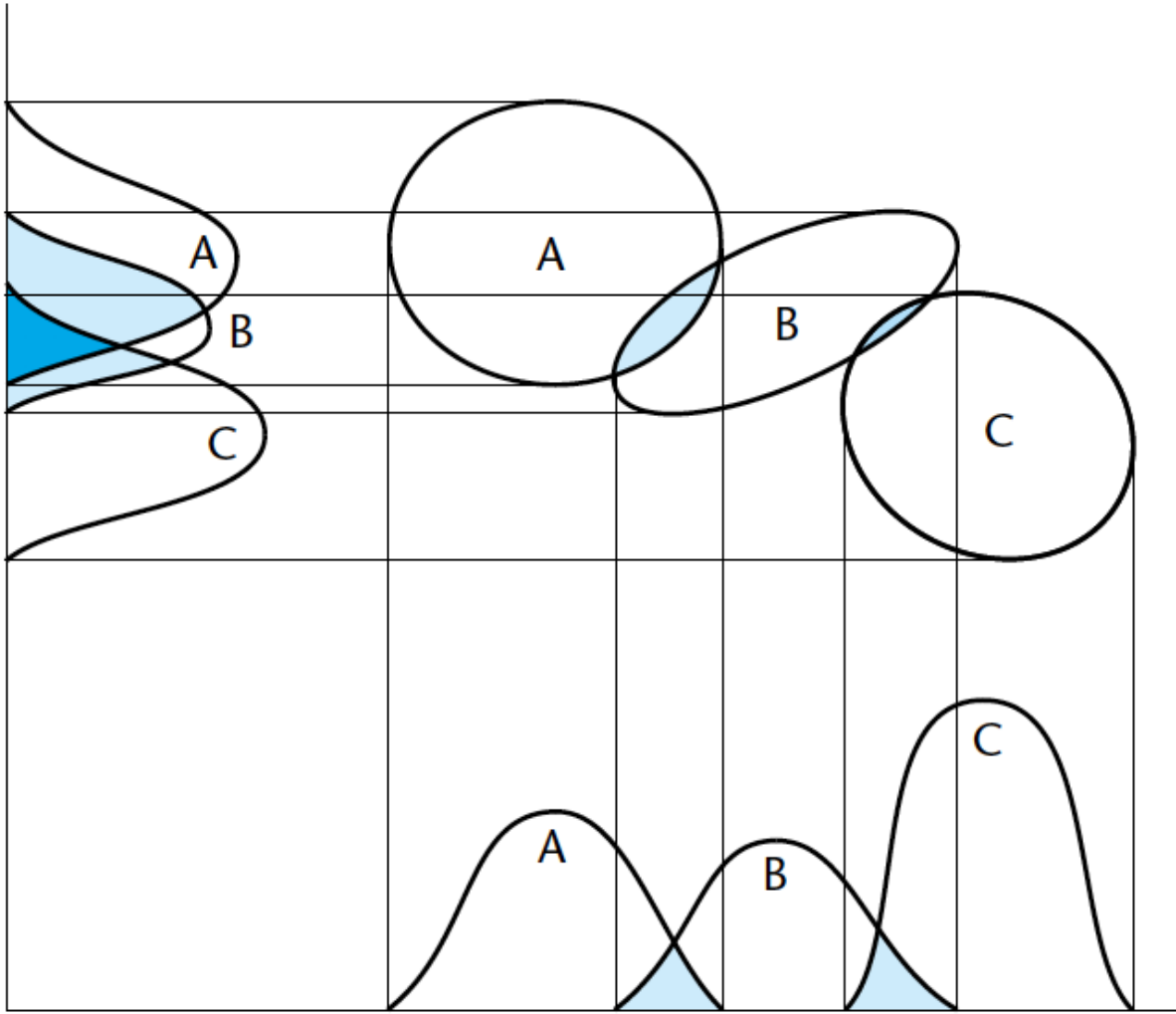


Temperature

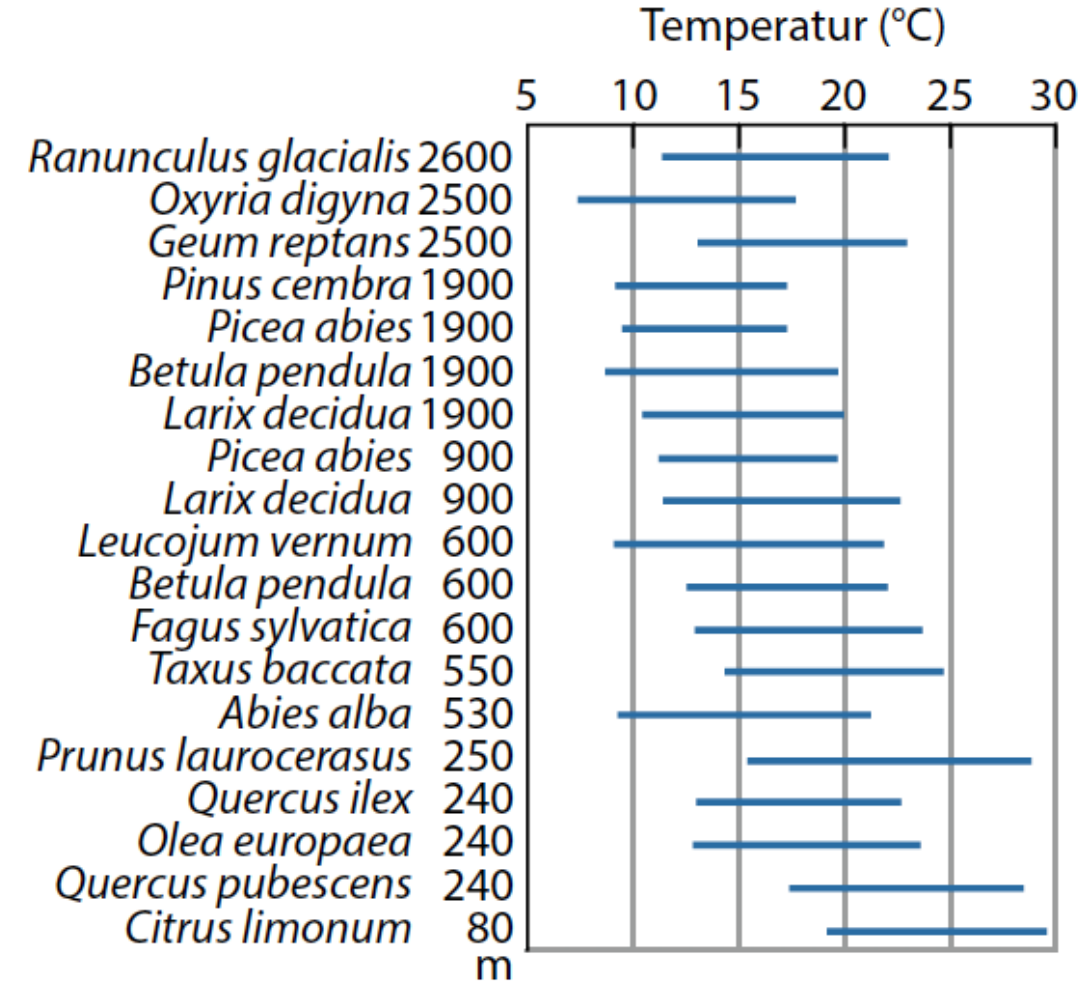
Environmental niche

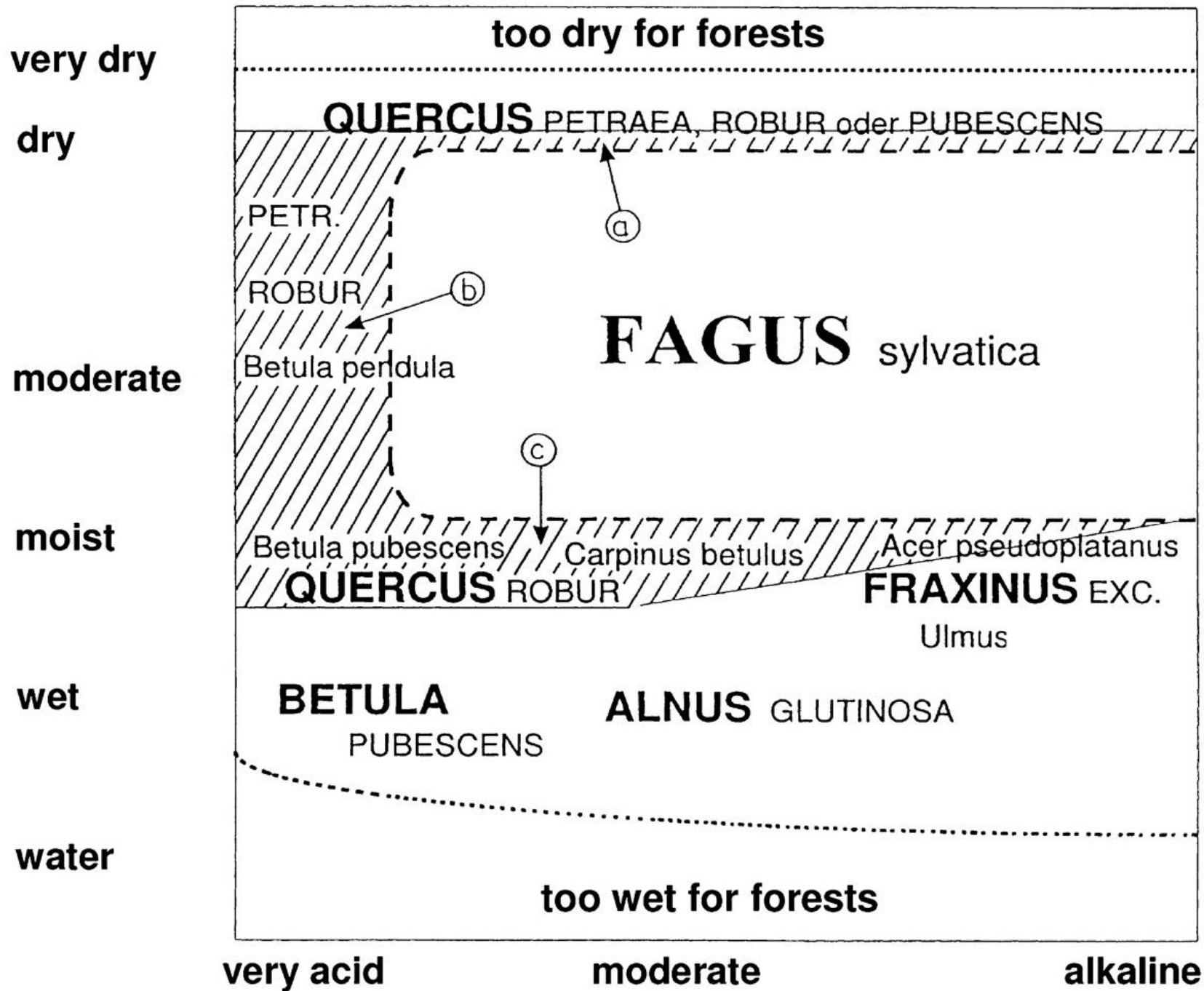


Environmental gradient 2

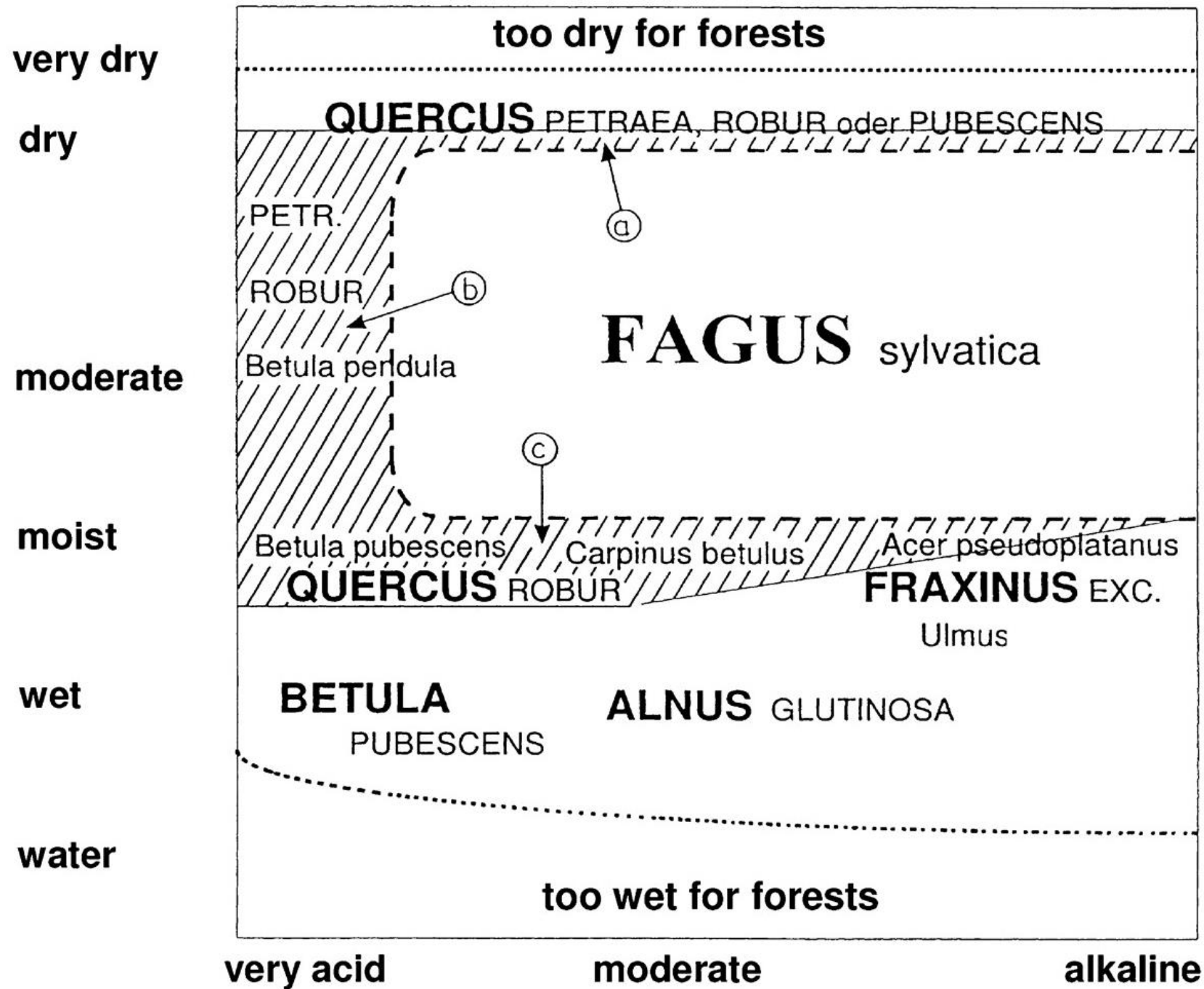


Environmental gradient 1

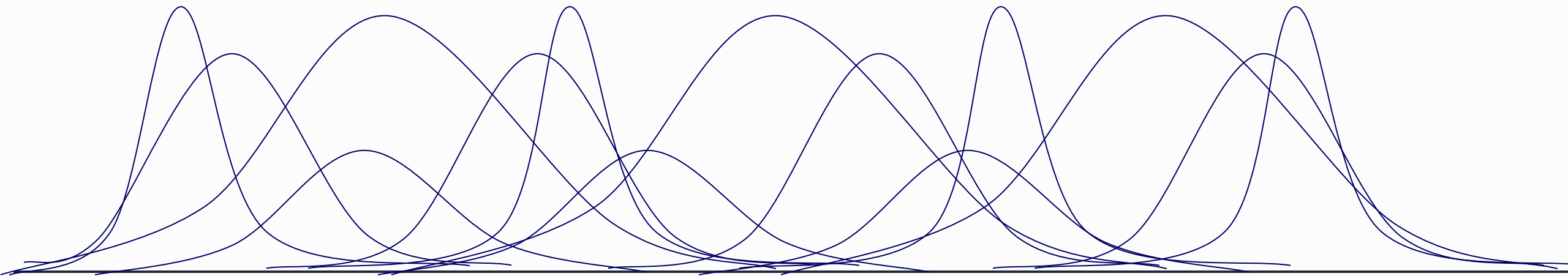




(from Ellenberg 2001)

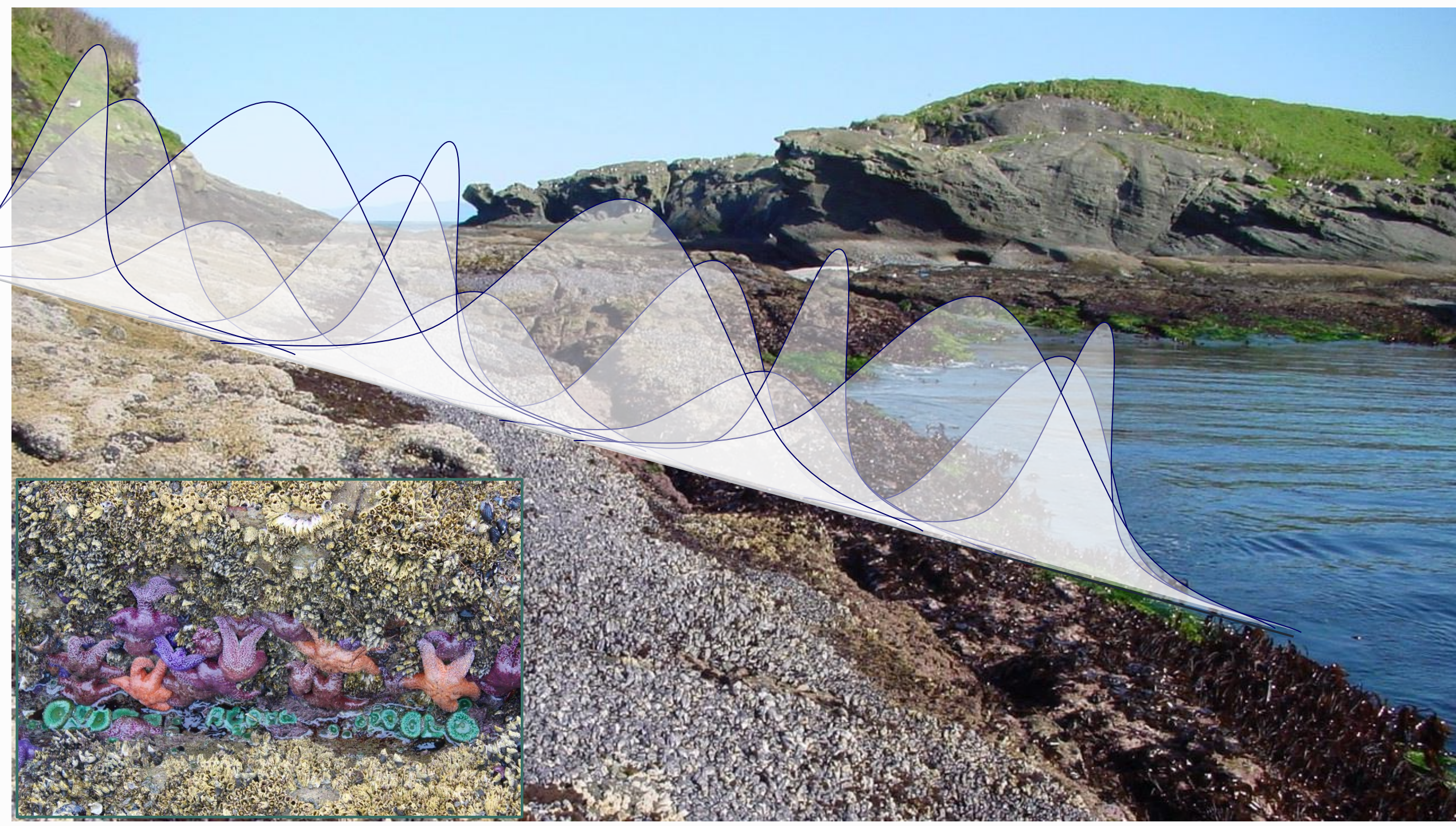


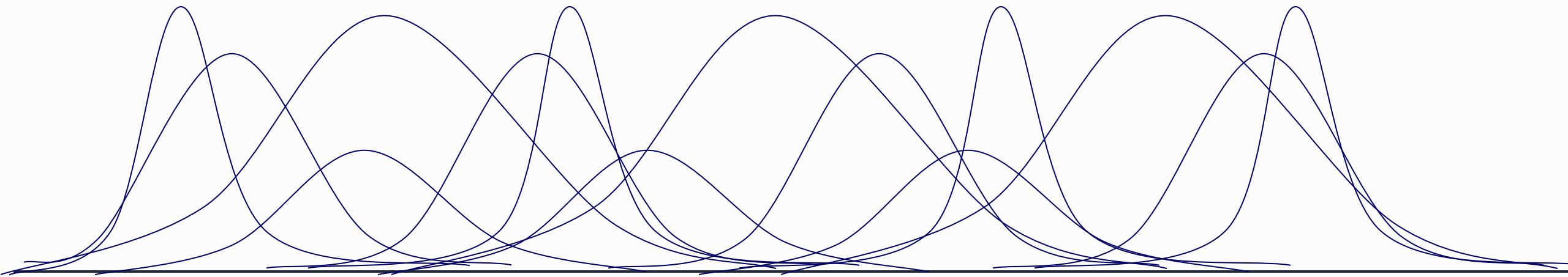
(from Ellenberg 2001)



Environmental gradient







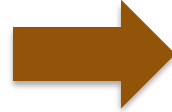
Environmental gradient

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Agrocani							4	5									6			1														
Agrocapi	3	4	2													8				1	1						4		5	3				
Agrorepe		1	1		7	5			4		4					9	3			3	5					5			9	5				
Agrostol						6			4	12										5		1				2	3					8		
Alopgeni									4	1										2					3		5					6		
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6					5							1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat				2			2	1					1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1			7	2			4	3				1	2				2						1	
Careprae		2	2		1	1										2		1								2		1	3	1				
Carevesi			4			1	4										1	7				2												
Carevulp								1		1		1	1						3		1				1									
Cirsarve							1							4						3											1			
Cirslanc																							1											
Cniddubi																													2					
Desccesp															8		1		5	4									3	1				
Eleounig											1	2							1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																					1					
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3										4														4		1	4				
Glycflui									1										2						7						2			
Glycmaxi									1			2											1	2	3		1				1	11	10	
Holclana						4	2						4				1		2	6										1				
Junceffu			1			1	1						1								4	1				1								
Lathprat													1	1							1									1				
Lotucom			1	1		1															1									1				
Lychflos			1			2															1													
Phalarun			5			1	2	2	1	11	8	2					1		1	2	1	1	6			1					1			
Planinte									3	3	1									1						1								
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Poa praA	3	2		1	5	6	1						3		1				1		4		2		2			4	2					
Poa triv			1		2	1	1	4												2		2	3		3	1								
Poteanse			1			2									1			1																
Poterept								1											1							1								
Ranuflam				1			1	3					1								1							3						
Ranurepe			4	1	1	1	2	4	1	4			1	3			3		5	4	4	2	6		9	9								
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Rumeacet				1			4														4							1			1			
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Siumlati				1				2				1						3					2	1			1							
Stelpalu				2			1				1	1	1					1	1	2			1	1										
Sympoffi																							3	1										
TaraoffA			1	1	1	1					1					1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9		4																		1	2		
Vicilath		1																																
Vicitetr		1	1										2														1	1						

Species

Sites

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
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Careprae		2	2		1	1										2		1							2		1	3	1					
Carevesi			4			1	4									1	7				2													
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Cirslanc																						1												
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Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																					1					
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TaraoffA			1	1	1	1					1					1			1	1	1				2				1	1				
Trifrepe			2			3													4	4	1	1												
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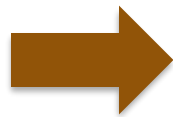
Species

Sites

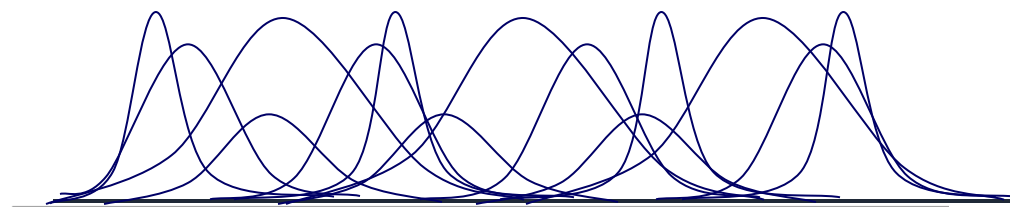
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Vicitepr		1	1	1				1				2																						
Cniddubi							2																											
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Lotucom				1		1					1		1							1														
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Cirsarve						1						4		3		1																		
Rumeacet			1			1						4	4			4			1															
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Festprat							1									2																		
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Lychflos												1	1			2			1						1									
Trirepe													1	4		3		4	2	1														
Caredist													2											1										
Junceffu												1	4			1			1	1														
Carevulp												1	1			1		1	3				1	1										
Poterept																1			1				1											
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Ranurepe									1		1	3	4	4	9	1	4	5	4	2	3	4	1			6	2	9	1					
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Agrocani												1				4					6						5							
Cirslanc																					1													
Sympoffi																					3					1								
Caltpalu																				1	6	1												
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Carevesi																	1			4	2	1			7		4							
Ranufllam												1					1			1				1			3	3						
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Agrostol											6					2			5		1		4					3	12	8				
Galipalu												2		1	1			1		1			1	3	3	3	2	2			1	1	1	
Alopgeni																		2					4					5	1	6				
Roriampb																																		
Glycflui																																		
Glycmaxi																																		

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																			1					3				
Agrocani							4	5									6			1														
Agrocapi	3	4	2													8				1	1						4		5	3				
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Agrostol						6			4		12								5			1				2	3					8		
Alopgeni									4		1								2					3		5					6			
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6						5						1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat			2				2	1					1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1			7	2			4	3				1	2			2							1	
Careprae	2	2			1	1										2			1						2		1	3	1					
Carevesi			4				1	4									1	7				2												
Carevulp									1		1		1	1					3		1				1									
Cirsarve							1							4						3										1				
Cirslanc																							1											
Cniddubi																													2					
Desccesp													8				1		5		4								3	1				
Eleounig											1		2						1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																					1					
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3										4														4		1	4				
Glycflui							1												2					7								2		
Glycmaxi							1				2											1	2	3		1					1	11	10	
Holclana							4	2					4				1			2	6										1			
Junceffu			1				1	1					1							4	1					1								
Lathprat												1	1								1									1				
Lotucorn		1	1		1																	1								1				
Lychflos			1			2							1								1			1										
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Planinte									3	3	1								1							1								
Poa palu			1		1	1		11		4		1		1			1		6	2		4	1		6	3								
Poa praA	3	2	1	5	6	1							3		1				1		4		2		2			4	2					
Poa triv			1		2	1	1	4												2		2	3		3	1								
Poteanse			1			2						1						1			2													
Poterept									1										1							1								
Ranuflam			1			1	3						1									1					3							
Ranurepe			4	1	1	1	2	4	1	4		1	3				3		5	4	4	2	6		9	9								
Roriamph									4		7													1			1							
RorisylA							1	2	1						1				1				3		4									
Rumeacet				1			4							4								4										1		
Rumethyr	1	5	2		3	5										3										2		4			1			
Siumlati				1			2				1						3					2	1			1								
Stelpalu				2			1				1		1	1			1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA			1	1	1	1					1						1		1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac			1			1					9		4																					
Vicilath	1																												1	2				
Vicitetr		1	1											2													1	1						



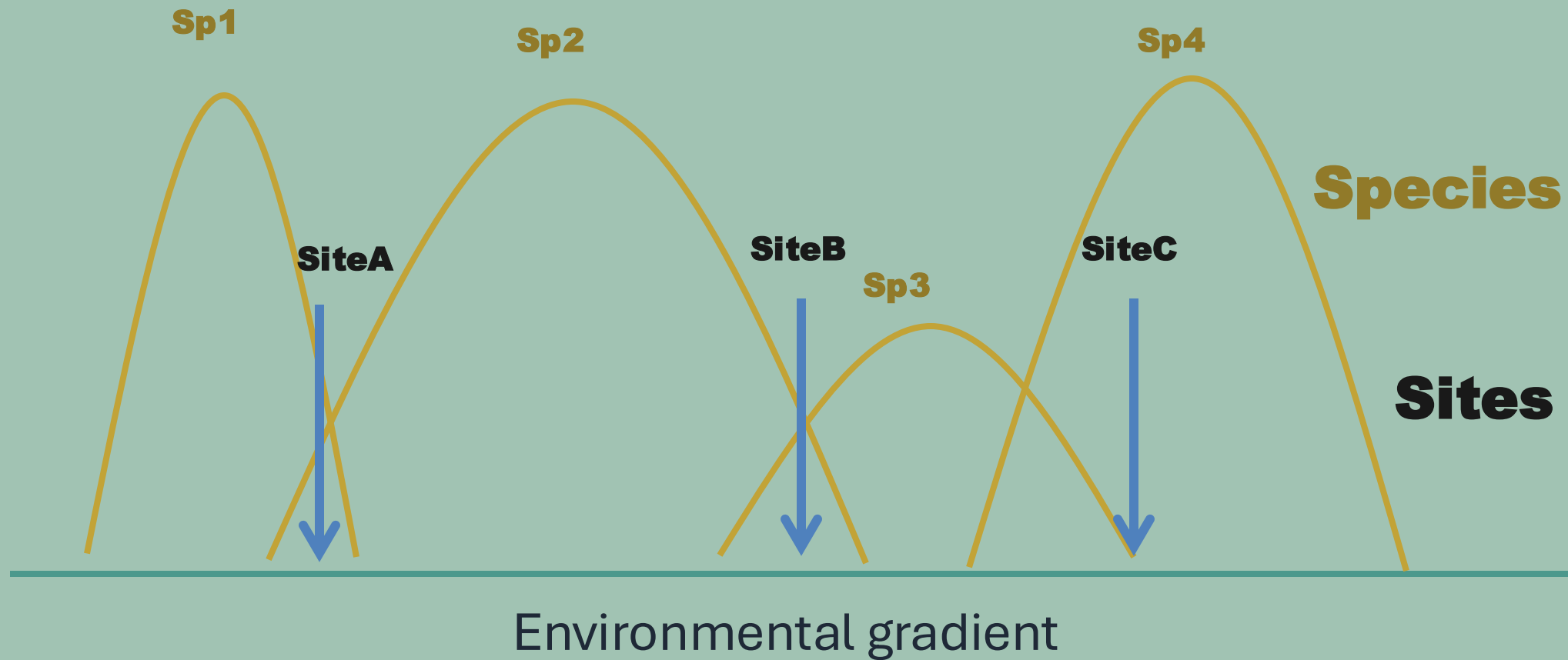
Species

[illegible]



Ordination - Correspondence Analysis

What is it trying to do?



If you knew the species or the sites you could position the other

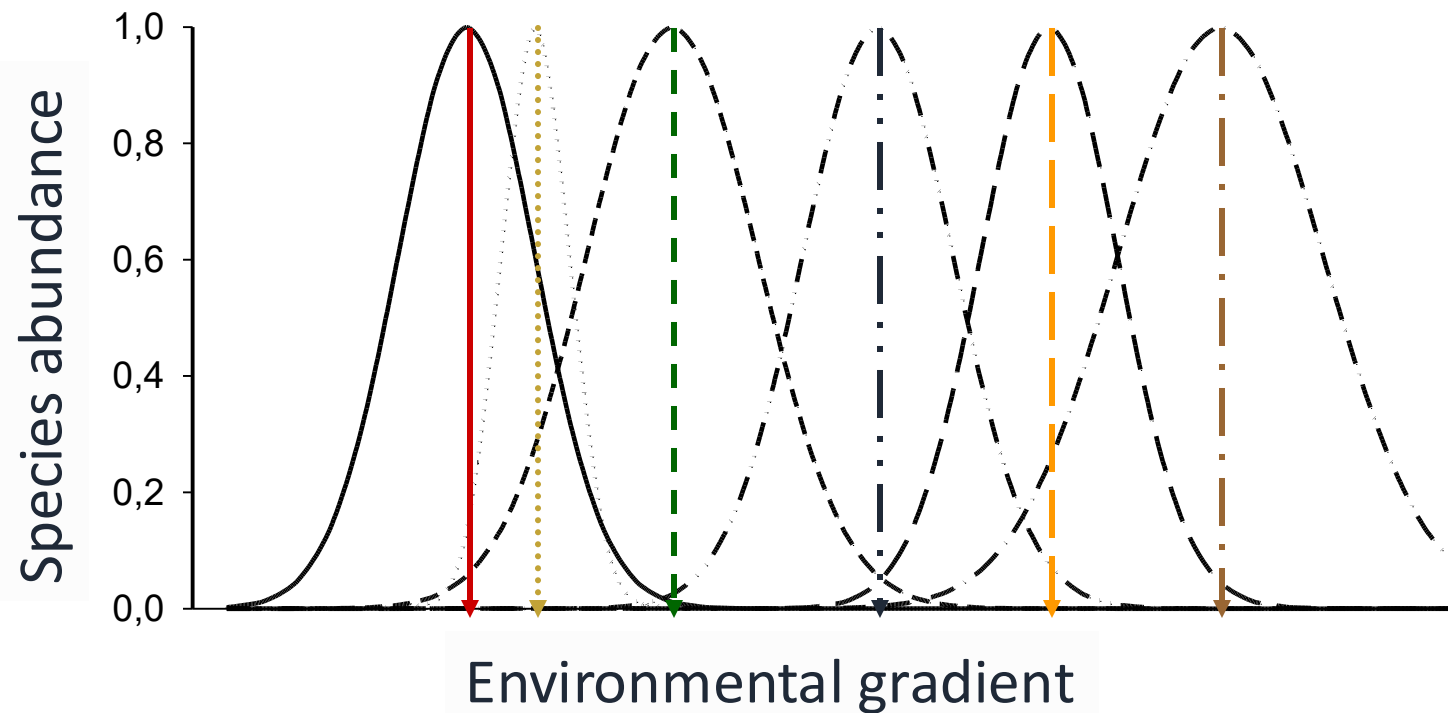
Correspondence Analysis



Assumes a **unimodal** relationship of species along important gradients

The principle:

Estimates the **optimum** of a species using a **weighted average**



[illegible]

Correspondence Analysis



	1	2	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	8	26	10	31	12	24	33	32	GM	
Vicilath	1																																-1.441	
Achimill	4	1								1			1																				-1.218	
Agrocapi	3	4	4	2	8	3	5					1								1													-1.217	
Euphesul			1			1	1																										-1.194	
GaliverA	1	1	4	3		4	1				4																						-1.141	
Eropvern	1	1						1		1																							-1.117	
Rumethyr	1	5	4	2	3	1			3	5				2																			-1.083	
Vicitetr	1	1	1					1			2																						-1.027	
Cniddubi						2																											-0.967	
Careprae	2	1	2	2		1	3	1		1				2			1																-0.955	
Anthodor	6	4		4						1	1				5						1												-0.946	
Lotucorn			1		1					1	1								1														-0.653	
Poa praA	3	2	1		2	4	5		6	3	4		2	1		1								2									-0.633	
Agrorepe	1	1	3		5	9	7	9	5			5	5		4	3						4											-0.574	
TaraoffA			1	1	1		1		1			1	2		1	1	1	1															-0.492	
Cirsarve				1						4	3		1																					-0.440
Rumeacet		1		1						4	4		4				1																-0.400	
Lathprat				1						1	1											1											-0.391	
Alopprat	1	1	2		2	3	7	4	7	1		5	5	1	5	4		3		2	1		2		1								-0.378	
Descocesp				1	3					8	5									4	1												-0.357	
Vicicrac				2	1					4			1	9		1																	-0.287	
Holclana				1						4	6	2	4								1					2							-0.219	
Festprat					1										2											1							-0.186	
Cardprat				1						3	2	1	2		1	2	1				1				1								-0.117	
Poteanse										1	2		2				1						1										-0.091	
Lychflos										1	1				2									1									-0.059	
Trifrepe										1	4		3		4	2	1																-0.034	
Caredist										2													1										-0.015	
Junceffu										1	4			1			1	1								1	1						-0.001	
Carevulp										1	1		1	3							1	1											0.021	
Poterept												1				1					1												0.072	
Stelpalu										1	1		1	1	2	2	1	1				1	1										0.097	
Poa triv										2			2	3	1			1	2		4			3	1	1							0.099	
Poa palu										1	1		2	6	1	4	6	1	4	1	1	1		1		3							0.117	
Eleounig												1				1	1					2											0.124	
Ranurepe										1	3	4	4	9	1	4	5	4	2	3	4	1		6	2	9	1						0.131	
RorisylA										1			4		1	1					1		3			2							0.164	
Agrocani											1			4						6					5								0.217	
Cirsanc																				1													0.225	
Sympoffi																				3				1									0.295	
Caltpalu																				1	6	1		5									0.322	
Planinte																1	1					3											0.361	
Carevesi																1			4	2	1		7		4								0.363	
Ranufiam											1				1							1			3	3							0.383	
Caregrac											2			5			4	1	4		7	3	2	8	2	1						1	0.399	
Phalarun												1	2		1	1	1	5	1	1	2	2		6	2	1	1	1	8			0.453		
Siumlati																								1	2	1							0.476	
Agrostol																																		0.512
Galipalu																																		0.577
Aloppeni																																		1.009
Roriamp																																		1.298
Glycflui																																		1.753
Glycmaxi																																		3.117

The principle:

Calculate the **weighted average of species** from **site** values (current order of sites)

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

x_j = Weighted average of species j

n = Number of sites

y_{ij} = Abundance of species j in site i

x_i = Site score of site i



CA – How it Works

Raw data

samples	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>Achmil</i>	1	3			2	2	2			4							2			
<i>Agrsto</i>			4	8				4	3			4	5	4	4	7				5
<i>Airpra</i>																		2	3	
<i>Alogen</i>		2	7	2				5	3			8	5				4			
<i>Antodo</i>					4	3	2			4								4	4	
<i>Belper</i>		3	2	2	2					2									2	
<i>Brohor</i>		4		3	2		2			4										
<i>Chealb</i>													1							
<i>Cirarv</i>				2																
<i>Elepal</i>								4						4	5	8				4
<i>Elyrep</i>	4	4	4	4	4				6											
<i>Empnig</i>																			2	
<i>Hyprad</i>											2							2	5	
<i>Junart</i>								4	4						3	3				4
<i>Junbuf</i>							2		4			4	3							
<i>Leoaut</i>		5	2	2	3	3	3	3	2	3	5	2	2	2	2		2	5	6	2
<i>Lolper</i>	7	5	6	5	2	6	6	4	2	6	7								2	
<i>Plalan</i>					5	5	5			3	3							2	3	
<i>Poapra</i>	4	4	5	4	2	3	4	4	4	4	4		2				1	3		
<i>Poatri</i>	2	7	6	5	6	4	5	4	5	4		4	9			2				
<i>Potpal</i>														2	2					
<i>Ranfla</i>								2					2	2	2	2				4
<i>Rumace</i>					5	6	3		2			2								
<i>Sagpro</i>				5				2	2		2	4	2						3	
<i>Salrep</i>																		3	3	5
<i>Tripra</i>					2	5	2													
<i>Trirep</i>		5	2	1	2	5	2	2	3	6	3	3	2	6	1			2	2	
<i>Violat</i>										1	2							1		
<i>Brarut</i>			2	2	2	6	2	2	2	2	4	4			4	4		6	3	4
<i>Calcus</i>														4		3				3
<i>Hiprha</i>									1									2	1	
<i>Poaann</i>	3	3		4			2		2		2							4		
<i>Ranacr</i>						3	2		2	3			3	1	1					



CA – How it Works

Calculating weighted mean of **species**

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20
Achmil	1	3	2	2	2					4	2									
Agrsto								4	8			3	4	4	5	4	4	7		5
Airpra											2								3	
Alogen		2						7	2			3	8	5	5			4		
Antodo			4	3	2					4	4								4	
Belper		3	2				2	2	2	2										
Brohor		4	2		2				3	4										
Chealb															1					
Cirarv									2											
Elepal														4		4	5	8		4
Elyrep	4	4	4					4	4			6								
Empnig																			2	
Hyprad						2					2								5	
Junart												4		4			3	3		4
Junbuf															3					
Leoaut															2	2	2		6	2
Lolper	7																			
Plalan																				
Poapra	4	4													2					
Poatri	2														9			2		
Potpal																	2	2		
Ranfla														2	2	2	2	2		4
Rumace			5	6	3							2	2							
Sagpro						2			5			2	4	2	2				3	
Salrep							3												3	5
Tripra			2	5	2															
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2	
Violat						2	1			1										
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4
Calcus																4		3		3
Hiprha							2					1							1	
Poaann	3	3			2	2	4		4			2								
Ranacr				3	2					3		2			3	1	1			

1. Iteration
Site score is random =
e.g. order of plots

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

5,9
27
4,6
16
12
4,3
4,2
0,7
0,4
19
5,7
1,9
7,6
12
6,8
30
18
12
17
23
2,9
11
6,2
10
11
2,7
22
2,5
29
8,2
3,2
7,6
7,4



CA – How it Works

Sort species by weighted species scores

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20	
Cirarv								2													0,4
Chealb															1						0,7
Empnig																			2		1,9
Viclat						2	1			1											2,5
Tripra			2	5	2																2,7
Potpal																	2	2			2,9
Hiprha							2					1							1		3,2
Brohor		4	2		2				3	4											4,2
Belper		3	2				2	2	2	2											4,3
Airpra											2								3		4,6
Elyrep	4	4	4					4	4			6									5,7
Achmil	1	3	2	2	2					4	2										5,9
Rumace			5	6	3							2	2								6,2
Junbuf					2							4	4			3					6,9
Ranacr				3	2					3		2			3	1	1				7,4
Hyprad						2					2								5		7,6
Poaann	3	3			2	2	4		4			2									7,6
Calcus																4		3	3		8,2
Sagpro						2			5			2	4	2	2				3		10
Salrep							3												3	5	11
Ranfla														2	2	2	2	2	4	4	11
Antodo			4	3	2					4	4								4		12
Junart												4		4			3	3	4		12
Plalan			5	5	5	3	3			3	2										12
Alogen		2						7	2			3	8	5	5			4			16
Poapra	4	4	2	3	4	4	3	5	4	4	1	4		4	2						17
Lolper	7	5	2	6	6	7	2	6	5	6		2		4							18
Elepal														4		4	5	8		4	19
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2		22
Poatri	2	7	6	4	5			6	5	4		5	4	4	9			2			23
Agrsto								4	8			3	4	4	5	4	4	7		5	27
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4	29
Leoaut		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2		6	2	30



CA – How it Works

Calculating weighted mean of **sites**

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20	
Cirarv								2													0,4
Chealb															1						0,7
Empnig																			2		1,9
Viclat						2	1			1											2,5
Tripra			2	5	2																2,7
Potpal																	2	2			2,9
Hiprha							2					1							1		3,2
Brohor		4	2		2				3	4											4,2
Belper		3	2				2	2	2	2											4,3
Airpra											2								3		4,6
Elyrep	4	4	4					4	4			6									5,7
Achmil	1	3	2	2	2					4	2										5,9
Rumace			5	6	3							2	2								6,2
Junbuf					2							4	4			3					6,9
Ranacr				3	2					3		2			3	1	1				7,4
Hyprad						2					2								5		7,6
Poaann	3	3			2	2	4		4			2									7,6
Calcus																4		3	3		8,2
Sagpro						2			5			2	4	2	2				3		10
Salrep							3												3	5	11
Ranfla														2	2	2	2	2	4		11
Antodo			4	3	2					4	4								4		12
Junart												4		4			3	3	4		12
Plalan			5	5	5	3	3			3	2										12
Alogen		2						7	2			3	8	5	5			4			16
Poapra	4	4	2	3	4	4	3	5	4	4	1	4		4	2						17
Lolper	7	5	2	6	6	7	2	6	5	6		2		4							18
Elepal														4		4	5	8		4	19
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2		22
Poatri	2	7	6	4	5			6	5	4		5	4	4	9			2			23
Agrsto								4	8			3	4	4	5	4	4	7		5	27
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4	29
Leoaut		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2		6	2	30
	1,6	2,8	2,7	2,6	2,7	1,7	2,4	1,0	3,8	2,2	1,1	3,9	2,5	0,6	1,9	1,4	0,4	0,7	3,7	2,3	

$$\tilde{x}_i = \frac{\sum_{j=1}^n \tilde{x}_j y_{ij}}{\sum_{j=1}^n y_{ij}}$$



CA – How it Works

Sort by weighted site scores

samples	15	8	16	3	17	14	1	11	13	10	20	18	12	6	7	5	2	19	4	9	
Cirarv																			2		0,2
Chealb									1												0,3
Empnig																		2			0,3
Viclat								2		1		1									0,5
Tripra														5	2	2					0,5
Potpal	2					2															0,6
Hiprha												2						1		1	1
Brohor										4					2	2	4		3		1
Belper				2						2		2				2	3		2		1
Airpra					2													3			1,1
Elyrep				4			4									4	4		4	6	1,1
Achmil					2		1			4					2	2	2	3			1,5
Rumace													2	6	3	5				2	1,5
Junbuf									3				4		2				4		1,6
Ranacr	1					1			3	3				3	2				2		1,6
Hyprad					2			2										5			2,2
Poaann							3	2				4			2		3		4	2	2,2
Calcus			3			4					3										2,3
Sagpro		2						2	2				4					3	5	2	2,5
Salrep											5	3						3			2,5
Ranfla	2	2	2			2			2		4										2,6
Antodo					4					4					3	2	4		4		3,5
Junart	3	4	3								4									4	3,6
Plalan					2			3		3		3		5	5	5					4,3
Alogen		5	4	7					5				8				2		2	3	4,9
Poapra		4		5	1		4	4	2	4		3		3	4	2	4		4	4	4,9
Lolper		4		6			7	7		6		2		6	6	2	5		5	2	6,3
Elepal	5	4	8			4					4										6,3
Trirep	1	2		2		6		3	2	6		2	3	5	2	2	5	2	1	3	6,7
Poatri		4	2	6			2		9	4			4	4	5	6	7		5	5	7,1
Agrsto	4	4	7	4		4			5		5		4						8	3	7,2
Braruf	4	2	4	2				4		2	4	6	4	6	2	2		3	2	2	8,3
Leoaut	2	3		2	2	2		5	2	3	2	5	2	3	3	3	5	6	2	2	9,9
	0,4	0,6	0,7	1,0	1,1	1,4	1,6	1,7	1,9	2,2	2,3	2,4	2,5	2,6	2,7	2,7	2,8	3,7	3,8	3,9	



CA – How it Works

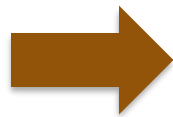
Further iterations until no changes

samples	17	15	14	20	19	16	1	18	12	11	13	5	8	7	9	10	6	4	2	3	
Emprig					2																0,5
Chealb											1										0,6
Potpal		2	2																		1
Airpra	2				3																1,1
Cirarv																		2			1,4
Hiprha					1			2							1						1,7
Viclat								1		2						1					2,1
Hyprad	2				5					2											2,7
Calcus			4	3		3															2,9
Salrep				5	3			3													3,4
Ranfla																					-
Tripra																					
Junbuf																					
Junart																					
Belper																					
Elepal																					
Ranacr																					
Achmil																					
Antodo																					
Brohor																					
Sagpro					3				4	2	2										10
Rumace									2			5		3	2		6				11
Poaann							3	4		2					2			4	3		11
Plalan	2							3		3		5		5		3	5				15
Elyrep							4					4			6			4	4	4	16
Alogen					4				8		5		5		3			2	2	7	21
Agrsto		4	4	5		7			4		5		4		3		8		4		23
Brarut		4		4	3	4			6	4	4		2	2	2	2	6		2		24
Trirep		1	6		2				2	3	3	2	2	2	2	3	6	5	1	5	26
Leoaut	2	2	2	2	6				5	2	5	2	3	3	3	2	3	2	5	2	28
Poapra	1						4	3		4	2	2	4	4	4	4	3	4	4	5	29
Lolper							7	2		7		2	4	6	2	6	6	5	5	6	36
Poatri						2	2		4		9	6	4	5	5	4	4	5	7	6	39
	5,4	11	11	12	12	16	17	19	23	24	25	26	27	29	29	30	32	32	32	32	

Definition of first axis: further axes are calculated on the same principle but for remaining unexplained variance

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																			1					3				
Agrocani							4	5									6			1														
Agrocapi	3	4	2													8				1	1						4		5	3				
Agrorepe	1	1		7	5				4		4				9	3			3	5					5			9	5					
Agrostol					6				4		12								5			1			2	3						8		
Alopgeni									4	1									2					3		5					6			
Alopprat		1		7	7	1			2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6					5							1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat			2			2	1						1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4		5	8		1				7	2			4	3				1	2			2							1	
Careprae	2	2		1	1											2			1						2		1	3	1					
Carevesi			4			1	4										1	7				2												
Carevulp								1		1		1	1						3		1				1									
Cirsarve						1								4						3										1				
Cirslanc																							1											
Cniddubi																													2					
Descesp													8				1		5		4								3	1				
Eleounig											1		2						1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat						2	1																						1					
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3										4														4		1	4				
Glycflui						1													2					7								2		
Glycmaxi						1					2											1	2	3		1					1	11	10	
Holclana						4	2						4				1			2	6													
Junceffu			1			1	1						1							4	1					1								
Lathprat												1	1								1									1				
Lotucorn		1	1		1																	1								1				
Lychflos			1			2							1								1													
Phalarun			5			1	2	2	1	11	8	2					1		1	2	1	1	6			1						1		
Planinte									3	3	1								1							1								
Poa palu			1		1	1		11		4		1		1			1		6	2		4	1		6	3								
Poa praA	3	2	1	5	6	1							3		1				1		4		2		2			4	2					
Poa triv			1		2	1	1	4												2		2	3		3	1								
Poteanse			1		2							1						1			2													
Poterept								1										1								1								
Ranuflam			1			1	3						1									1					3							
Ranurepe			4	1	1	1	2	4	1	4		1	3				3		5	4	4	2	6		9	9								
Roriamph								4		7														1			1							
RorisylA							1	2	1						1				1				3		4									
Rumeacet				1		4								4								4						1				1		
Rumethyr	1	5	2		3	5										3										2		4			1			
Siumlati				1			2				1						3						2	1			1							
Stelpalu				2			1				1		1	1			1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA			1	1	1	1					1						1		1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9			4																				
Vicilath		1																												1	2			
Vicitetr		1	1											2													1	1						



Species

	1	2	7	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	8	26	10	31	12	24	33	32		
Vicilath	1																																		
Achimill	4		4		3						1				1																				
Agrocapi	3	4	4	2	8	3	5						1							1															
Euphesul				1		1	1																												
GaliverA	1	1	4	3		4	1					4																							
Eropvern	1		1					1			1																								
Rumethyr	1	5	4	2	3	1			3		5				2																				
Vicitetr		1	1	1				1				2																							
Cniddubi							2																												
Careprae		2	1	2	2		1	3	1		1				2			1																	
Anthodor	6		4			4						1	1			5					1														
Lotucorn				1		1					1		1							1															
Poa praA	3		2	1		2	4	5		6	3	4						1	1							2									
Agrorepe	1		1	3		5	9	7	9	5				5	5		4	3					4												
TaraoffA			1	1		1		1		1				1	2		1	1	1	1															
Cirsarve					1						4		3			1																			
Rumeacet			1			1					4	4				4			1																
Lathprat					1							1		1										1											
Alopprat	1	1		2		2	3	7	4	7				5	5	1	5	4		3		2	1		2		1								
Desccesp					1	3						8		5							4	1													
Vicicrac					2	1						4				1	9		1																
Holclana					1							4	6	2		4					1						2								
Festprat						1																					1								
Cardprat					1							3	2	1		2		1	2	1			1			1									
Poteanse											1		2			2			1						1										
Lychflos											1	1				2			1						1										
Trifrepe												1	4			3		4	2	1															
Caredist											2													1											
Junceffu											1	4				1			1	1							1	1							
Carevulp											1	1		1		1	3					1	1												
Poterept														1			1					1													
Stelpalu											1	1			1	1	2	2	1	1	1		1	1											
Poa triv										2			2	3	1				1	2		4			3	1	1								
Poa palu										1	1			2	6	1	4	6	1	4	1	1	1		1		3								
Eleounig												1					1	1					2												
Ranurepe								1	1	3	4	4	9	1	4	5	4	2	3	4	1			6	2	9	1								
RorisylA								1					4		1	1					1			3			2								
Agrocani												1			4						6					5									
Cirslanc																				1															
Sympoffi																				3					1										
Caltpalu																			1	6	1			5											
Planinte														1		1	1					3						3							
Carevesi													1						4	2	1			7		4									
Ranufflam											1			1					1				1			3	3								
Caregrac											2			5				4	1	4			7	3	2	8	2	1					1		
Phalarun												1	2		1	1	1	5	1	1	2	2		6	2	1	1	1	8						
Siumlati																		1	2	3					1	2	1						1		
Agrostol											6				2			5		1		4					3	12	8						
Galipalu											2		1	1			1	1		1	1		1	3	3	3	2			1	1		1		
Alopgeni																	2					4					5	1	6		3				
RoriampH																												1	4		7				
Glycflui																		2												2		7			
Glycmaxi																					1					2	1	1		1	2	3	10	11	



Correspondence Analysis



	1	2	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	8	26	10	31	12	24	33	32	GM	
Vicilath	1																																-1.441	
Achimill	4	1								1			1																				-1.218	
Agrocapi	3	4	4	2	8	3	5					1								1													-1.217	
Euphesul			1			1		1																									-1.194	
GaliverA	1	1	4	3		4	1				4																						-1.141	
Eropvern	1	1						1		1																							-1.117	
Rumethyr	1	5	4	2	3	1			3	5				2																			-1.083	
Vicitetr	1	1	1					1			2																						-1.027	
Cniddubi						2																											-0.967	
Careprae	2	1	2	2		1	3	1		1				2			1																-0.955	
Anthodor	6	4		4						1	1				5						1												-0.946	
Lotucorn			1		1					1	1								1														-0.653	
Poa praA	3	2	1		2	4	5		6	3	4		2	1		1								2									-0.633	
Agrorepe	1	1	3		5	9	7	9	5			5	5	4	3							4											-0.574	
TaraoffA			1	1	1		1		1			1	2		1	1	1	1															-0.492	
Cirsarve				1						4	3		1																					-0.440
Rumeacet		1		1						4	4		4					1															-0.400	
Lathprat				1						1	1											1											-0.391	
Alopprat	1	1	2		2	3	7	4	7	1		5	5	1	5	4		3		2	1		2		1								-0.378	
Desccesp				1	3					8	5								4	1													-0.357	
Vicicrac				2	1					4			1	9		1																	-0.287	
Holclana				1						4	6	2	4							1						2							-0.219	
Festprat					1										2											1							-0.186	
Cardprat				1						3	2	1	2		1	2	1				1				1								-0.117	
Poteanse										1	2		2										1										-0.091	
Lychflos										1	1				2									1									-0.059	
Trifrepe											1	4	3		4	2	1																-0.034	
Caredist											2												1										-0.015	
Junceffu										1	4			1			1	1								1	1						-0.001	
Carevulp										1	1		1	1	3						1	1											0.021	
Poterept													1		1						1												0.072	
Stelpalu										1	1		1	1	2	2	1	1				1	1										0.097	
Poa triv										2			2	3	1			1	2		4			3	1	1							0.099	
Poa palu										1	1		2	6	1	4	6	1	4	1	1	1		1		3							0.117	
Eleounig												1			1	1						2											0.124	
Ranurepe										1	1	3	4	4	9	1	4	5	4	2	3	4	1		6	2	9	1					0.131	
RorisylA										1				4		1	1				1			3		2							0.164	
Agrocani											1			4						6					5								0.217	
Cirslanc																					1												0.225	
Sympoffi																						3			1								0.295	
Caltpalu																						1	6	1		5							0.322	
Planinte																1	1	1					3										0.361	
Carevesi																	1		4	2	1			7	4								0.363	
Ranufiam											1				1							1			3	3							0.383	
Caregrac											2			5				4	1	4			7	3	2	8	2	1			1		0.399	
Phalarun												1	2		1	1	5	1	1	2	2		6	2	1	1	1	8				0.453		
Siumlati																		1	2	3				1	2	1							0.476	
Agrostol												6		2			5		1		4					3	12	8					0.512	
Galipalu												2	1	1		1		1			1	3	3	3	2	2						1	0.577	
Aloppeni																	2				4												1.009	
Roriamp																																		1.298
Glycflui																																		1.753
Glycmaxi																																		3.117

The principle:

Calculate the **weighted average of species** from **site** values (current order of sites)

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

x_j = Weighted average of **species j**

n = Number of **sites**

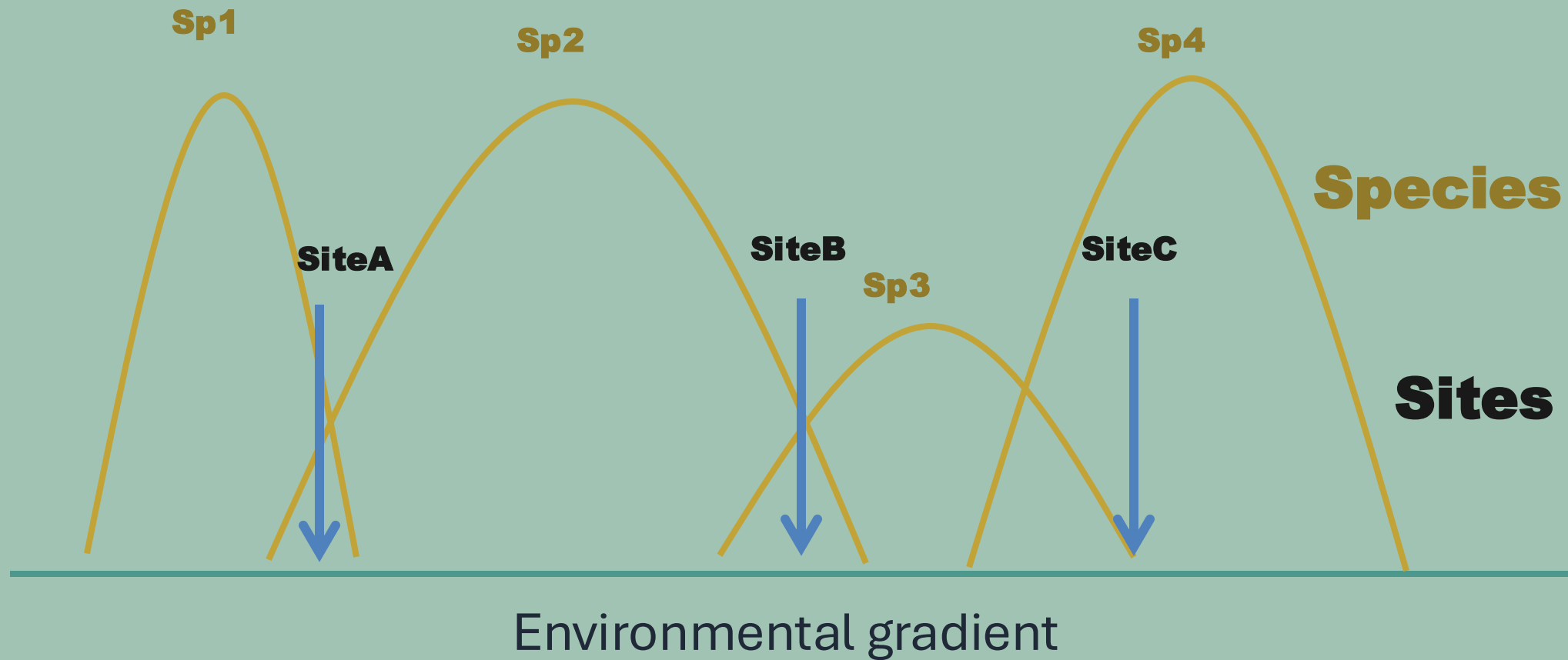
y_{ij} = Abundance of species j in **site i**

x_i = Site score of **site i**



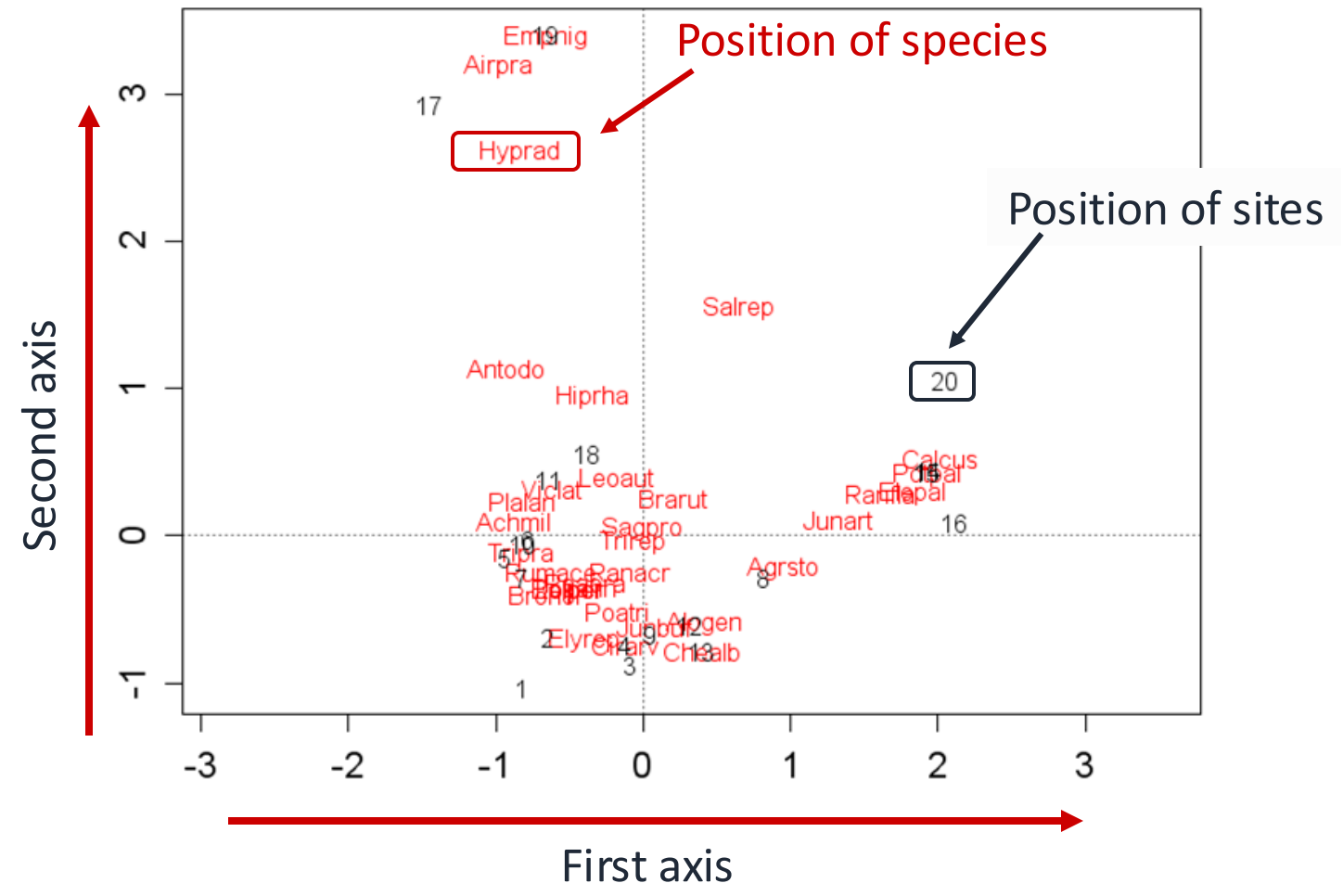
Ordination - Correspondence Analysis

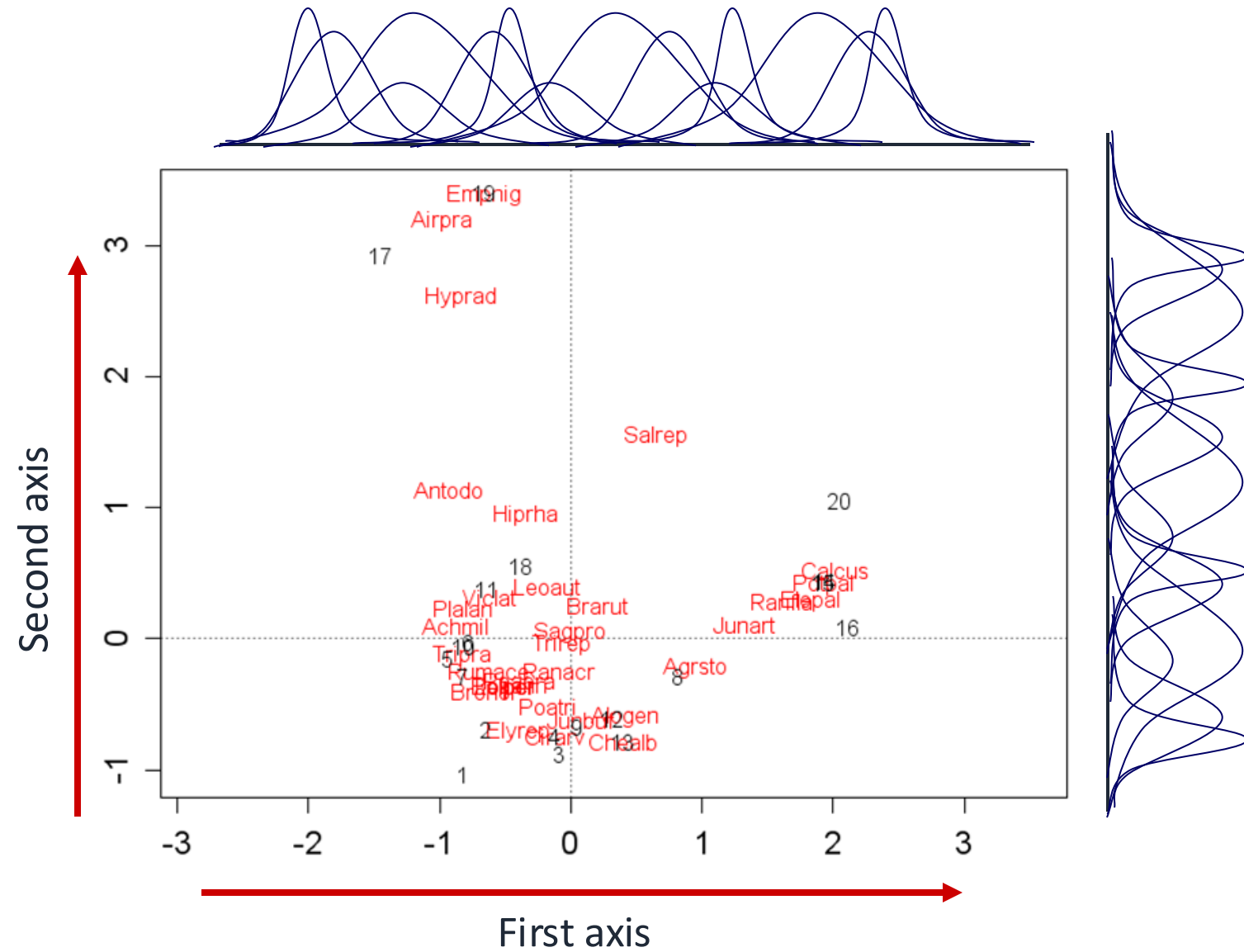
What is it trying to do?



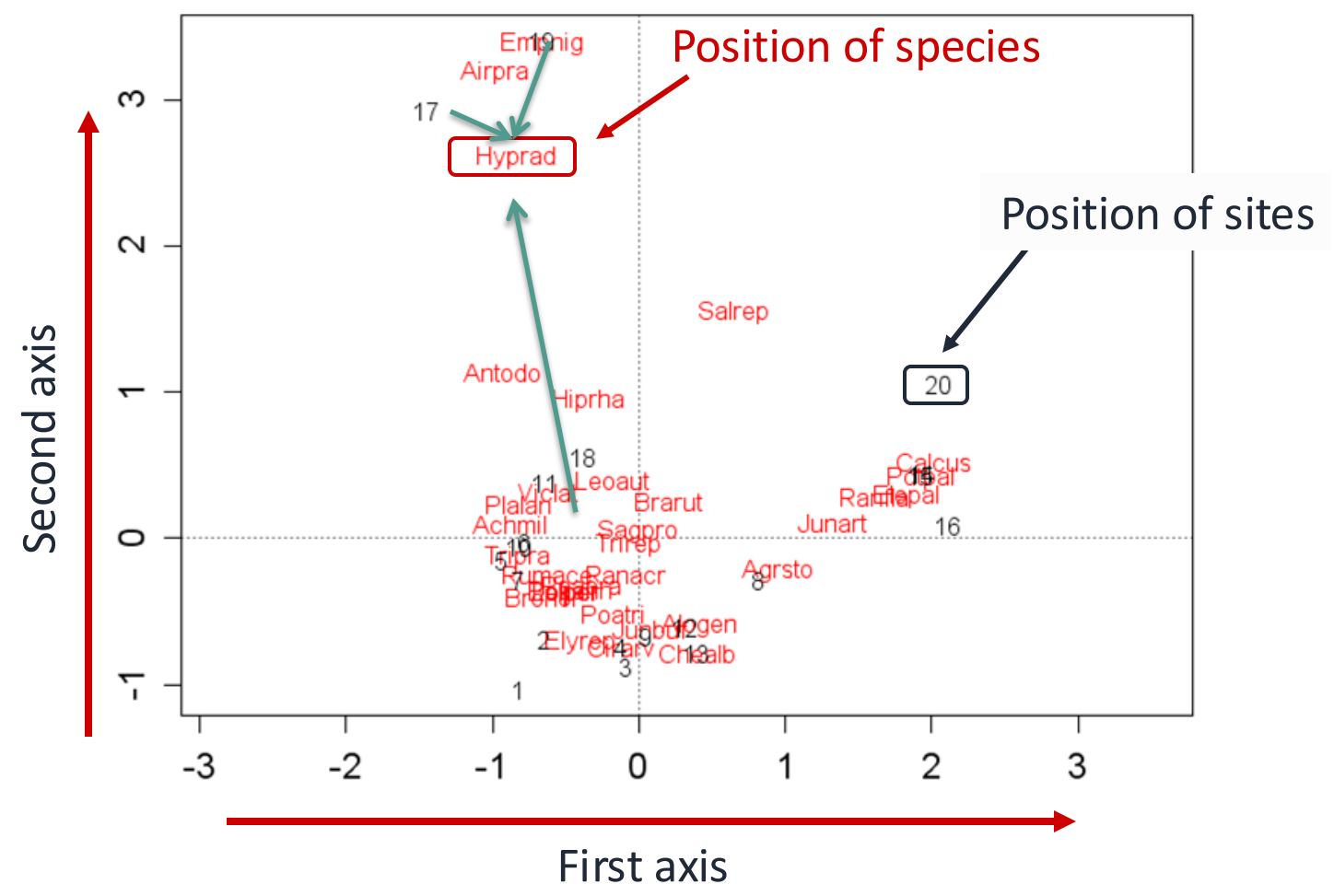
If you knew the species or the sites you could position the other

Correspondence Analysis





Correspondence Analysis





Correspondence Analysis

Each axis of the CA is an **eigenvector**

Each axis has an **eigenvalue**

- Measures the degree of differentiation between species along this axis (range 0-1)
- The larger, the more species can be differentiated along this axis

	Inertia	
Total	2.156	← total inertia: Overall variance of the data
Unconstrained	2.156 19	

Eigenvalues for unconstrained axes:

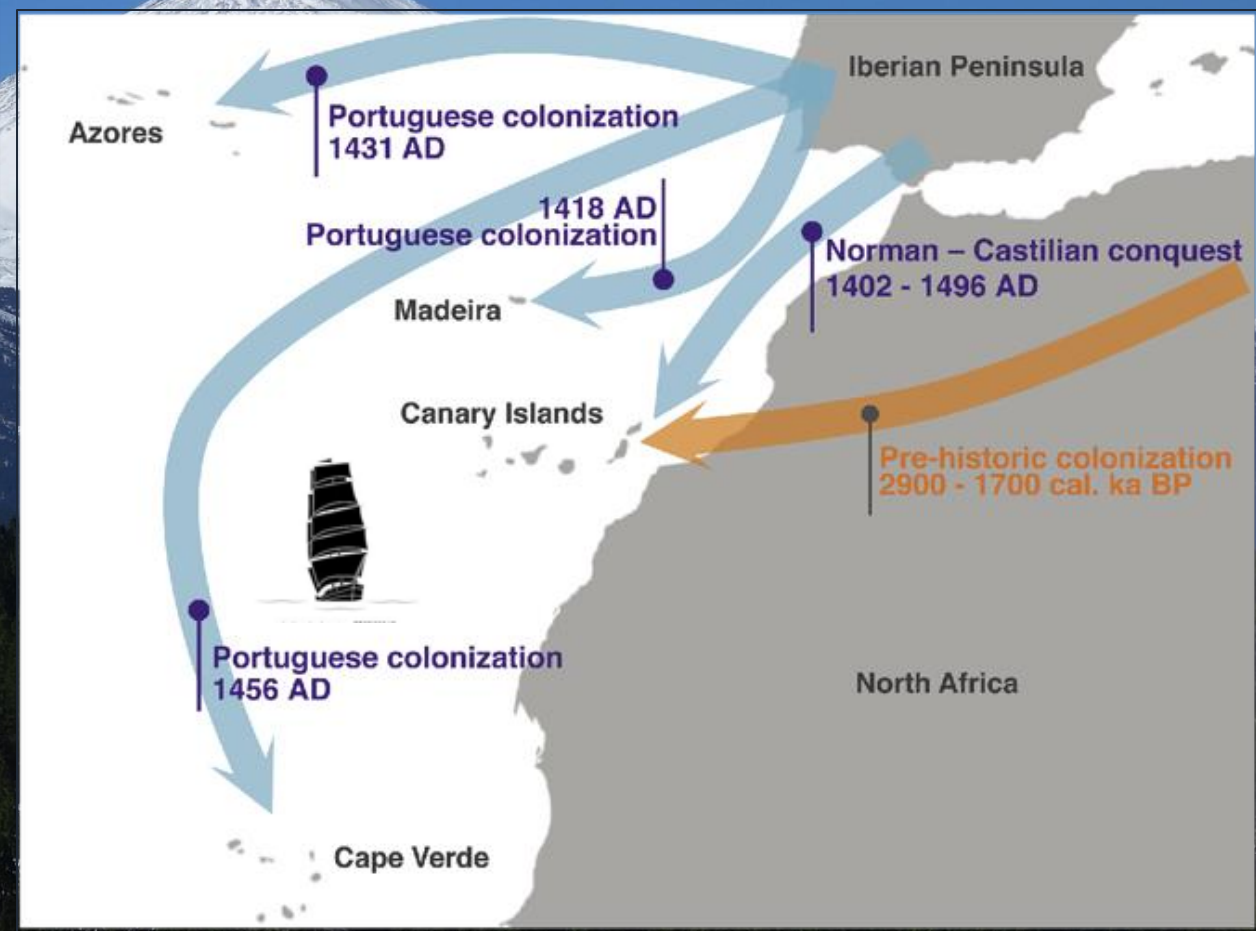
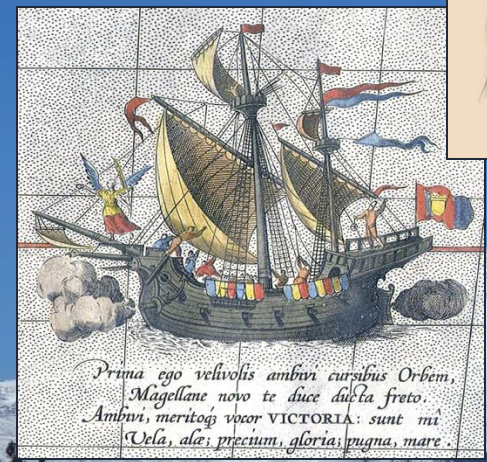
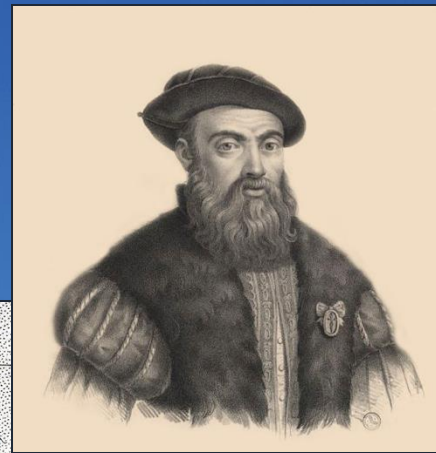
CA1	CA2	CA3	CA4	CA5	CA6
0.52722	0.39537	0.25162	0.20418	0.14553	0.10627

Decline in explained variance with increasing ordination axes

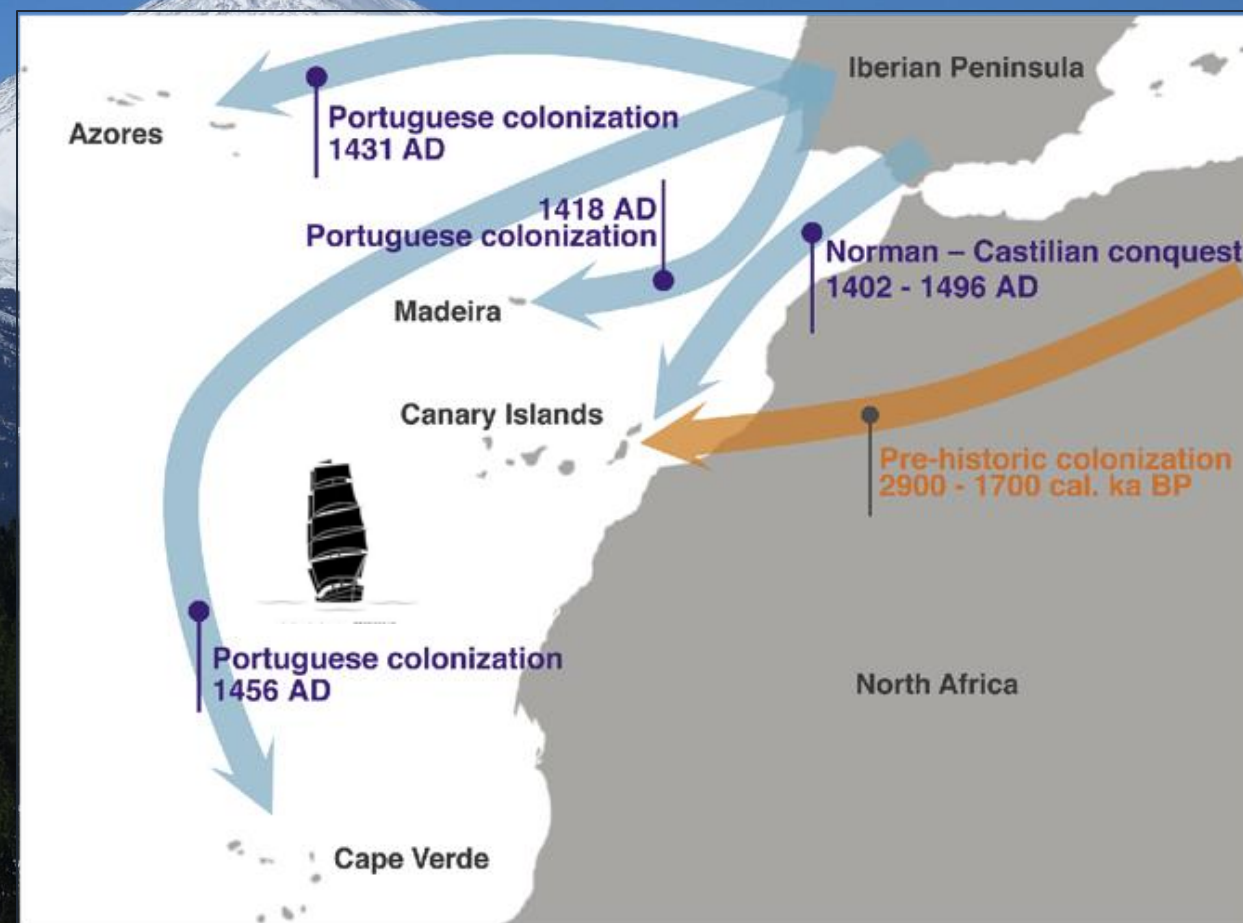


Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de

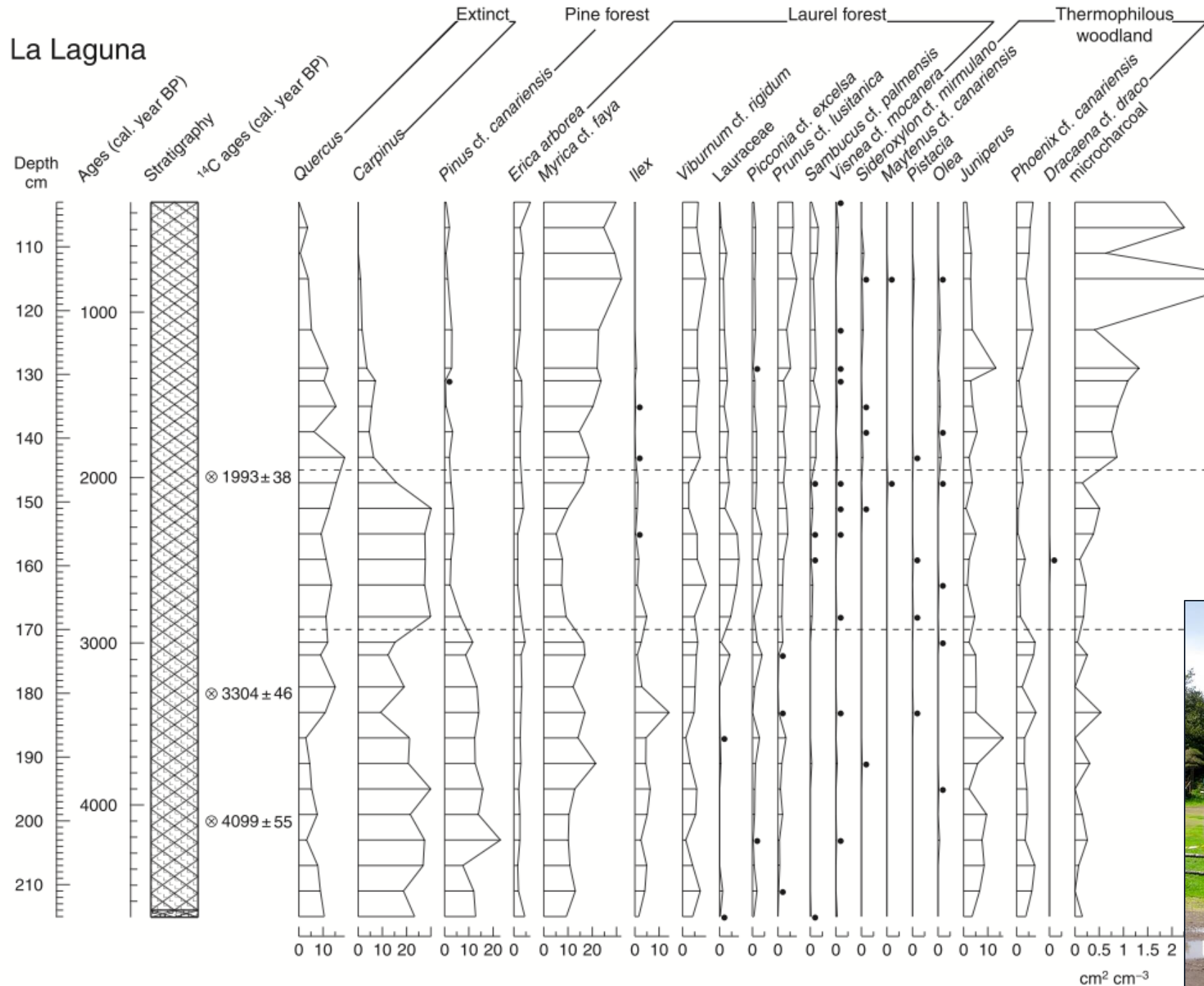
Islands



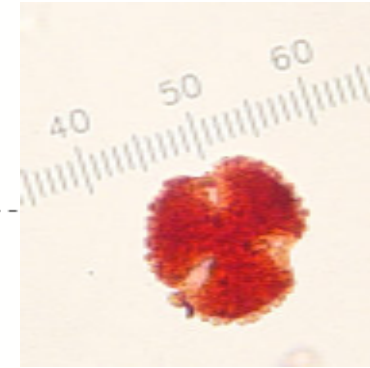
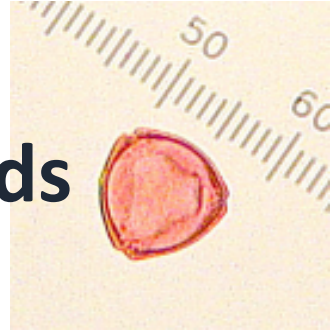
Islands



La Laguna

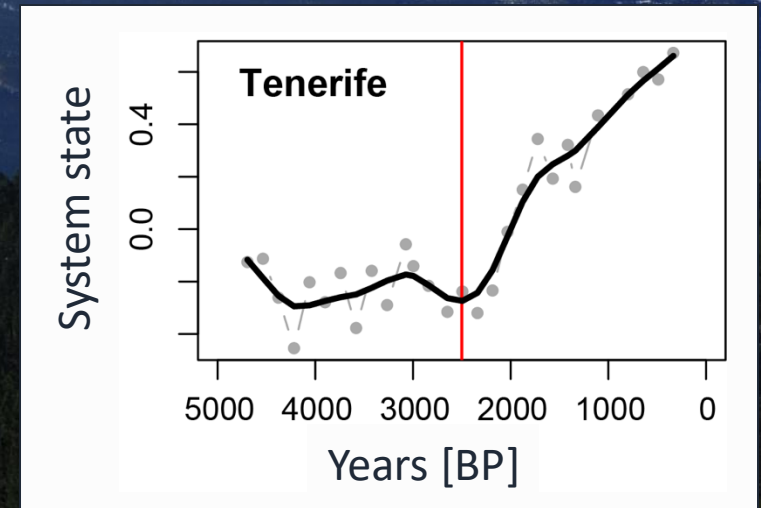


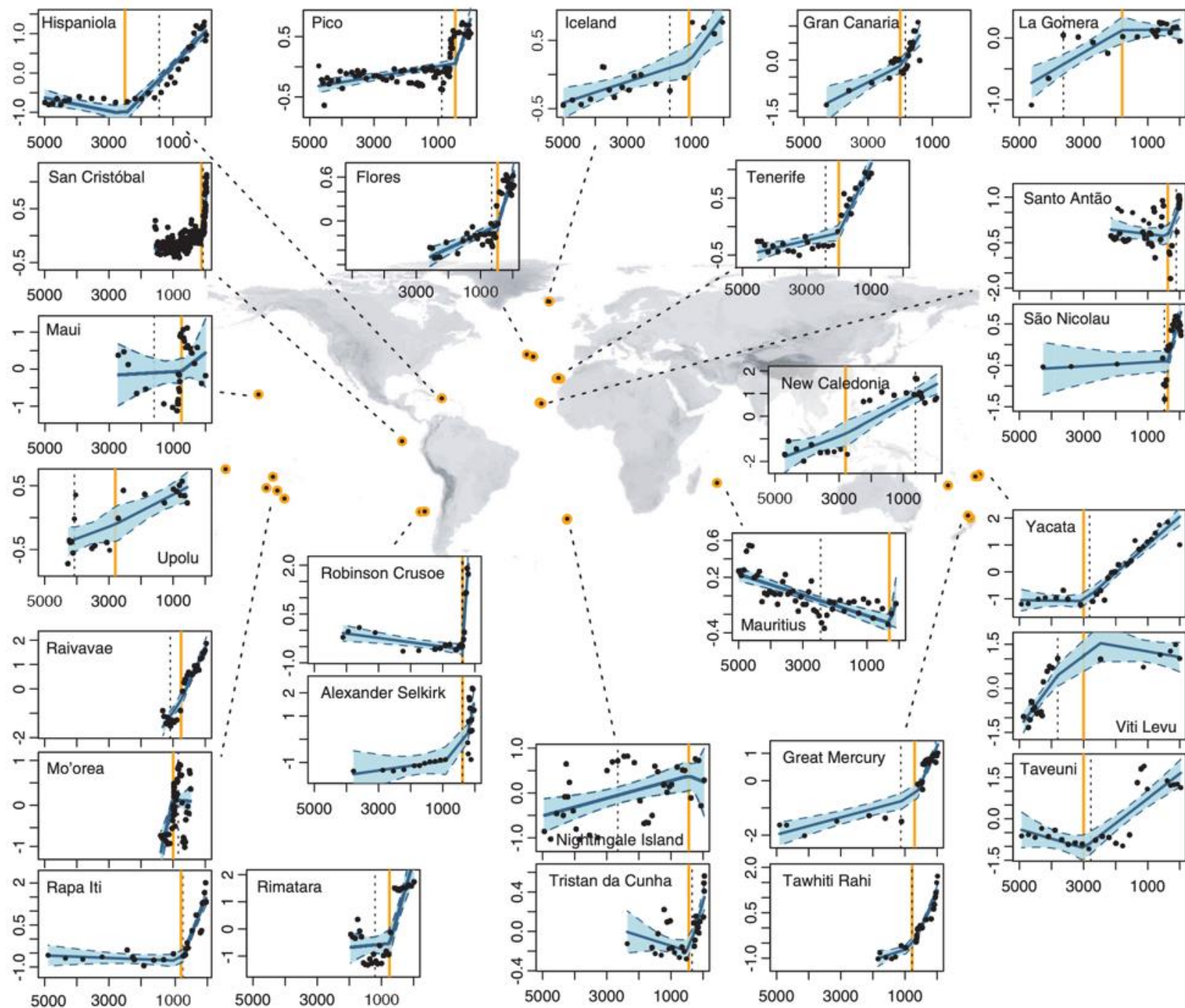
Tenerife, Canary Islands





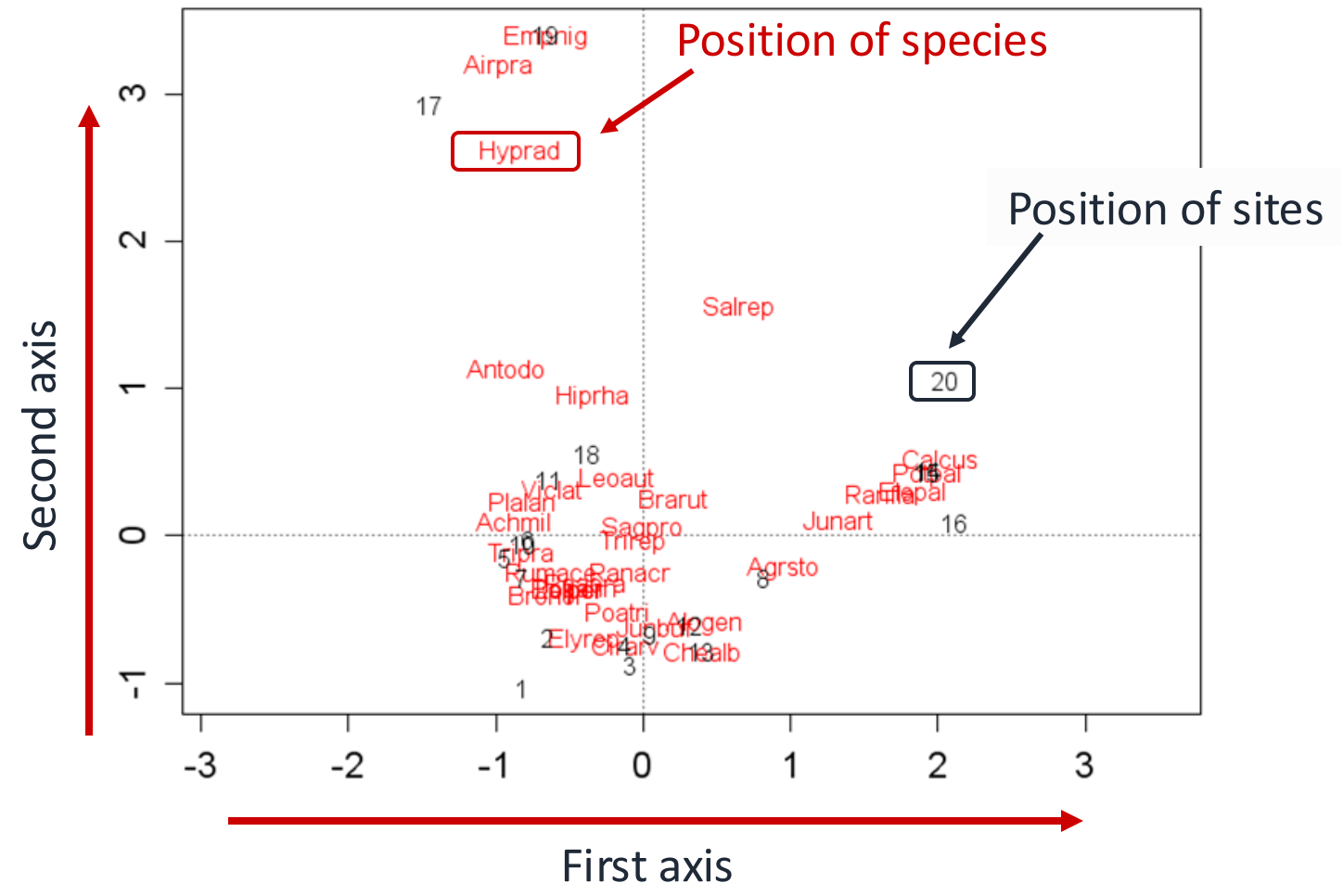
Islands





Nogue et al. (Science, 2021)

Correspondence Analysis

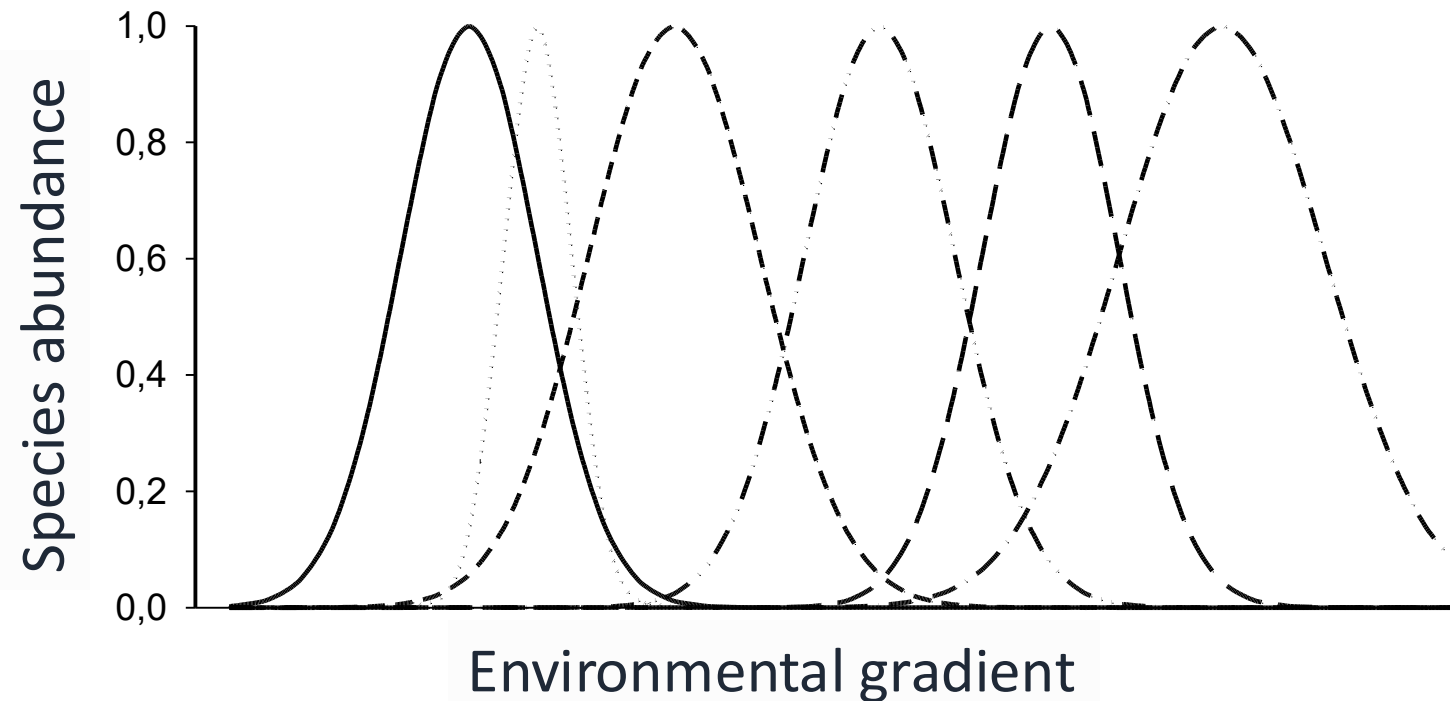


Ordination



Linear $\leftrightarrow$ unimodal Ordination methods?

- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“

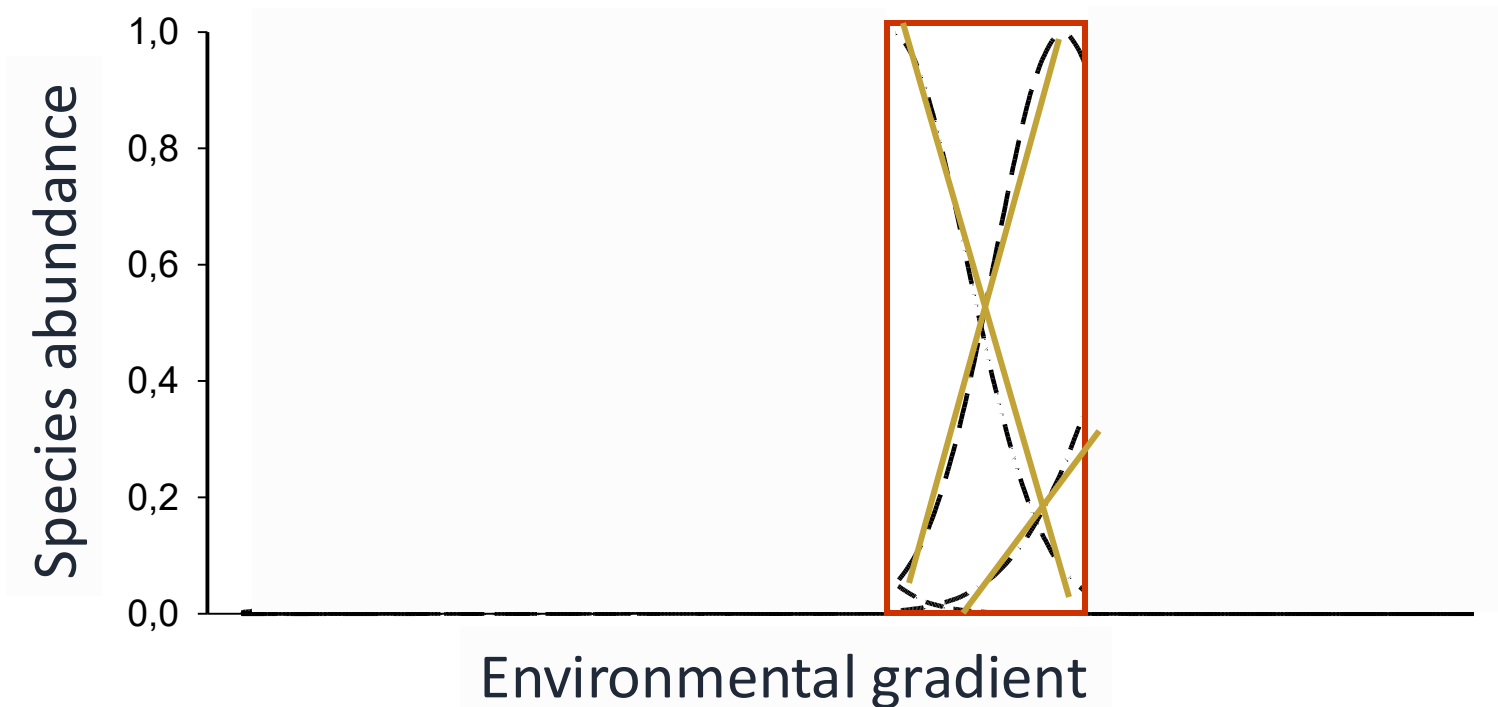


Ordination



Linear $\leftrightarrow$ unimodal Ordination methods?

- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“



Overview on ordination methods



	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis



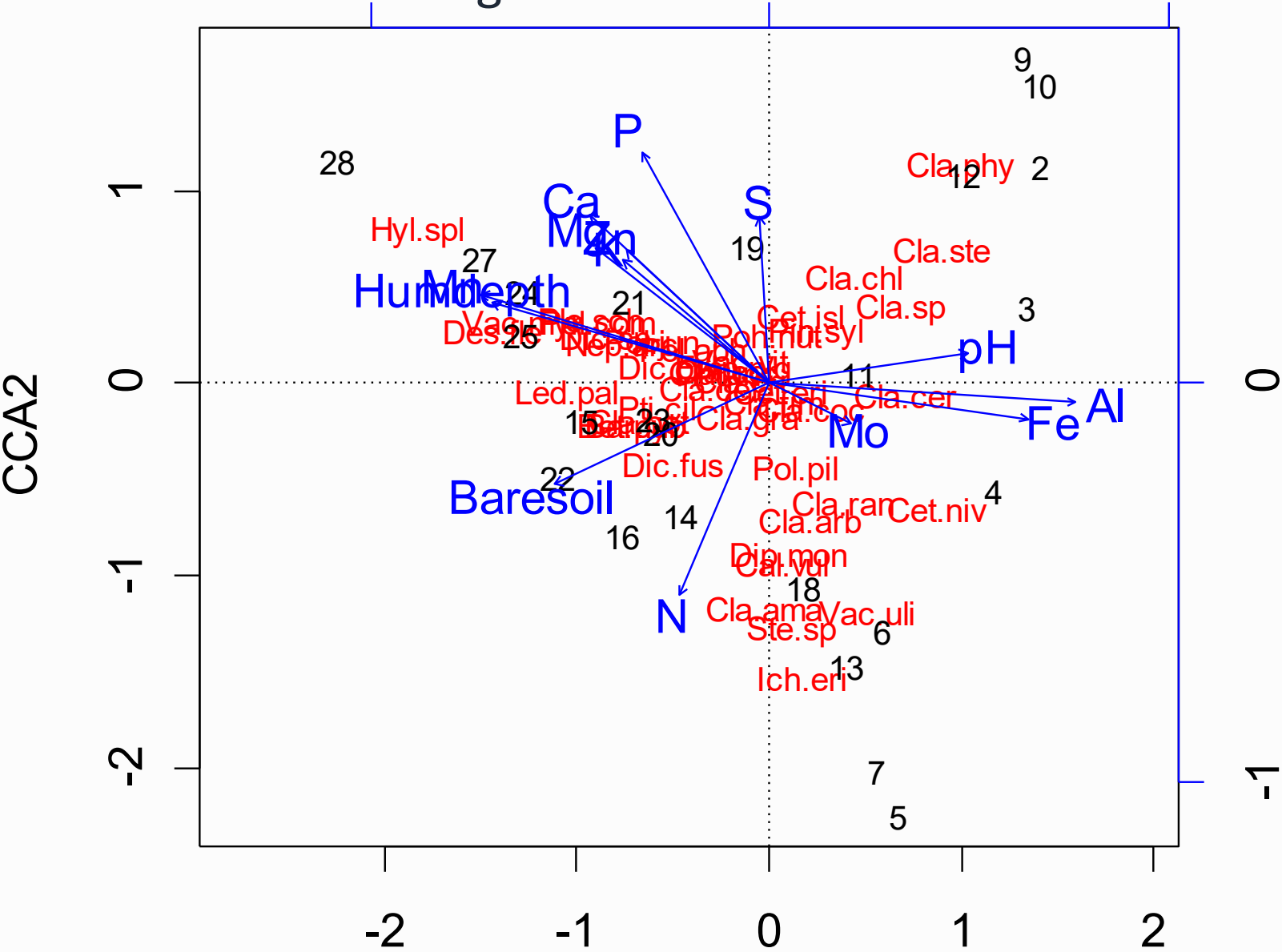
Overview on ordination methods

	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis
Any relation	NMDS (Non-metric multidimensional scaling)

Overview on ordination methods

	Indirect (no environmental data)	Direct (with environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis	RDA Redundancy Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis	CCA Canonical CA
Any relation	NMDS (Non-metric multidimensional scaling)	

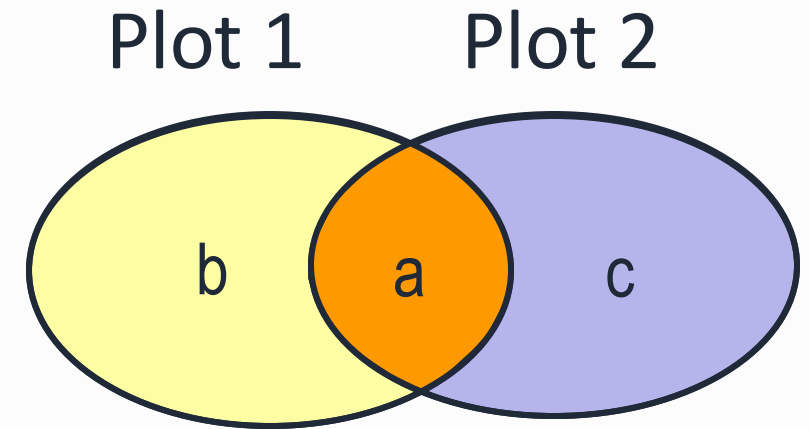
plot(cca(varespec,varechem))
CCA – with all environmental gradients





Similarity- and dissimilarity indices

		<i>Plot 1</i>	
		+	-
<i>Plot 2</i>	+	a	c
	-	b	d



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

e.g.: Sørensen similarity = $2a / (2a+b+c)$

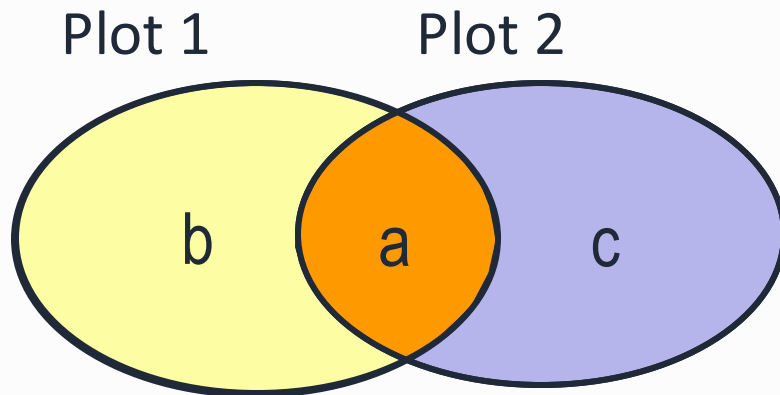
Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$sim = \frac{(a+b+c+d)(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

Yule & Kendall (1950) in Jurasinski (2012)



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

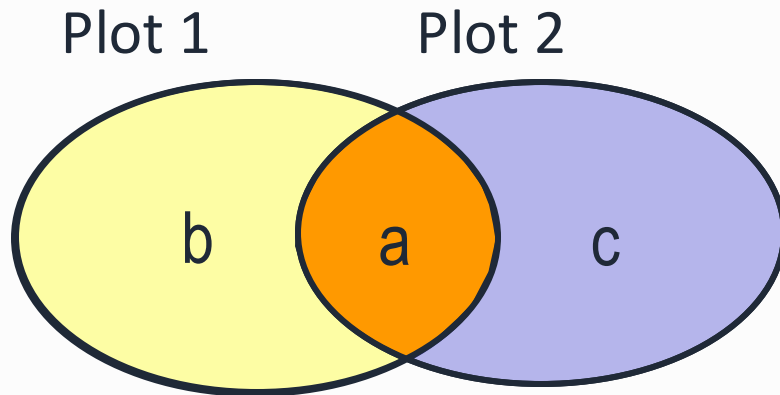
d: Species not in the two plots

Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

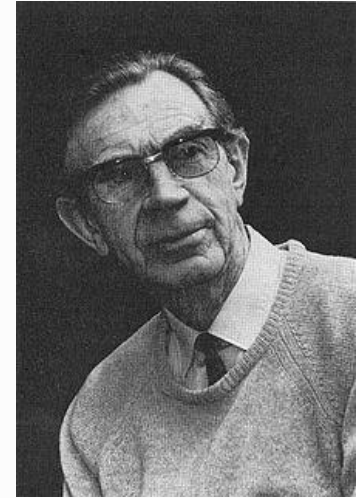
d: Species not in the two plots

Correspondence Analysis

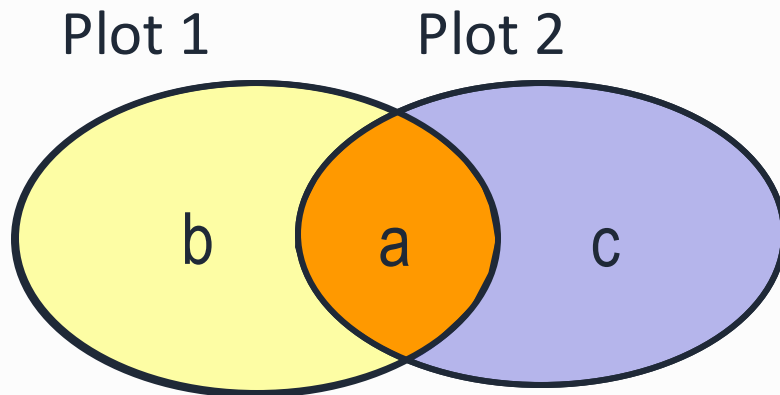


Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



Thorvald Sørensen



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Species not in the two plots



(Dis)Similarity Matrices

	V_1	V_2	V_3	V_4
O_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
O_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
O_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
O_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
O_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
O_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$

Raw data

dissimilarity
matrix

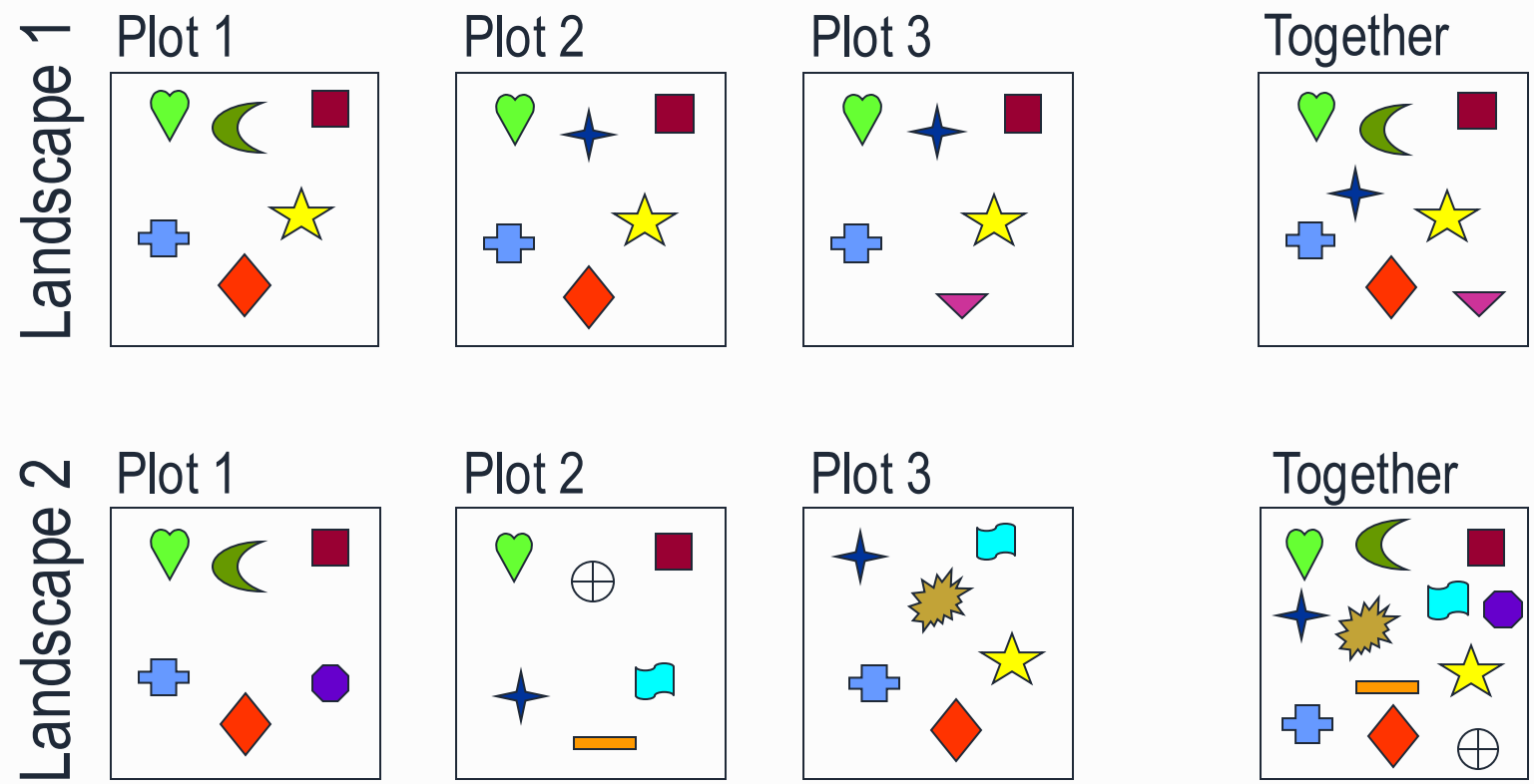


similarity
matrix

O_1	0					
O_2	$d_{2,1}$	0				
O_3	$d_{3,1}$	$d_{3,2}$	0			
O_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
O_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
O_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

	O_1	O_2	O_3	O_4	O_5	O_6
O_1	1					
O_2	$s_{2,1}$	1				
O_3	$s_{3,1}$	$s_{3,2}$	1			
O_4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	1		
O_5	$s_{5,1}$	$s_{5,2}$	$s_{5,3}$	$s_{5,4}$	1	
O_6	$s_{6,1}$	$s_{6,2}$	$s_{6,3}$	$s_{6,4}$	$s_{6,5}$	1

Why do we need these measures?



	α - Plots	α - Landscape	$\beta_{\text{SÖR}}$
Landscape 1	6	8	0,83
Landscape 2	6	12	0,33

Diversity of indices



$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$

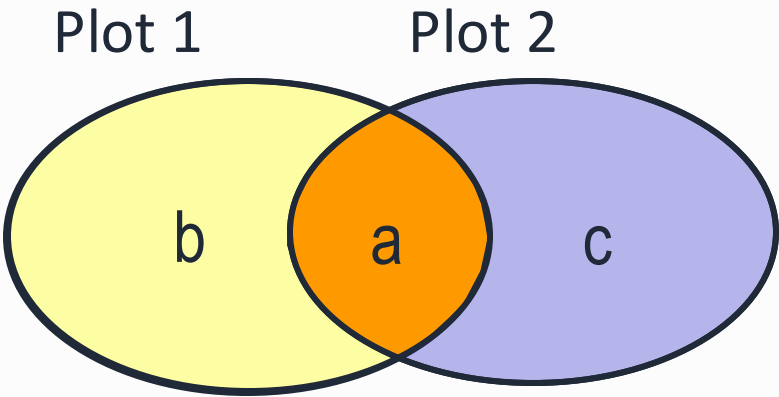
$$\beta_{\text{Williams}} = \frac{\min(b, c)}{a + b + c}$$

$$\beta_{\text{Jaccard}} = \frac{a}{a + b + c}$$

$$\beta_{\text{Lande}} = \frac{b + c}{2}$$

$$\beta_{\text{Wilson \& Shmida}} = \frac{b + c}{2a + b + c}$$

Diversity of indices

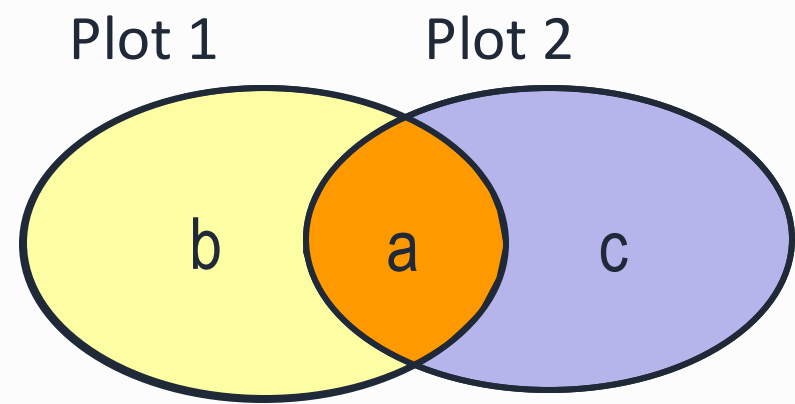


soerensen
jaccard
ochiai
mountford
whittaker
lande
wilsonshmida
cocogaston
magurran
harrison
cody
williams
williams2
harte
simpson
lennon
weiher
ruggiero
lennon2
routledge
routledge2
routledge3
sokal1
dice
kulczlinsky

$$\begin{aligned} sim &= \frac{2a}{2a+b+c} \\ sim &= \frac{a}{a+b+c} \\ sim &= \frac{a}{\sqrt{(a+b)(a+c)}} \\ sim &= \frac{2a}{(a(b+c)+2bc)} \\ sim &= \frac{a+b+c}{2a+b+c} - 1 \\ sim &= \frac{b+c}{2} \\ sim &= \frac{b+c}{2a+b+c} \\ sim &= \frac{b+c}{a+b+c} \\ sim &= (2a+b+c)(1 - \frac{a}{a+b+c}) \\ sim &= \frac{\min(b,c)}{\max(b,c)+a} \\ sim &= 1 - \frac{a(2a+b+c)}{2(a+b)(a+c)} \\ sim &= \frac{\min(b,c)}{a+b+c} \\ sim &= \frac{(bc)+1}{(a+b+c)^2 - (a+b+c)} \\ 1 - \frac{2a}{2a+b+c} \\ sim &= \frac{\min(b,c)}{\min(b,c)+a} \\ sim &= \frac{2|b-c|}{2a+b+c} \\ sim &= b+c \\ sim &= \frac{a}{a+c} \\ sim &= 1 - \left[\frac{\log(\frac{2a+b+c}{a+b+c})}{\log 2} \right] \\ sim &= \frac{(a+b+c)^2}{(a+b+c)^2 - 2bc} \\ sim &= \text{toolong, see below} \\ sim &= e^{routledge2} - 1 \\ sim &= \frac{a}{a+2(b+c)} \\ sim &= \frac{a}{\min((b+a), (c+a))} \\ sim &= \frac{a}{b+c} \end{aligned}$$

Soerensen (1948)
Jaccard (1912)
Ochiai (1957), Shi (1993)
Mountford (1962), Shi (1993)
Whittaker (1960), Magurran (1988)
Lande (1996)
Wilson & Shmida (1984)
Colwell & Coddington (1948), Gaston et al. (2001)
Magurran (1988)
Harrison et al. (1992), Koleff et al. (2003)
Cody (1993)
Williams (1996), Koleff et al. (2003)
Williams (1996), Koleff et al. (2003)
Harte & Kinzig (1997), Koleff et al. (2003)
Simpson (1949), Koleff et al. (2003)
Lennon et al. (2001), Koleff et al. (2003)
Weiher & Boylen (1994)
Ruggiero et al. (1998), Koleff et al. (2003)
Lennon et al. (2001), Koleff et al. (2003)
Routledge (1977), Magurran (1988)
Routledge (1977), Wilson & Shmida (1984)
Routledge (1977)
Sokal & Sneath (1963)
Association index of Dice (1945), Wolda (1981)
Oosting (1956), Southwood (1978)

Diversity of indices



kulcz2insky	$sim = \frac{\frac{a}{2}(2a+b+c)}{(a+b)(a+c)}$
mcconnagh	$sim = \frac{\frac{a^2-bc}{a^2}}{(a+b)(a+c)}$
simpson2	$sim = \frac{\frac{a}{a+b}}{\frac{a}{3a+b+c}}$
legendre2	$sim = \frac{\frac{3a}{3a+b+c}}{\frac{1}{2*\sqrt{n_2}}}$
fager	$sim = \frac{\frac{2a-(b+c)}{2a+b+c}}{\frac{a}{2b}}$
maarel	$sim = \frac{\frac{a}{2b}}{\frac{a^2}{(a+b)(a+c)}}$
lamont	$sim = \frac{\frac{a}{2b}}{\frac{a}{a+b} + \frac{a}{a+c}}$
johnson	
sorgenfrei	
johnson2	

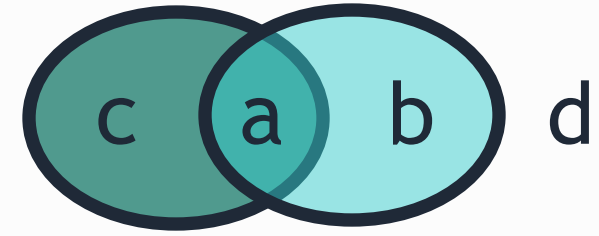
Oosting (1956), Southwood (1978)
 Hubalek (1982)
 Simpson (1960), Shi (1993)
 Legendre & Legendre (1998)
 Fager (1957), Shi (1993)
 van der Maarel (1969)
 Lamont and Grant (1979)
 Johnson (1971)
 Sorgenfrei (1959)
 Johnson (1967)

Computable symmetric indices (including unshared species):

manhattan	$sim = \frac{\frac{b+c}{a+b+c+d}}{\frac{a+d}{a+b+c+d}}$
simplematching	$sim = \frac{\frac{a}{a+b+c+d}}{\frac{a}{(a+b)(a+c)}}$
margaleff	$sim = \frac{\frac{ad-bc}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}}{\frac{a+d}{a+2(b+c)+d}}$
pearson	$sim = \frac{\frac{\sqrt{ad+c}}{\sqrt{ad+a+b+c}}}{\frac{ad-bc}{\sqrt{(a+b+c+d)(a+b)(a+c)}}}$
roger	$sim = \frac{\frac{(a+b+c+d)(-\frac{a}{2})^2}{(a+b)(a+c)}}{\frac{a-(b+c)+d}{a+b+c+d}}$
baroni	$sim = \frac{\frac{a}{a+b+c+d}}{\frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}}$
dennis	$sim = \frac{\frac{2a+2d}{(a+d+(a+b+c+d))}}{\frac{a+d}{b+c}}$
fossum	$sim = \frac{\frac{(a+b+c+d)(ad-bc -\frac{a+b+c+d}{2})^2}{(a+b)(a+c)(b+d)(c+d)}}{\frac{ad-bc}{ad+bc}}$
gower	$sim = \frac{\frac{4(ad-bc)}{(a+b)^2+(b+c)^2}}{\frac{4(ad-bc)}{(a+d)-(b+c)}}$
legendre	$sim = \frac{\frac{N}{(aN-2n_2)}}{\frac{2(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}}$
sokal2	$sim = \frac{\frac{(a+b)(a+c)}{(a+d)-(b+c)}}{\frac{a-(a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}}$
sokal3	
sokal4	
stiles	
yule	
michael	
hamann	
forbes	
chisquare	
peirce	
eyraud	

Mean Manhattan, Legendre & Legendre (1998)
 Sokal & Michener 1958
 Clifford & Stevenson (1975)
 Phi of Pearson, Gower & Legendre (1986)
 Rogers & Tanimoto (1960), Gower & Legendre (1986)
 Baroni-Urbani & Buser (1976), Wolda (1981)
 Holliday et al. (2002)
 Holliday et al. (2002)
 Gower & Legendre (1986)
 Gower & Legendre (1986)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Stiles (1946)
 Yule & Kendall (1973)
 Michael (1920), Shi (1993)
 Hamann (1961)
 Forbes (1925), Shi (1993)
 Yule & Kendall (1950)
 Peirce (1984) in Shi (1993)
 Eyraud (1936) in Shi (1993)

Diversity of indices



$$\beta_{ij} = (\varepsilon a + \delta d) (\varepsilon a + \delta d + \lambda(b + c))^{-1}$$

β_{ij} = Similarity Value for Data Sets i, j

a = Species common in i and j

b = Species only in i

c = Species only in j

d = Species neither in i nor in j but in the area

$\varepsilon, \delta, \lambda$ = coefficients

Sørensen's Coefficient is binary



		Object 1	
		1	0
Object 2	1	a	b
	0	c	d

$$S = \frac{2a}{2a + b + c}$$

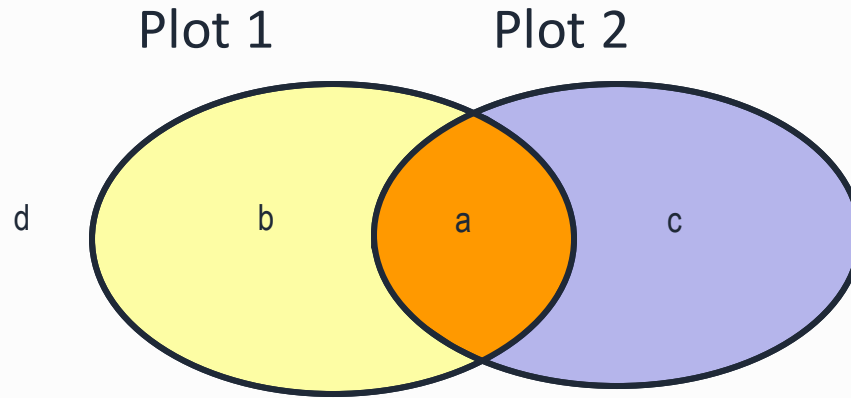
Bray-Curtis (=Steinhaus' Coefficient)

Bray-Curtis is Sørensen coefficient but including abundance information:

$$BC_{ij} = \frac{2C_{ij}}{S_i + S_j}$$

C = sum of the lesser abundance values for shared species
S = total abundance of species counted at both sites

Similarity- and dissimilarity indices



a: Shared species

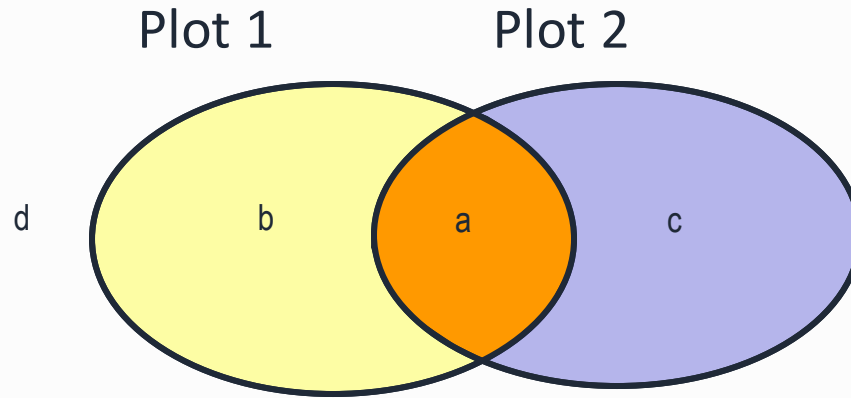
b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species



Similarity- and dissimilarity indices



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species

e.g.: Sørensen similarity = $2a / (2a+b+c)$

	V_1	V_2	V_3	V_4
O_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
O_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
O_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
O_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
O_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
O_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$



similarity
matrix

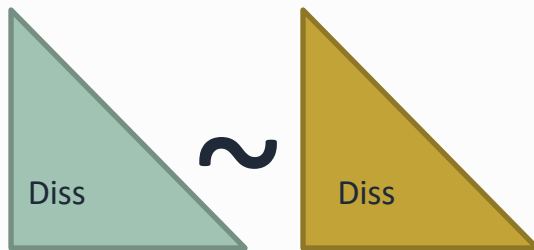
	O_1	O_2	O_3	O_4	O_5	O_6
O_1	1					
O_2	$s_{2,1}$	1				
O_3	$s_{3,1}$	$s_{3,2}$	1			
O_4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	1		
O_5	$s_{5,1}$	$s_{5,2}$	$s_{5,3}$	$s_{5,4}$	1	
O_6	$s_{6,1}$	$s_{6,2}$	$s_{6,3}$	$s_{6,4}$	$s_{6,5}$	1

Raw data



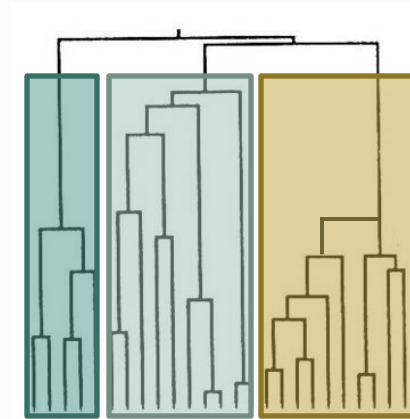
	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆
O ₁	0					
O ₂	d _{2,1}	0				
O ₃	d _{3,1}	d _{3,2}	0			
O ₄	d _{4,1}	d _{4,2}	d _{4,3}	0		
O ₅	d _{5,1}	d _{5,2}	d _{5,3}	d _{5,4}	0	
O ₆	d _{6,1}	d _{6,2}	d _{6,3}	d _{6,4}	d _{6,5}	0

Dissimilarity matrix



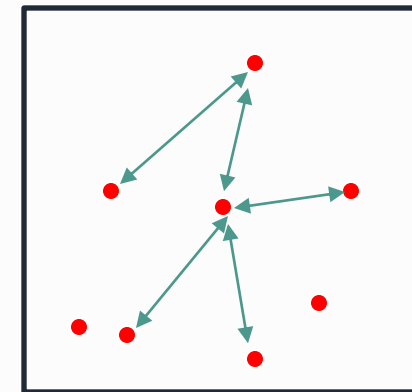
Relate to other
dissimilarity matrix

or



Classification

or

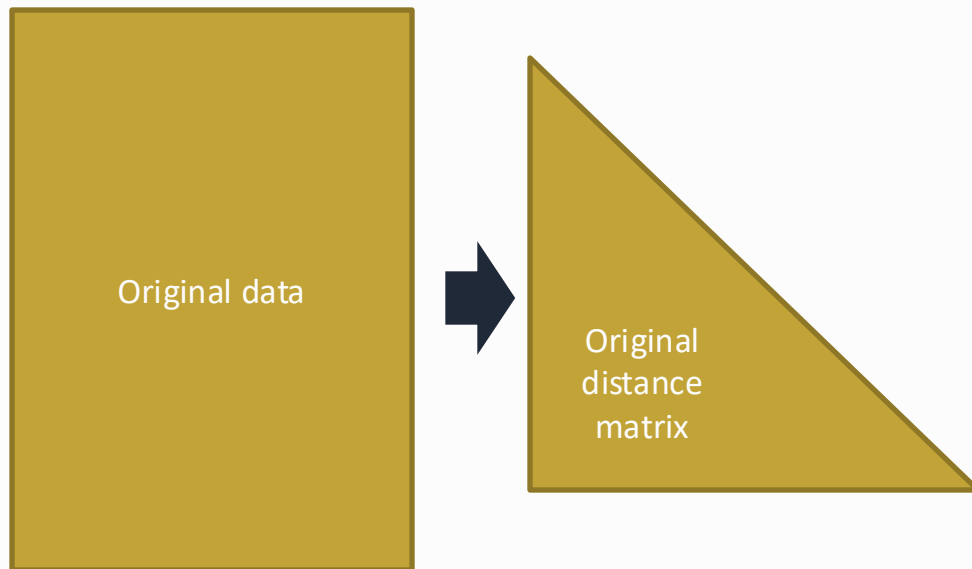


Dimension reduction
(e.g. NMDS)



(N)MDS - The goal

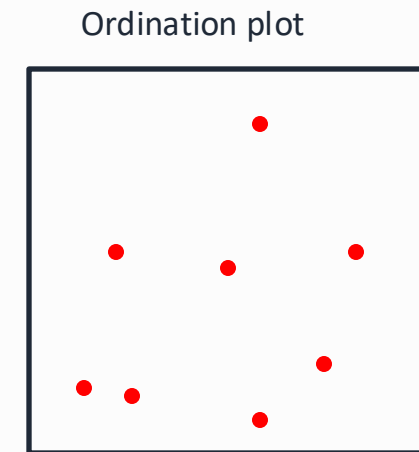
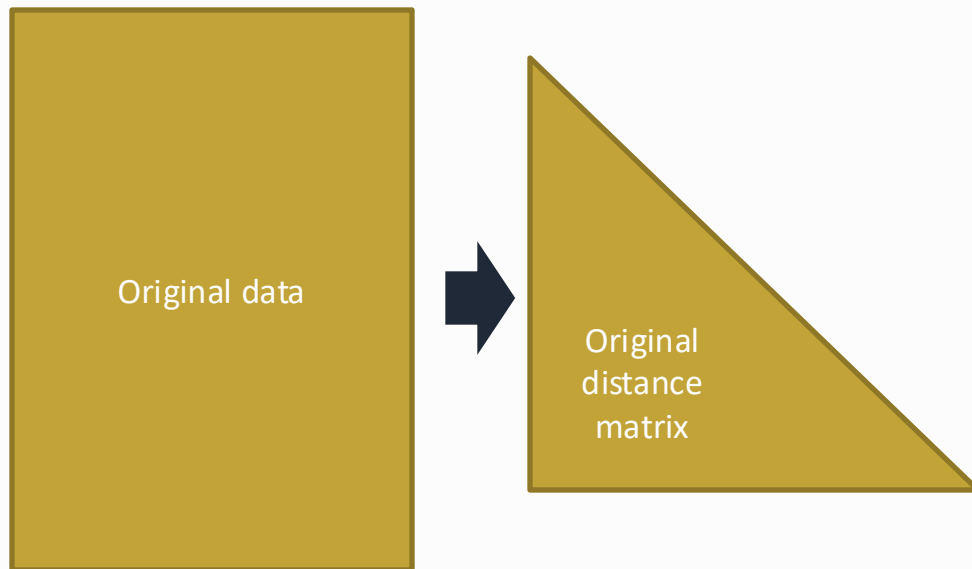
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





(N)MDS - The goal

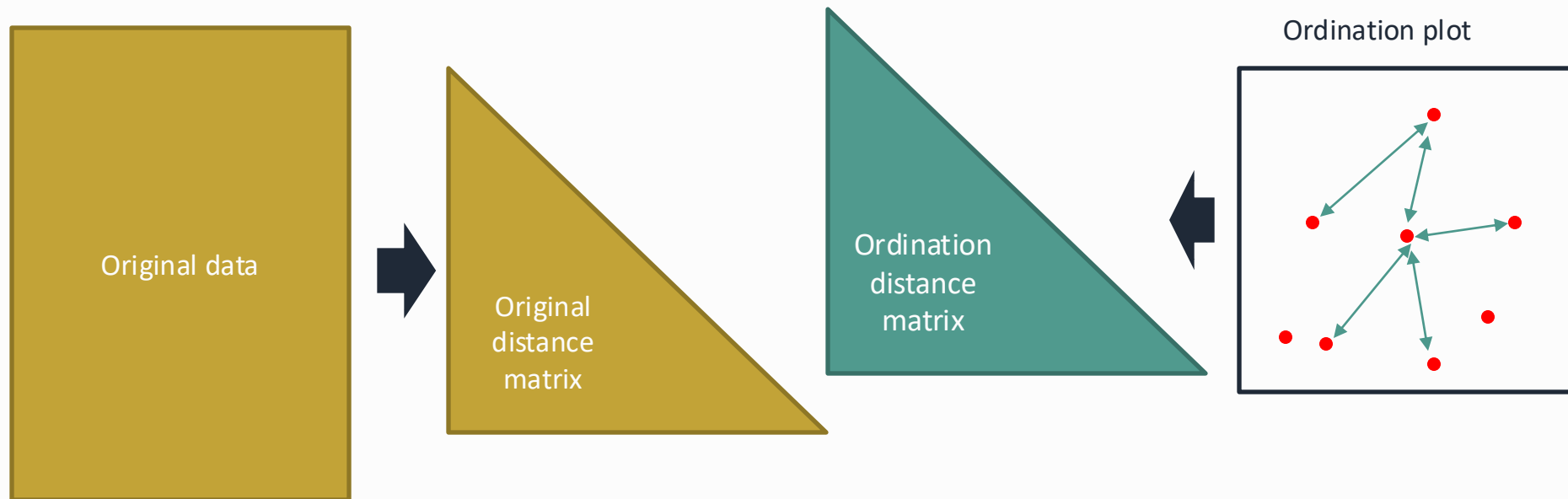
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





(N)MDS - The goal

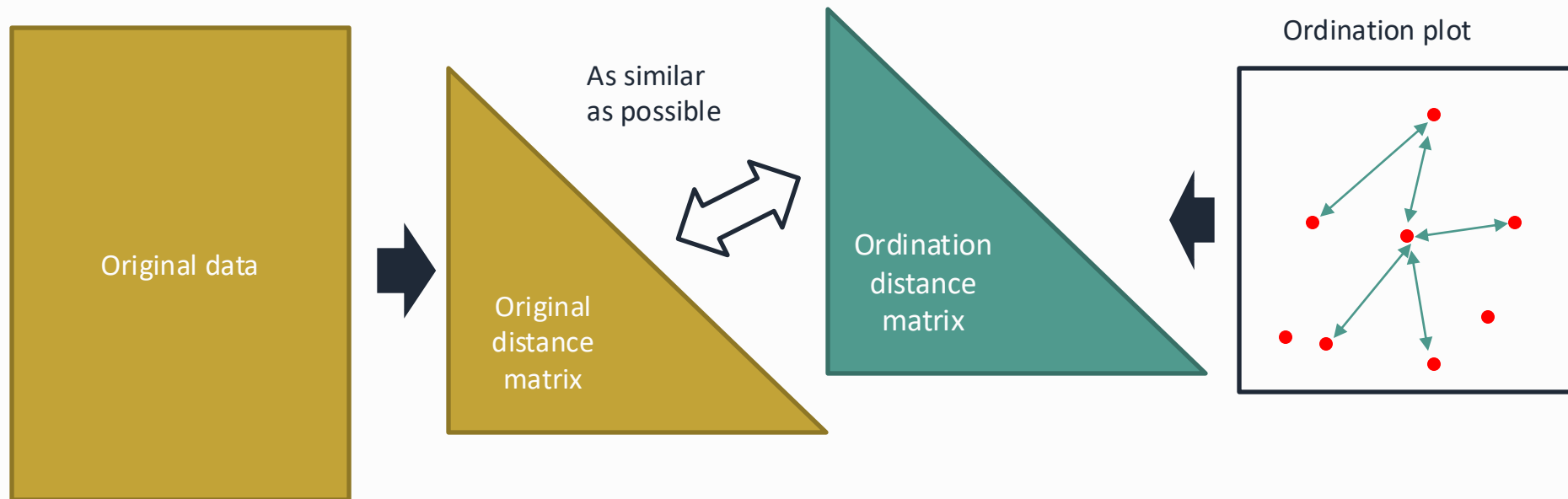
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





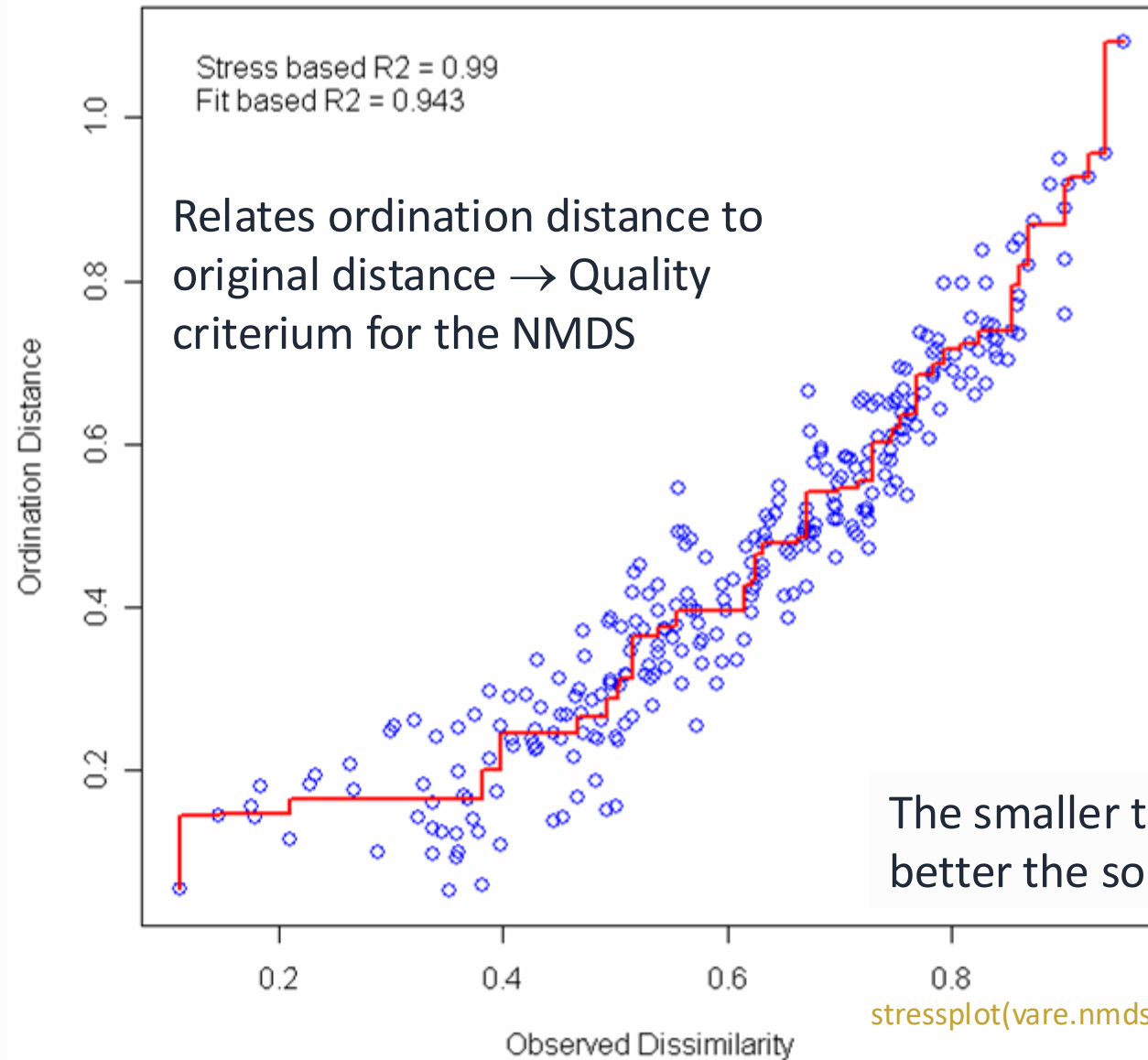
(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





NMDS – stressplot

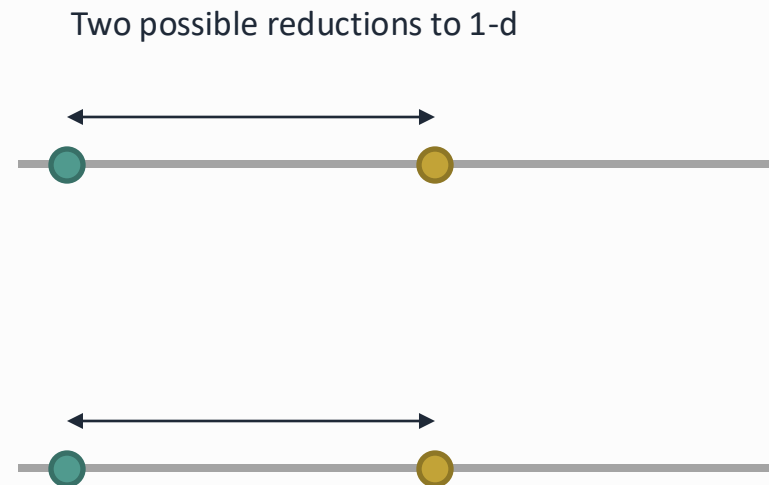
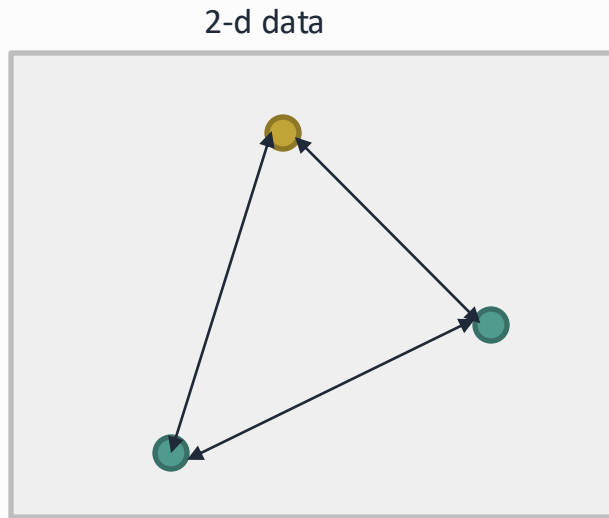


Ordination



Distortion:

- Reducing high dimensional data to low dimensional data causes loss of information:

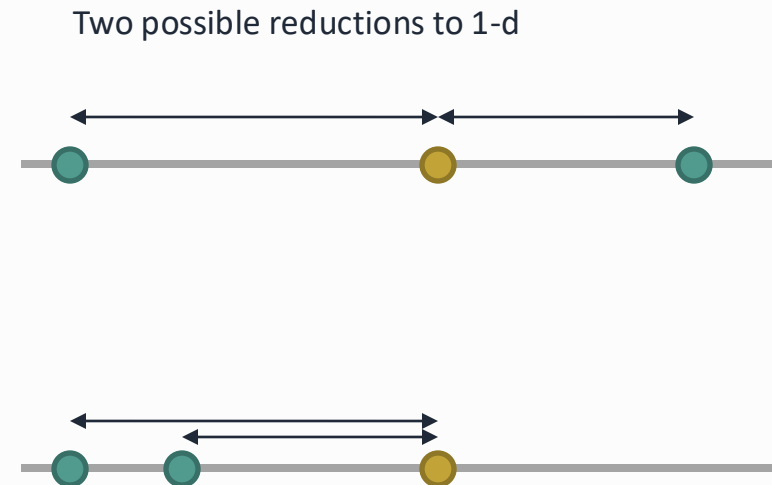
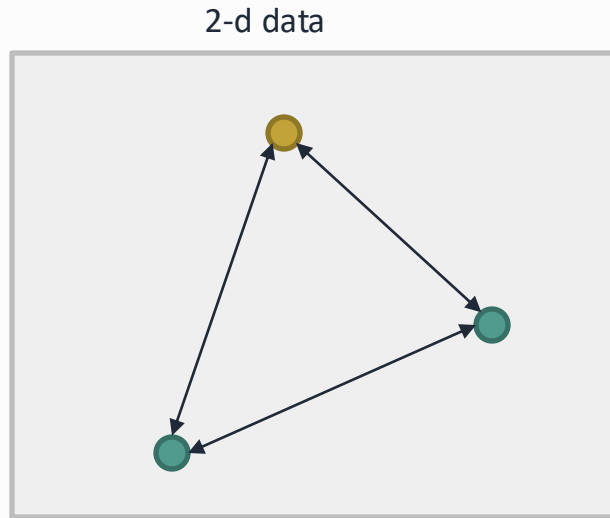


Ordination



Distortion:

- Reducing high dimensional data to low dimensional data causes loss of information:

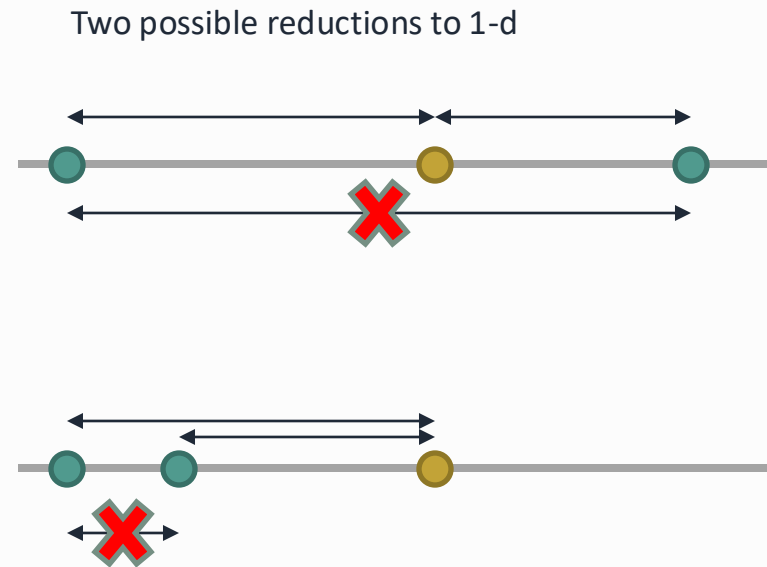
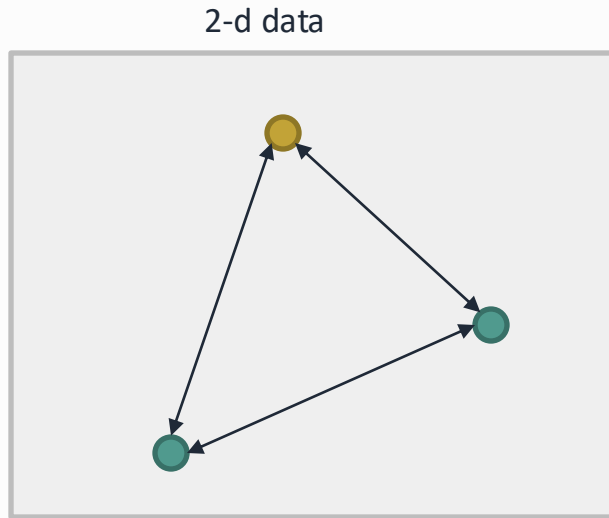


Ordination



Distortion:

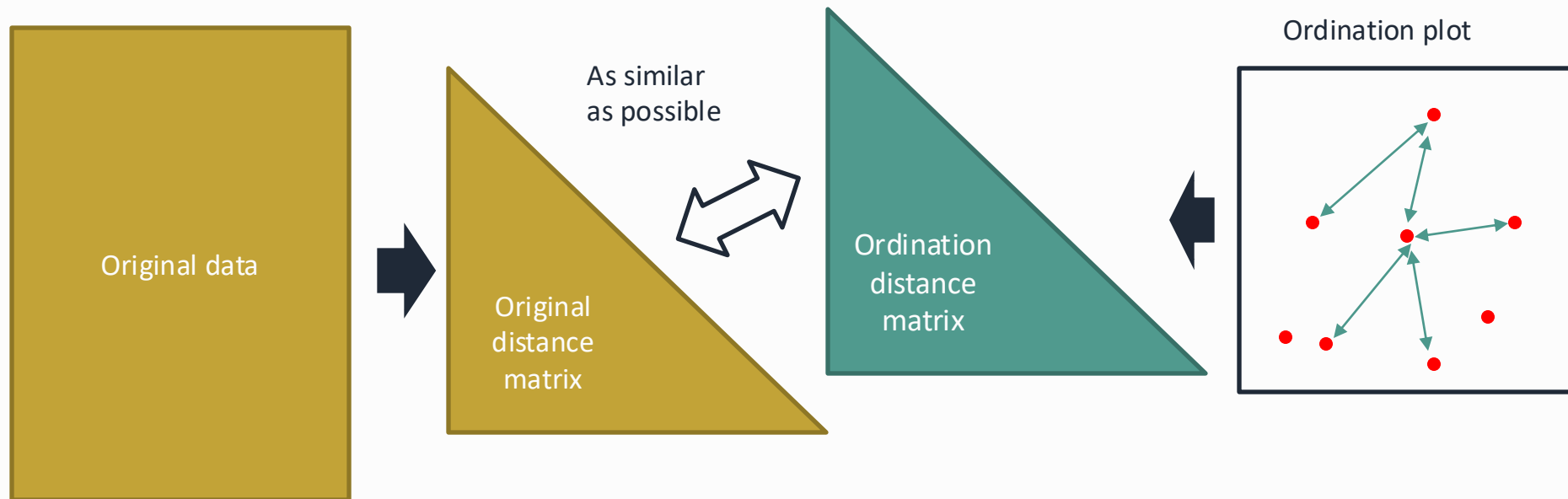
- Reducing high dimensional data to low dimensional data causes loss of information:





(N)MDS

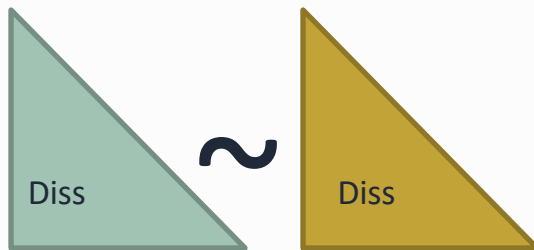
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





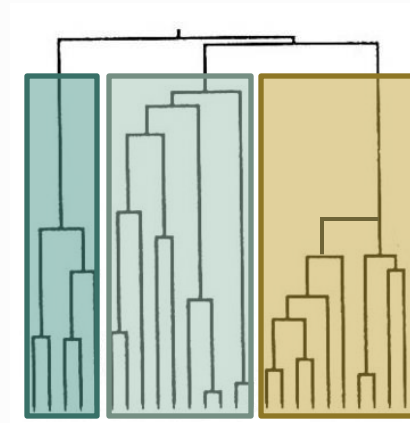
	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆
O ₁	0					
O ₂	d _{2,1}	0				
O ₃	d _{3,1}	d _{3,2}	0			
O ₄	d _{4,1}	d _{4,2}	d _{4,3}	0		
O ₅	d _{5,1}	d _{5,2}	d _{5,3}	d _{5,4}	0	
O ₆	d _{6,1}	d _{6,2}	d _{6,3}	d _{6,4}	d _{6,5}	0

Dissimilarity matrix



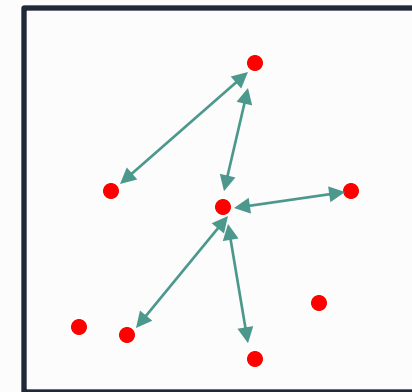
Relate to other
dissimilarity matrix

or



Classification

or



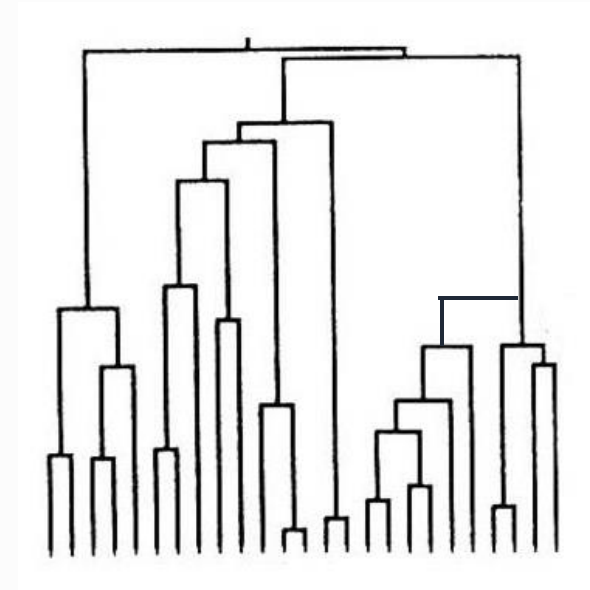
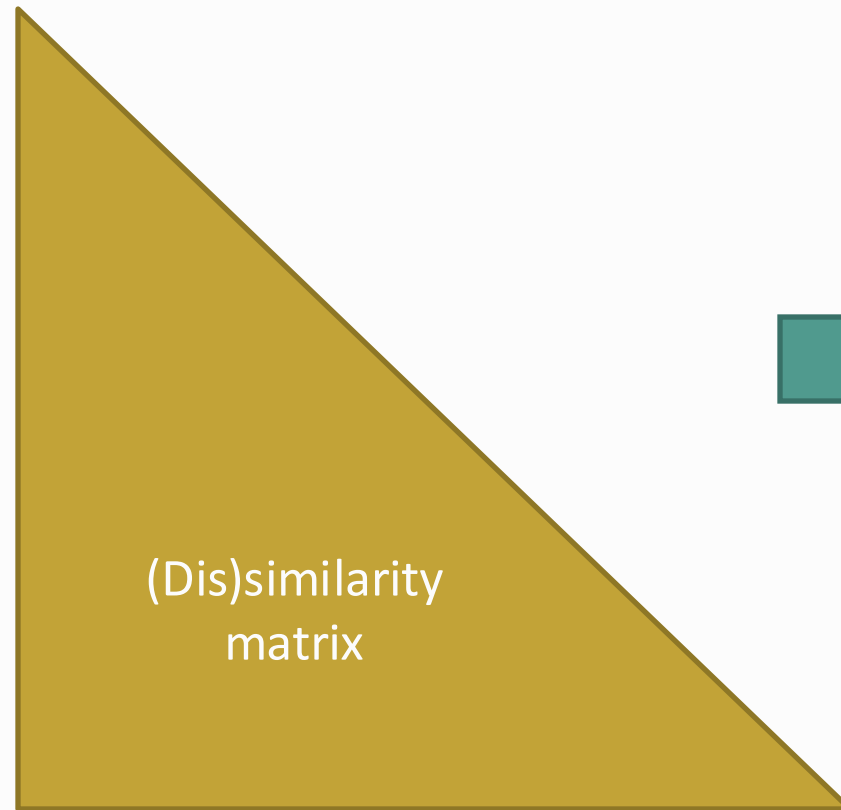
Dimension reduction
(e.g. NMDS)

Questions?

Cluster Analysis



Create a dendrogram by division or agglomeration

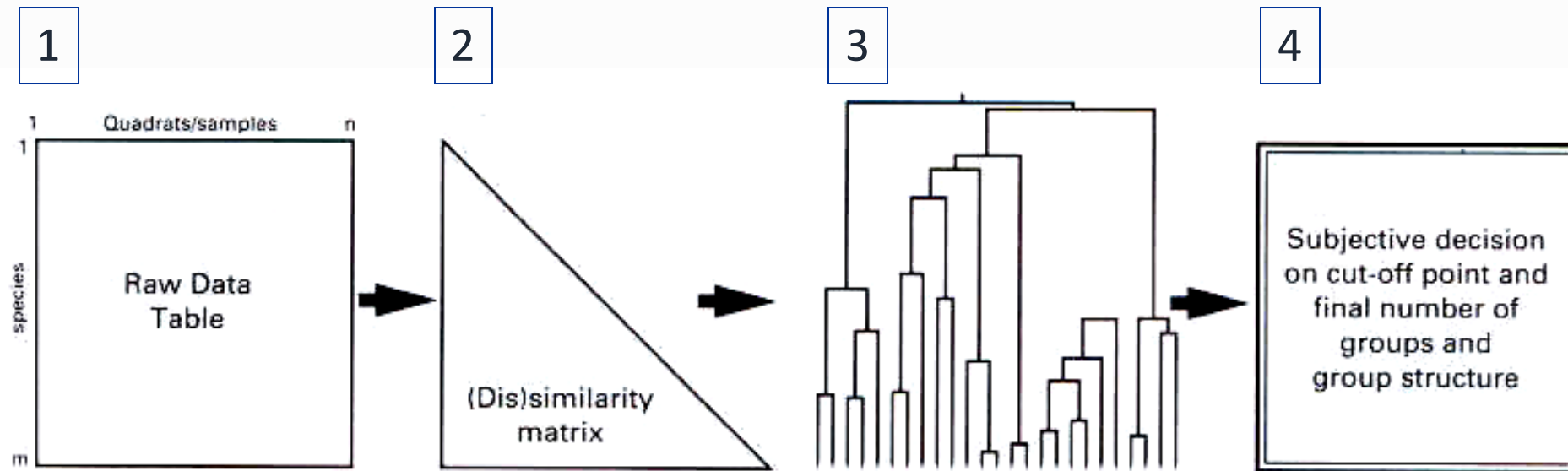




Cluster Analysis

Approach:

Raw data [1] to calculate a dissimilarity matrix [2], visualised in a cluster diagram [3], as baseline for a grouping structure [4]



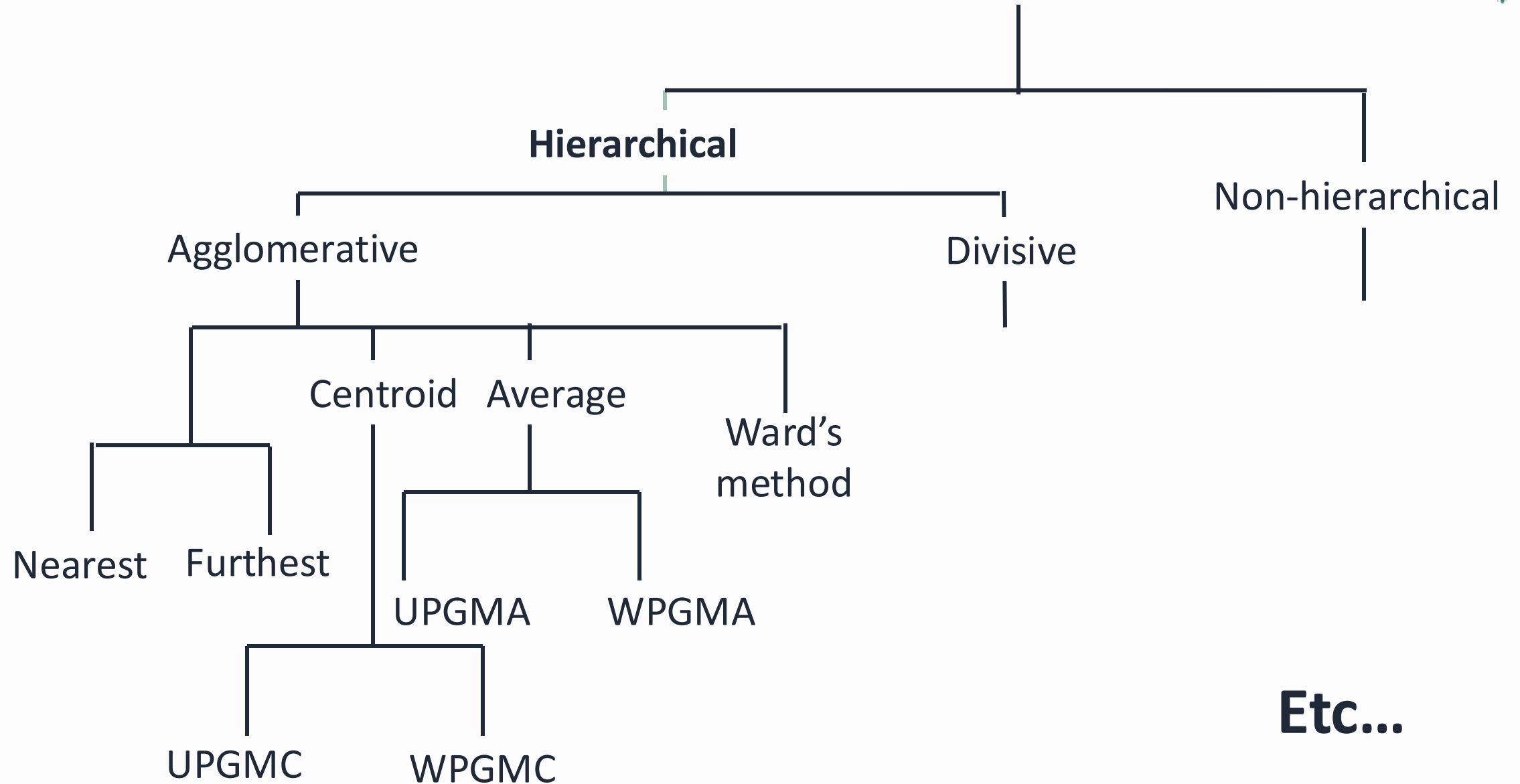


Clustering Algorithms

How are groups defined:

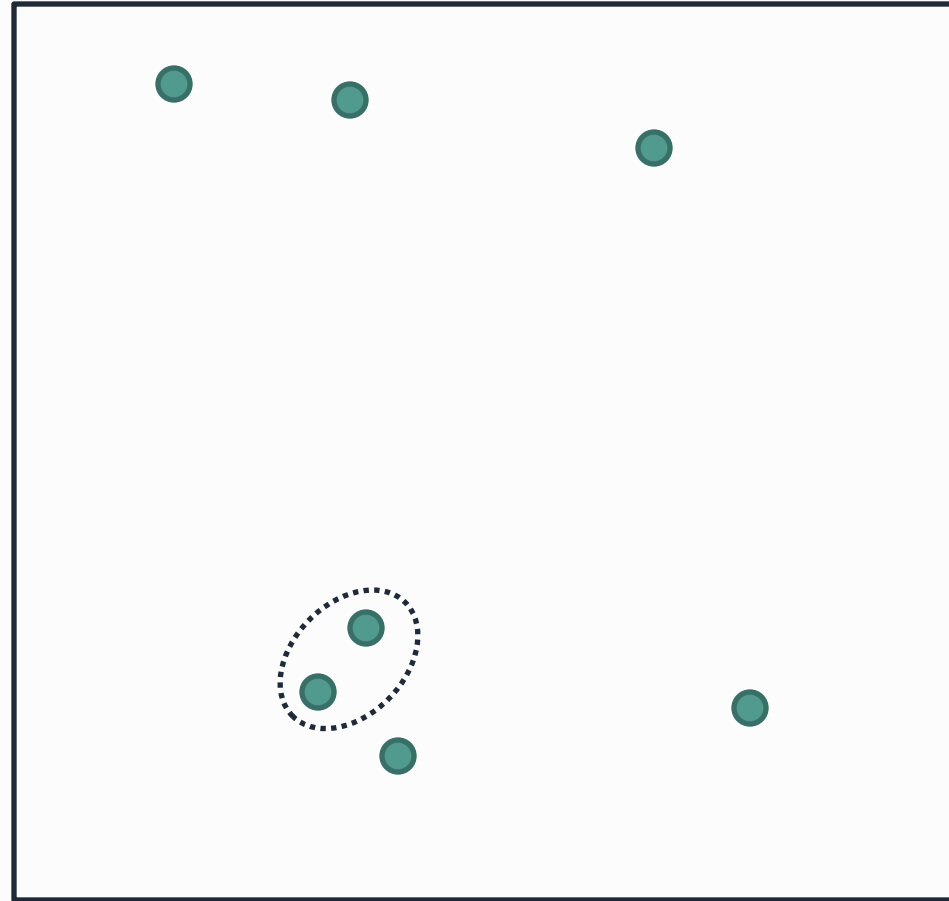
- **Hierarchical** algorithms: several smaller groups can be re-joined on a higher level.
- **Non-hierarchical** algorithms: Groups exist without any relation between each other.

Clustering Algorithms



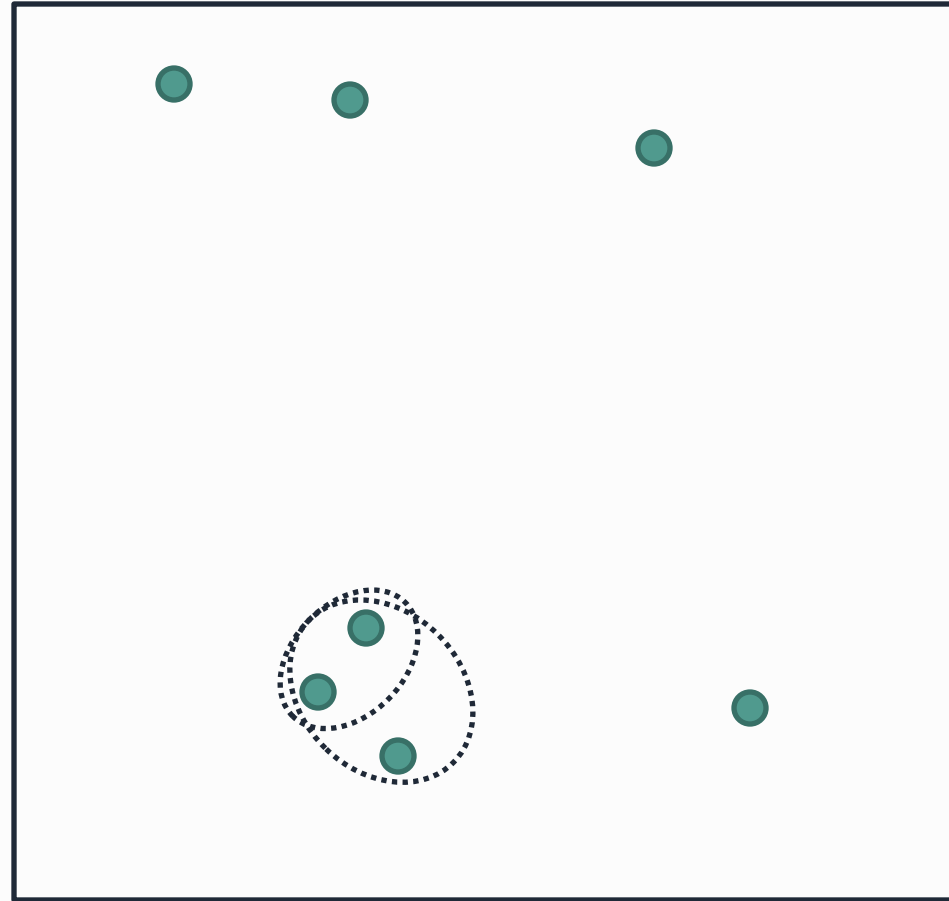


Linkage Algorithms: Single Linkage



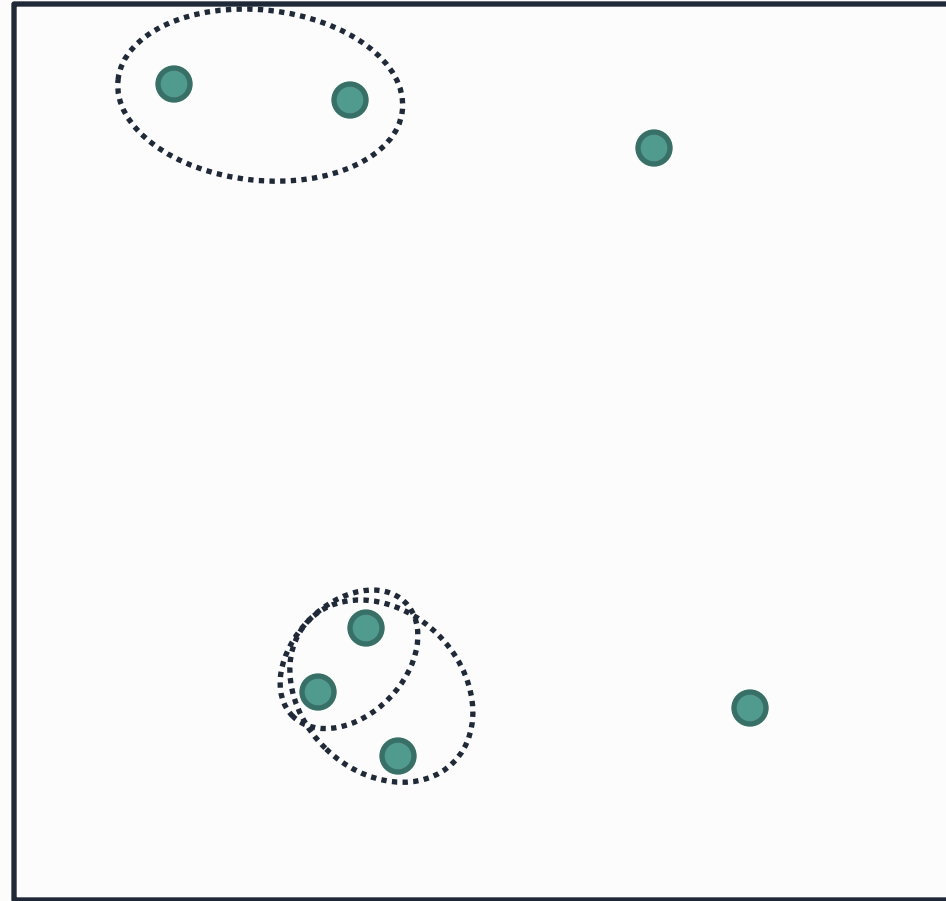


Linkage Algorithms: Single Linkage



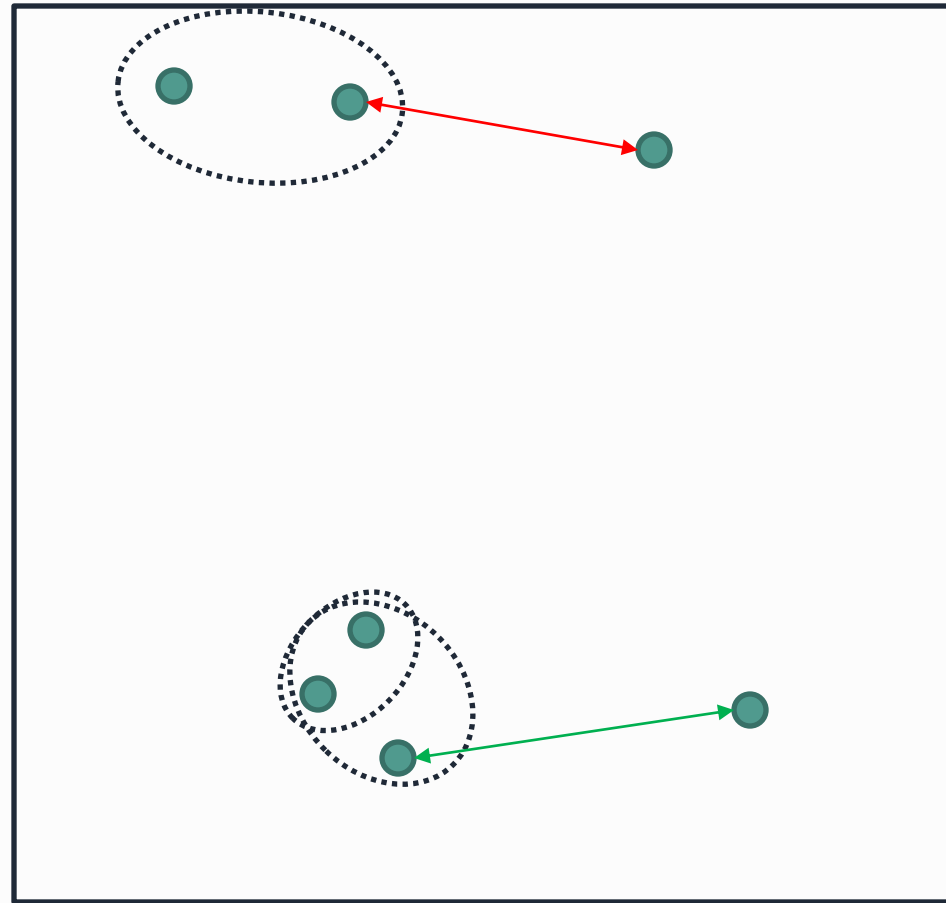


Linkage Algorithms: Single Linkage



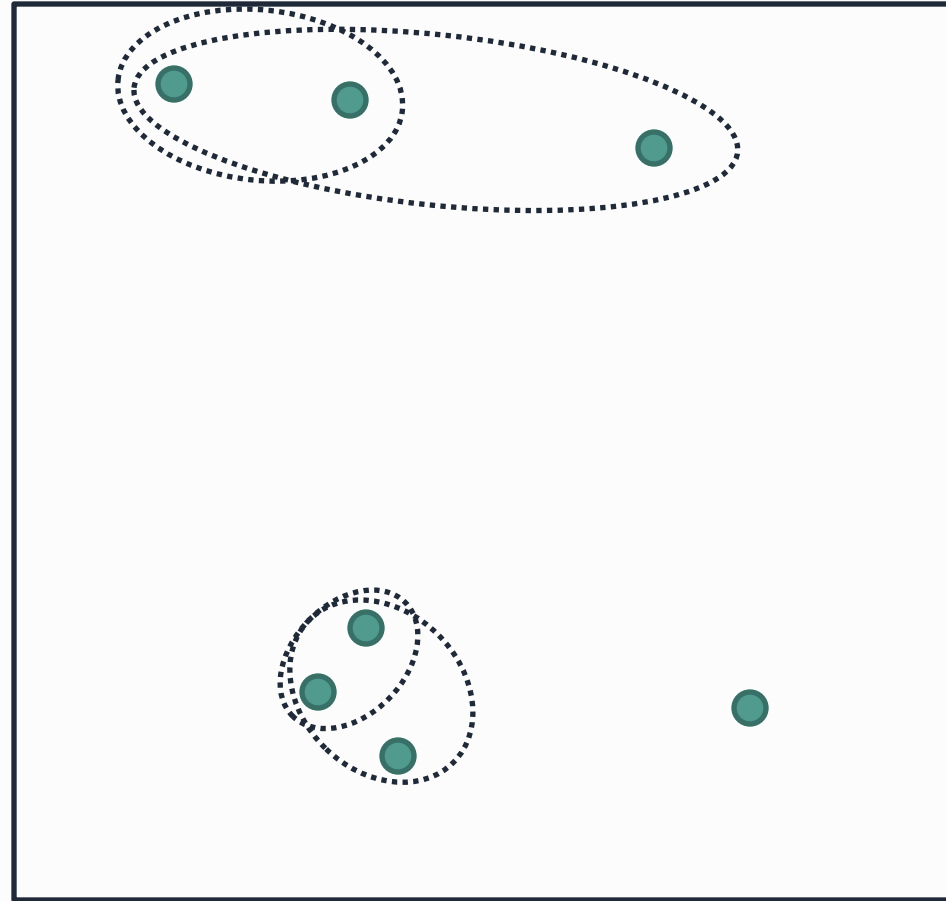


Linkage Algorithms: Single Linkage



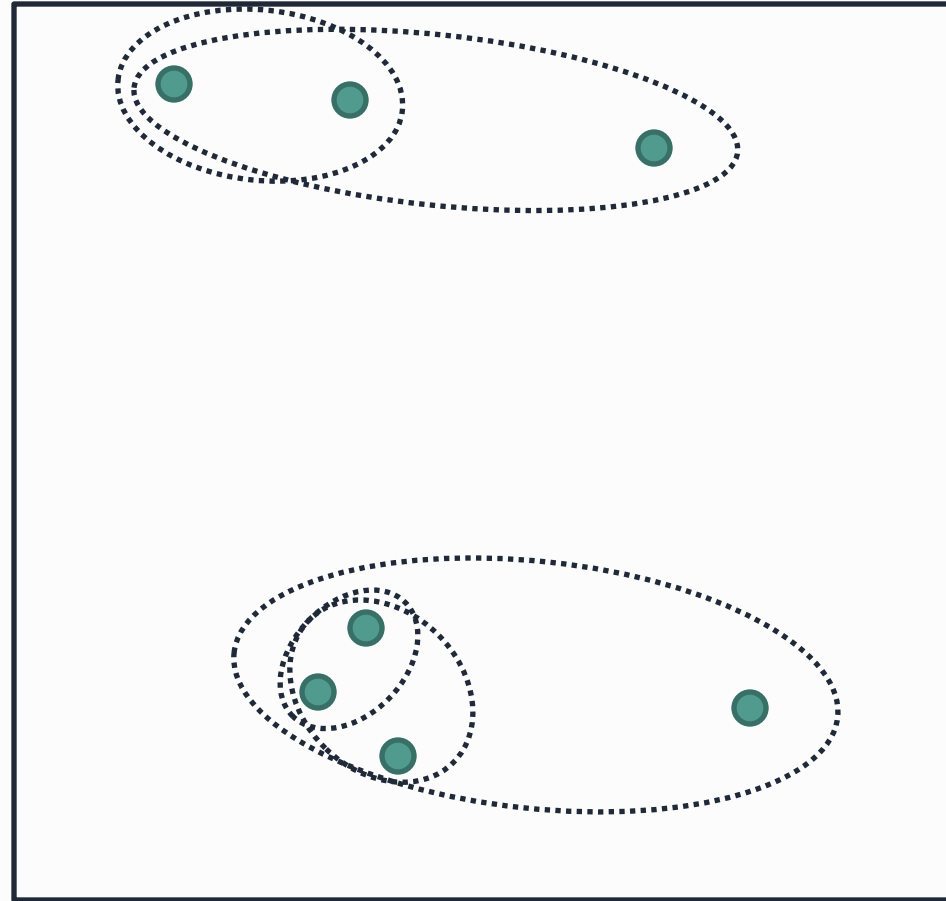


Linkage Algorithms: Single Linkage



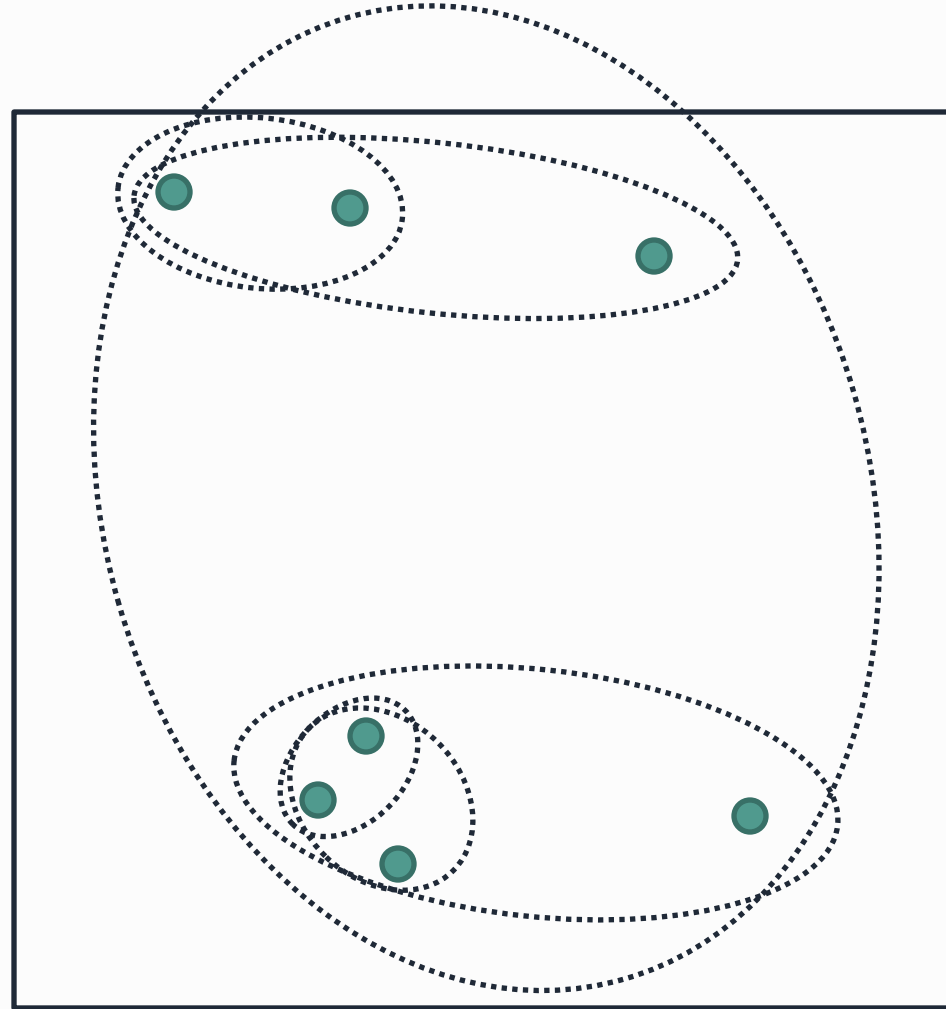


Linkage Algorithms: Single Linkage



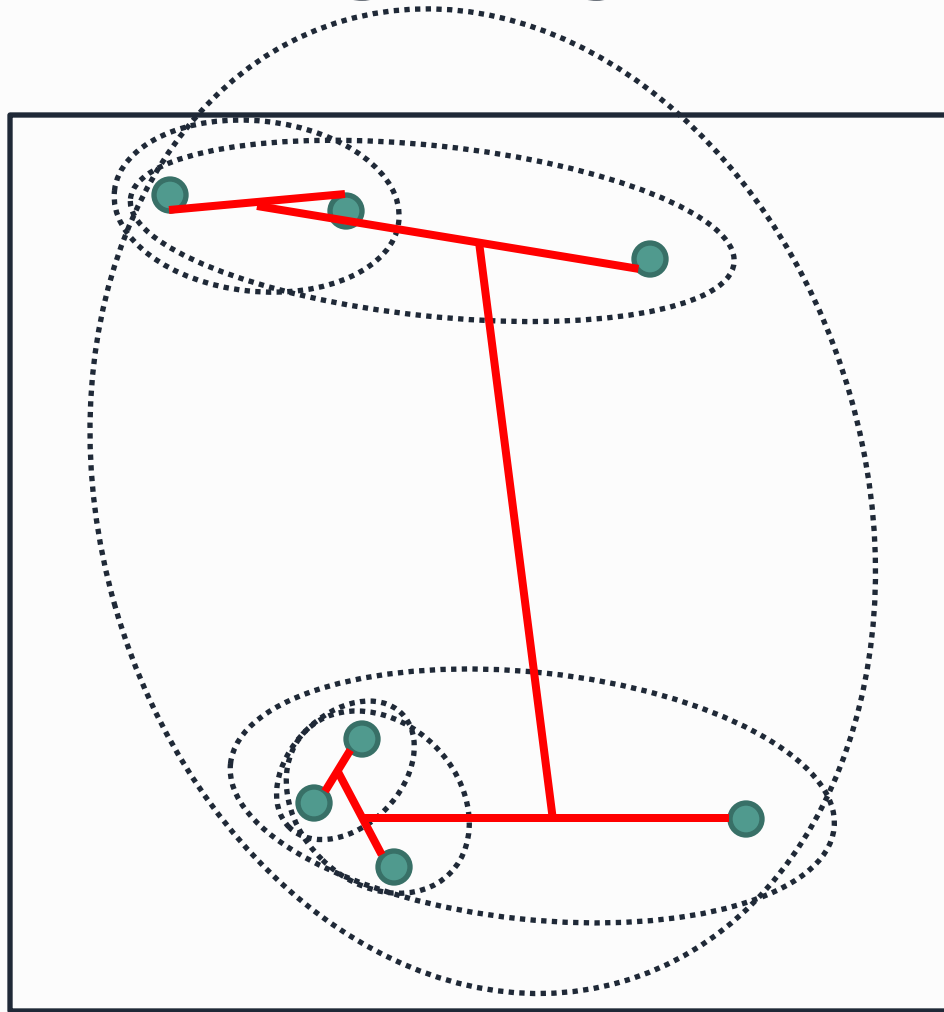


Linkage Algorithms: Single Linkage



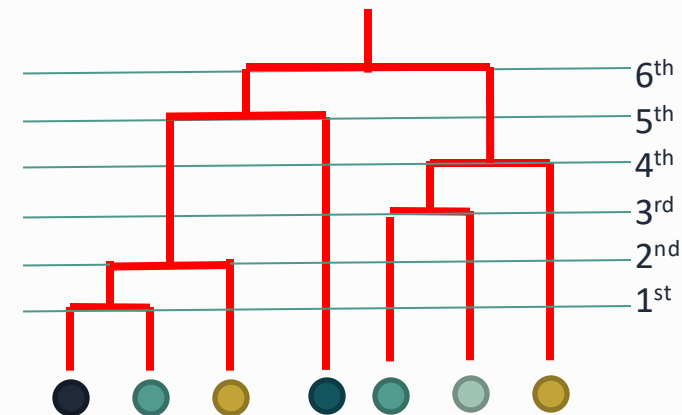
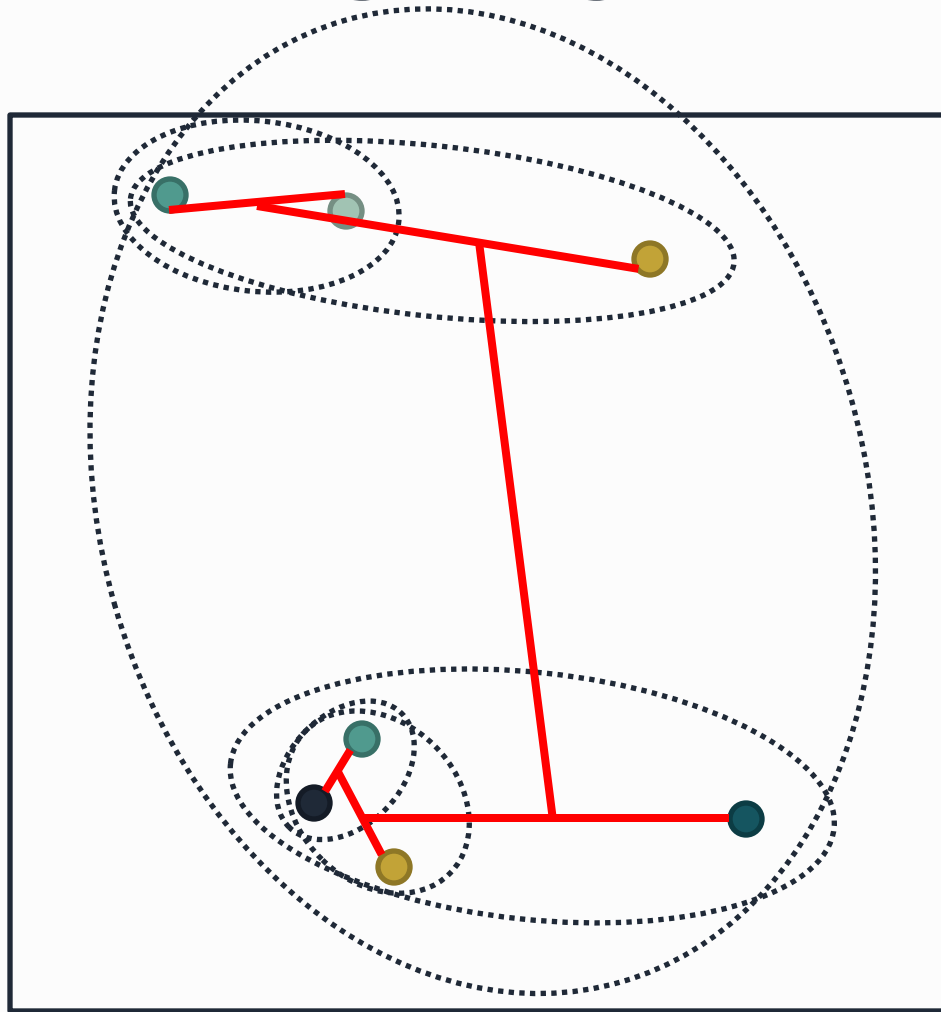


Linkage Algorithms: Single Linkage



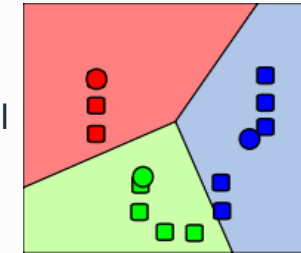
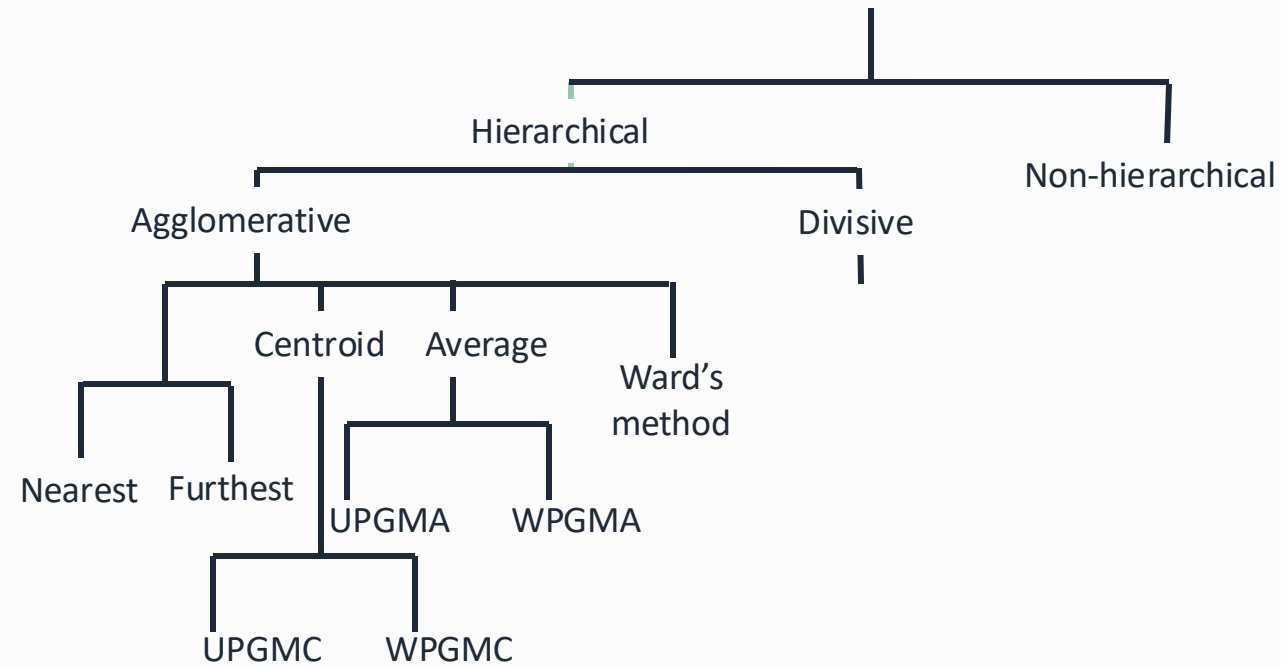


Linkage Algorithms: Single Linkage



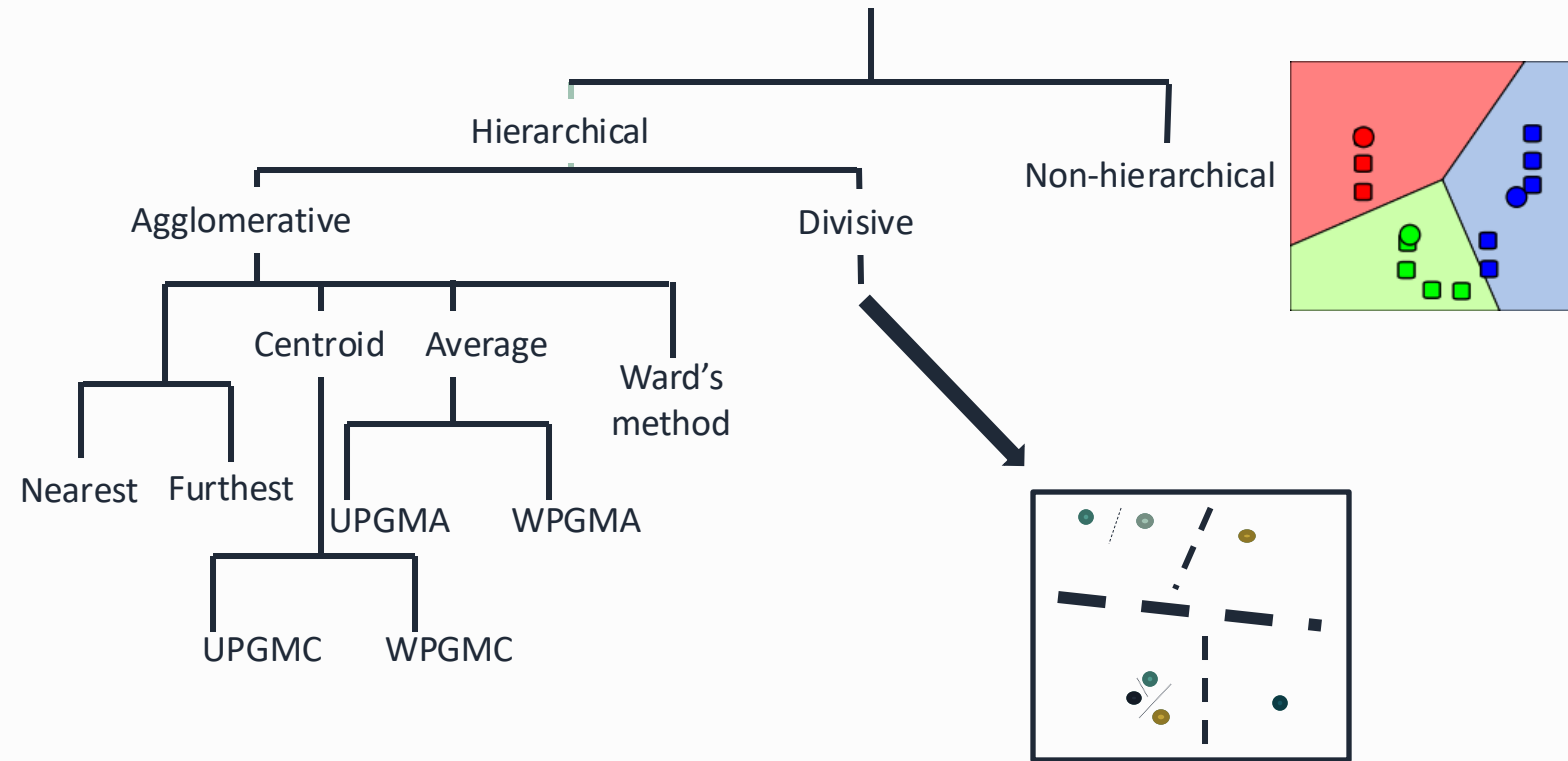


Clustering Algorithms



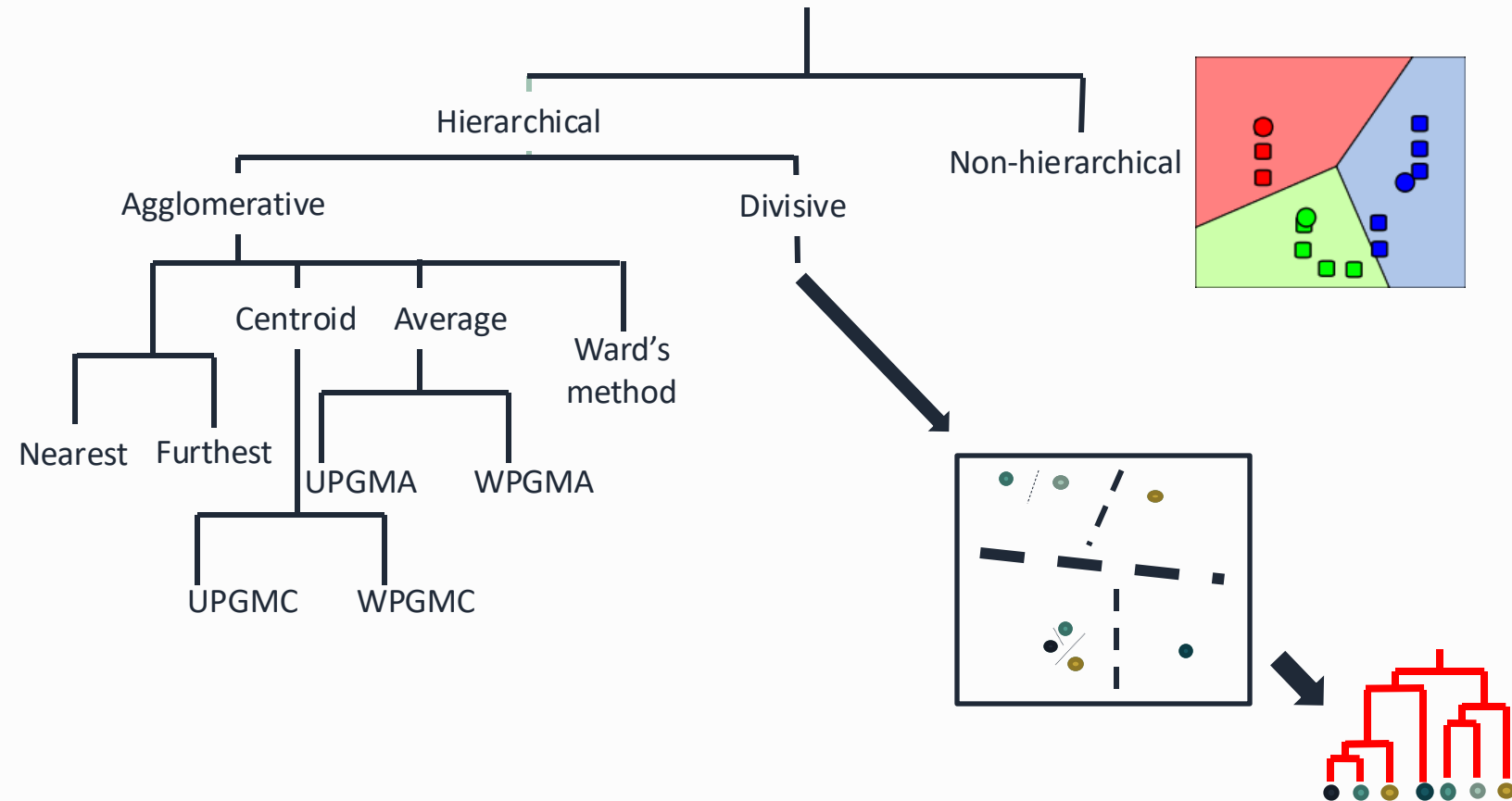


Clustering Algorithms



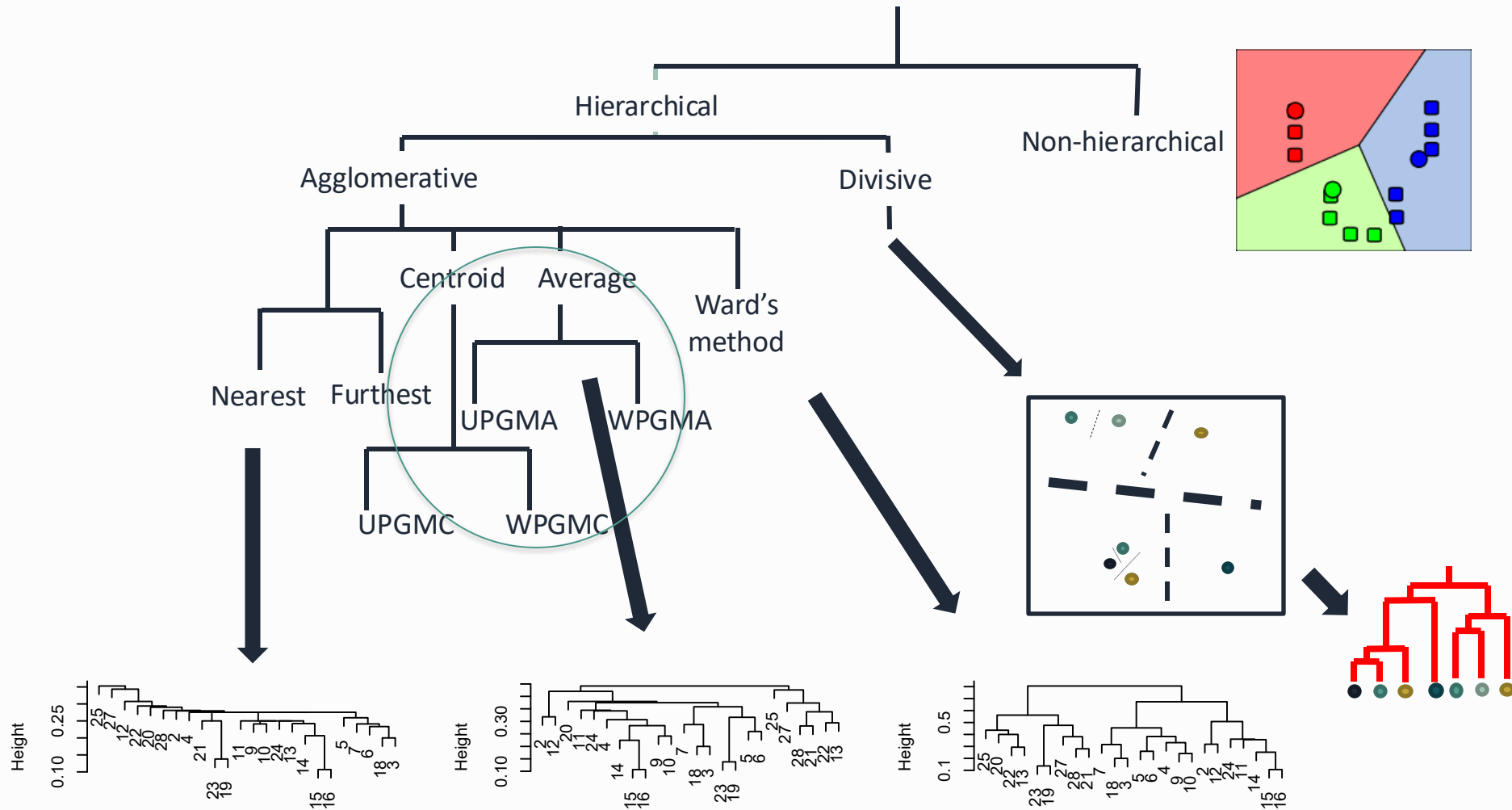


Clustering Algorithms





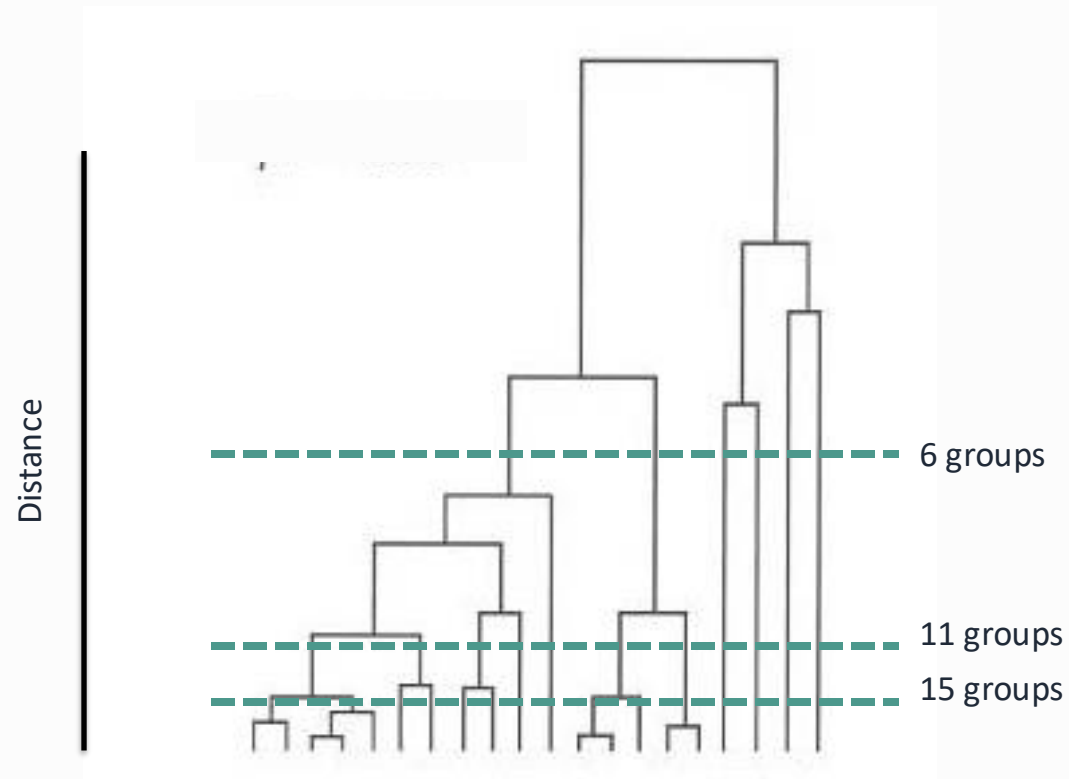
Clustering Algorithms





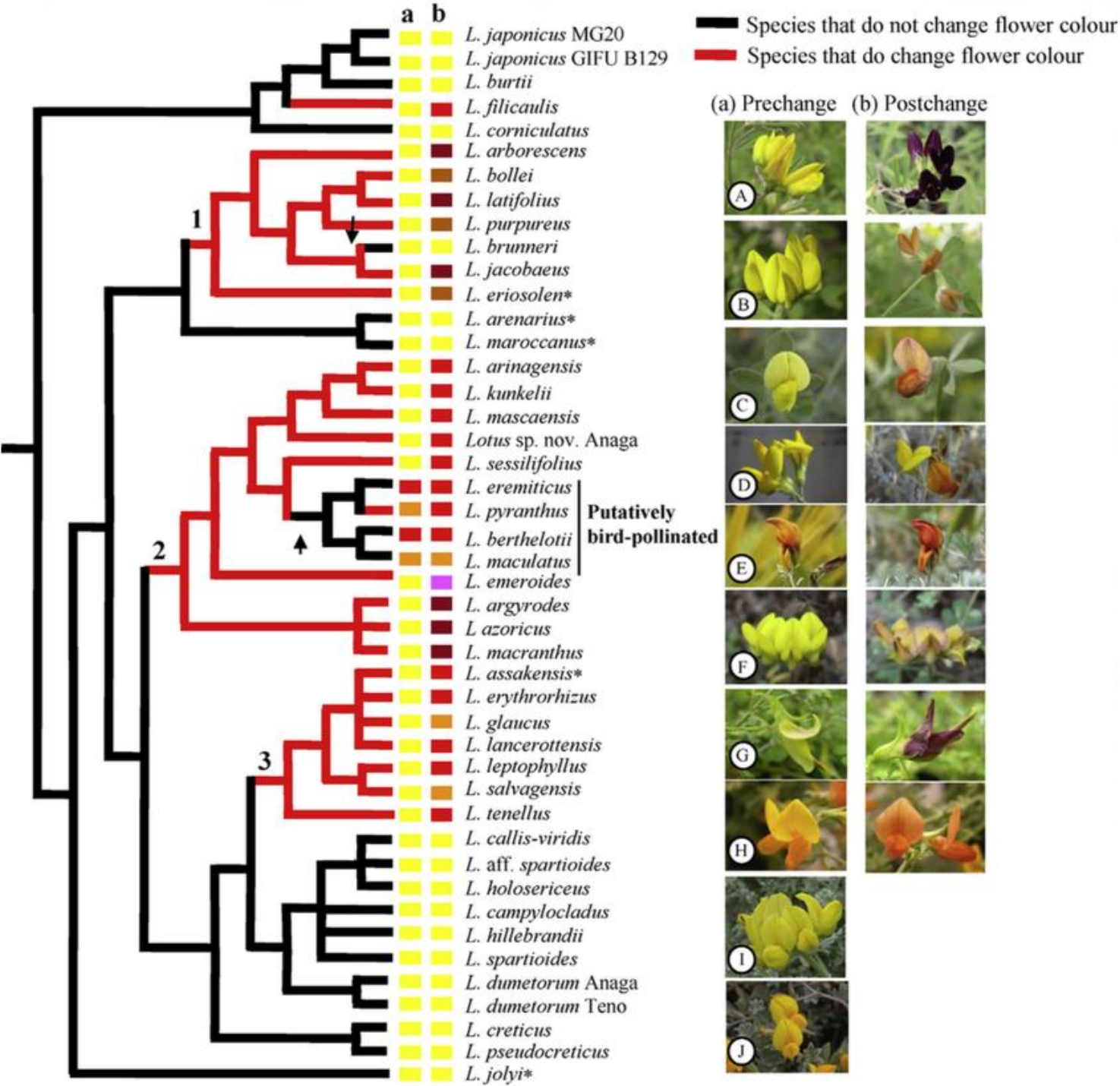
Define groups

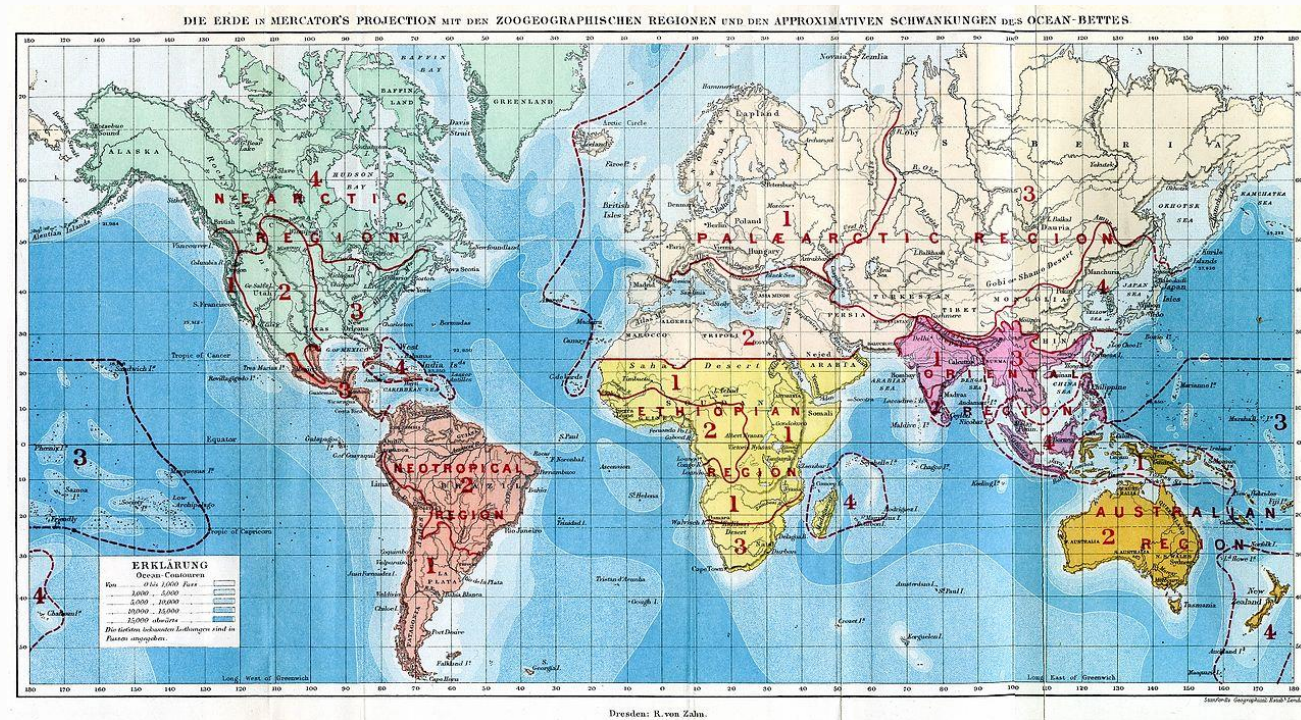
Groups can be defined at various levels (*cophenetic distances*)



The tree can also be the aim:
Phylogenetic tree of
Lotuses

Mort et al. (2002)
Systematic Botany

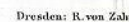




A black and white portrait of Alfred Russel Wallace, an elderly man with a long white beard and glasses, wearing a dark suit. The portrait is oval-shaped. Below the portrait is his signature, "Alfred R. Wallace", written in cursive.

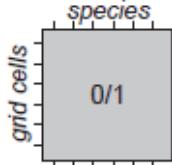
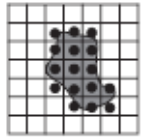
Alfred Wallace

³*Department of Ecology and Evolutionary Biology, Yale University, PO Box 208106, 165 Prospect Street, New Haven, CT 06520-8106, USA*



[1] Defining purpose and objectives

[2] Distribution data



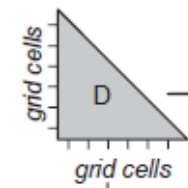
[3] Selection of distance metric



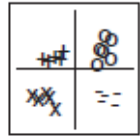
[4] Visualizing turnover



Distance matrix



[5] Ordination

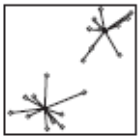


Map of results

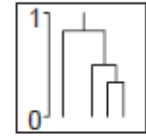


[6] Cluster analysis

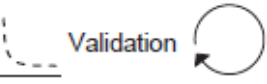
Non-hierarchical



Hierarchical

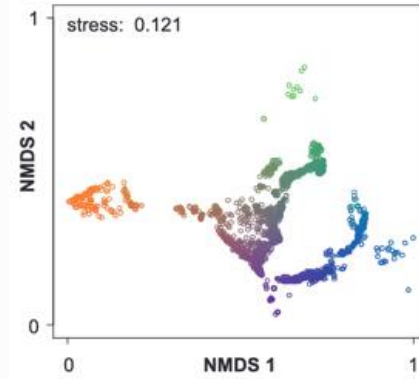


Validation



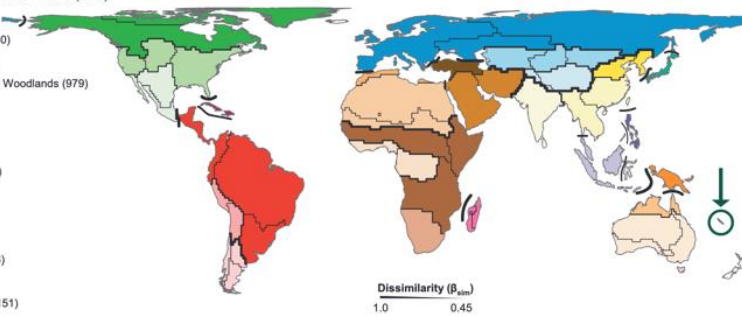
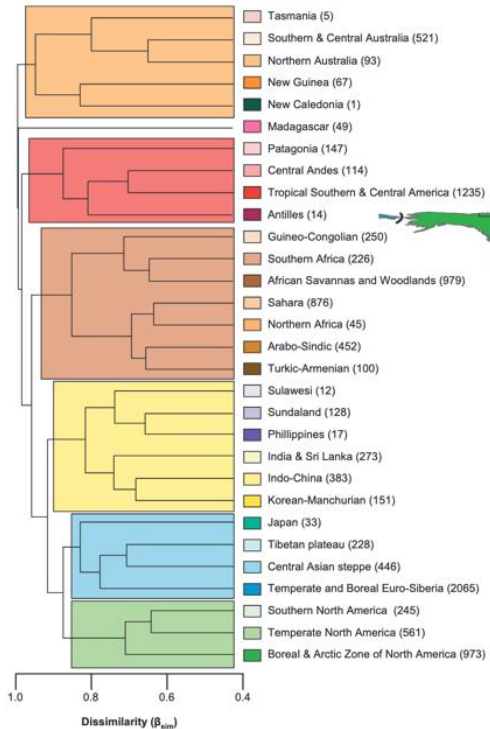
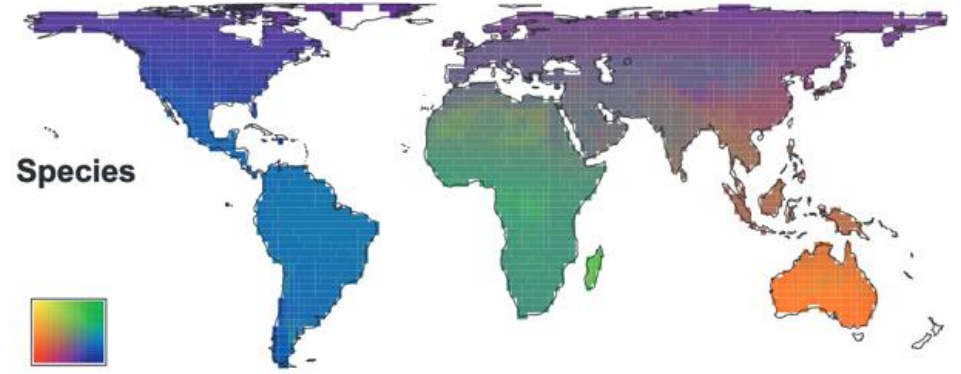
Map of results

Spatial framework for further historical, ecological, and conservation biogeographical studies



(f)

Species



Questions?





Tutorials

- Great web-page with resources on Community Data in R by David Zelený:
www.davidzeleny.net/anadat-r/doku.php/en:start
- How joint SDMs help analysing community data (Paper and R package) by Maximilian Pichler and Florian Hartig:
besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210X.13687

Thank you!